



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II Situation Publication - Common

Release 1.2rev0

Národní Dopravní Informační Centrum

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Summary

The specification defines format for publishing common traffic information (accidents etc.).

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [?]. For additional information see [?].

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [?]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [?], *ElaboratedDataPublication* [?], *PredefinedLocationsPublication* [?] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing general information about traffic situation. Common traffic information include, in particular, accidents, restrictions, closures, traffic conditions, road activities, etc.

2.1 Publication type SituationPublication

Format is based on DATEX II publication of type *SituationPublication* [?]. The profile uses most of data structures, provided by this type of publication.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
   ↳instance" xmlns="http://datex2.eu/schema/2/2_0" xsi:schemaLocation="http://datex2.
   ↳eu/schema/2/2_0 ../schema/schema.xsd">
3   <exchange>
4     <supplierIdentification>
5       <country>cz</country>
6       <nationalIdentifier>DIC</nationalIdentifier>
7     </supplierIdentification>
8   </exchange>
9   <payloadPublication xsi:type="SituationPublication" lang="cs">
10    <feedDescription>
11      <values>
12        <value lang="cs">x-source:cz-ndic_d2-common-v1.2</value>
13      </values>
14    </feedDescription>
15    <feedType>x-format:cz-ndic_d2-common-v1.2</feedType>
16    <publicationTime>2023-08-30T09:18:24+02:00</publicationTime>
17    <publicationCreator>
18      <country>cz</country>
19      <nationalIdentifier>NDIC</nationalIdentifier>
20    </publicationCreator>
```

Each file contains a unique identifier of a source in the *feedDescription* element and a unique identifier of a source format in the *feedType* element. Both of these identifications link to the national access point (<https://registr.dopravniinfo.cz/>).

2.2 Event description

Each event is described by the *situation* element, the described situation can evolve in time, so the element is versioned and has its unique identifier.

The *situationRecord* element can occur more than once in the *situation* element (but each time of a different type), it is usually information related to one event, situation. The *situationRecord* element can evolve over time, so the element is versioned and has its unique identifier derived from the identifier of the parent situation.

The traffic situation publication in particular includes:

- Increased traffic levels, i.e. queuing traffic, stationary traffic
- Accidents
- Activities at the roadside,
- Operating conditions
 - road conditions related to weather (snow, water, etc.)
 - other road condition (sand, oil, spilled load, etc.)
 - conditions given by the environment (precipitation, visibility, wind, etc.)
- Operator Actions
 - Roadworks
 - Network management
 - * Detours
 - * Speed limitation
 - * Changing of traffic organization

This publication does not include any status information.

Most of the structures provided by the *SituationPublication* class are used to describe events. See the W3C XML Schema or [?] as appropriate for the exact definition. e.g.

```
<situationRecord version="2" xsi:type="RoadOrCarriagewayOrLaneManagement" id=
  ↪"ff808181-83c8-5229-0186-cfcf07786763_RoadOrCarriagewayOrLaneManagement">
```

```
<situationRecord version="2" xsi:type="Accident" id="ff808181-83c8-5229-0186-
  ↪cfcf07786763_Accident">
```

The impact of an event is described in the *impact* element, either in terms of expected delay or an estimate of the remaining traffic capacity (when closure = 0).

```
<impact>
  <capacityRemaining>0</capacityRemaining>
</impact>
```

If the cause of the event is known, it is described by one value from the enumeration in the *cause* element or in the *accidentCause* element in the case of the *Accident* class.

```
<cause xsi:type="NonManagedCause">
  <causeType>earlierAccident</causeType>
</cause>
```

```
<accidentCause>limitedVisibility</accidentCause>
```

Any additional supplementary information on the event is given by an html link in the *urlLink* element.

```

56     <urlLink>
57         <urlLinkAddress>dopravniinfo.cz/te/ff808181-53b7-4de8-0154-d27e38fb58ea</
↪urlLinkAddress>
58         <urlLinkType>html</urlLinkType>
59     </urlLink>

```

The indication that this is a safety-related event according to the Delegated Regulation 886/2013 is given in the *safetyRelatedMessage* tag.

```

261 <situationRecordExtension>
262     <situationRecordExtendedApproved>
263         <safetyRelatedMessage>true</safetyRelatedMessage>
264     </situationRecordExtendedApproved>
265 </situationRecordExtension>

```

2.3 Location description

Each *SituationRecord* has its own position description, which may (or may not) differ within a *Situation*. For example, one *SituationRecord* describing a point source problem that resulted in a linear problem (the second *SituationRecord*).

A location tends to be described in a *SituationRecord* by multiple methods, all referring to the same slice of reality.

```

60 <groupOfLocations xsi:type="Linear">

```

The position description methods are further described in the chapter “Location description”.

TIME VALIDITY

Time validity is stated independently for every single *SituationRecord* described in a situation and is defined by a time interval between two time instants. The class *Validity* is used with an instance of class *OverallPeriod* in the role of *validityTimeSpecification* and elements *overallStartTime* and *overallStopTime* to define start and end of the interval.

Setting a week-day and recurring time validity within the specified period, e.g. for recurring intermittent nightly tunnel closure, is not allowed. For such information, splitting to more messages valid in respective times is used.

Used time validity intervals may fall into past, present or future.

```
36     <validity>
37         <validityStatus>definedByValidityTimeSpec</validityStatus>
38         <validityTimeSpecification>
39             <overallStartTime>2023-03-11T09:30:00+01:00</overallStartTime>
40             <overallEndTime>2023-11-11T12:35:00+01:00</overallEndTime>
41         </validityTimeSpecification>
42     </validity>
```

LOCATION DESCRIPTION

When describing locations, nested descriptions of instances of the *Location* class are **always** used, namely the descendants of the *NetworkLocation* class.

The *Linear* class is always used to describe a position, regardless of the fact that it may be a point position, e.g. an accident, expressible by a single coordinate. The reason for this is to unify descriptions between multiple methods. Area type position descriptions and separate point position descriptions are not used.

60

```
<groupOfLocations xsi:type="Linear">
```

Multiple methods can be used to describe a location, depending on their availability in a given location. At least one description must be given. Within a single situation, each *situationRecord* can have a different location, for example a point position for an accident and a line position for a caused queue or caused closure.

The following description lists the methods used for location description.

4.1 Location description by means of ALERT-C

The method describes the position using ALERT-C codes, class *alertCLinear* of type *AlertCMethod2Linear* is used. The use of this method depends on the presence of an ALERT-C location at a given location.

The *alertCLinear* method uses a primary location (*alertCMethod2PrimaryPointLocation* element), a secondary location (*alertCMethod2SecondaryPointLocation* element) to describe the event. Of course, the method also contains elements allowing to identify a specific location table.

The primary location is, according to the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the convoy or where the driver first enters the problem area.

In order to find out the direction of the situation, the *alertCDirectionCoded* element is used, which takes the value *positive* if the direction of the traffic flow agrees with the coding of the given route.

Important: WARNING, the meaning of this attribute is therefore exactly the opposite of the *direction* attribute in RDS-TMC, which is used to find the secondary location.

The *alertCLinear* element may occur multiple times in the situation description for long events that must be described by more than one ALERT-C path. This is because ALERT-C only allows locations to be chained within a single path.

Note: The *alertCLinear* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment). When primary and secondary locations are the same. The direction can be read from the *alertCDirectionCoded* element).

```

58 <alertCLinear xsi:type="AlertCMethod2Linear">
59   <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
60   <alertCLocationTableNumber>25</alertCLocationTableNumber>
61   <alertCLocationTableVersion>10.1</alertCLocationTableVersion>
62   <alertCDirection>
63     <alertCDirectionCoded>positive</alertCDirectionCoded>
64   </alertCDirection>
65   <alertCMethod2PrimaryPointLocation>
66     <alertCLocation>
67       <specificLocation>18231</specificLocation>
68     </alertCLocation>
69   </alertCMethod2PrimaryPointLocation>
70   <alertCMethod2SecondaryPointLocation>
71     <alertCLocation>
72       <specificLocation>10388</specificLocation>
73     </alertCLocation>
74   </alertCMethod2SecondaryPointLocation>
75 </alertCLinear>

```

4.2 Location description using mileage

The method uses *linearWithinLinearElement* for the description of the position using mileage. The *roadNumber* element describes the number of the affected road, the *directionRelativeOnLinearSection* element describes the direction of the affected road, and the *fromPoint* and *toPoint* elements describe the distance of the start and end of the event from the start of the road in meters.

Note: The *linearWithinLinearElement* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short section). When the offsets *fromPoint* and *toPoint* refer to a segment with zero length.

```

76 <linearExtension>
77   <openlrExtendedLinear>
78     <firstDirection>
79       <openlrLocationReferencePoint>
80         <openlrCoordinate>
81           <latitude>50.136887007413563</latitude>
82           <longitude>14.504980597632043</longitude>
83         </openlrCoordinate>
84         <openlrLineAttributes>
85           <openlrFunctionalRoadClass>FRC4</openlrFunctionalRoadClass>
86           <openlrFormOfWay>multipleCarriageway</openlrFormOfWay>
87           <openlrBearing>350</openlrBearing>
88         </openlrLineAttributes>
89         <openlrPathAttributes>
90           <openlrLowestFRCToNextLRPoint>FRC4</openlrLowestFRCToNextLRPoint>
91           <openlrDistanceToNextLRPoint>165</openlrDistanceToNextLRPoint>
92         </openlrPathAttributes>
93       </openlrLocationReferencePoint>
94       <openlrLastLocationReferencePoint>
95         <openlrCoordinate>
96           <latitude>50.138363834263082</latitude>
97           <longitude>14.504772075868614</longitude>
98         </openlrCoordinate>

```

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```

99      <openlrLineAttributes>
100        <openlrFunctionalRoadClass>FRC4</openlrFunctionalRoadClass>
101        <openlrFormOfWay>multipleCarriageway</openlrFormOfWay>
102        <openlrBearing>180</openlrBearing>
103      </openlrLineAttributes>
104    </openlrLastLocationReferencePoint>
105    <openlrOffsets>
106      <openlrPositiveOffset>16</openlrPositiveOffset>
107      <openlrNegativeOffset>4</openlrNegativeOffset>
108    </openlrOffsets>
109  </firstDirection>
110 </openlrExtendedLinear>

```

4.3 Location description using the Global Network extension

Native method of position input in NDIC using **Extension of DATEX II model of level B**.

The method describes, in the *globalNetworkLinear* element, the location using the segments *linearWithinLinearGNElement* of the Global Network (GN) road network, identified by the *networkVersion* version. Each segment contains a sequence, an identifier referring to the GN, and a relative truncation (for the first and last segment). Furthermore, the position is bounded by the elements *startPoint* and *endPoint* in coordinates (here S-JTSK).

Note: The *globalNetworkLinear* method is also used to describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment). The *startPoint* and *endPoint* positions contain the same coordinate.

Note: To determine the length of the affected sections, the *PercentageDistanceAlongLinearElement* class is used, it uses the distance from the end and the beginning of the section expressed as a percentage of the section length (where the value 100 = 100%, i.e. the end of the section).

```

121 <globalNetworkLinear>
122   <networkVersion>22.12</networkVersion>
123   <linearGeometryType>continuous</linearGeometryType>
124   <startPoint>
125     <sjtskPointCoordinates>
126       <sjtskX>-668325.9586937</sjtskX>
127       <sjtskY>-976774.250129909</sjtskY>
128     </sjtskPointCoordinates>
129   </startPoint>
130   <endPoint>
131     <sjtskPointCoordinates>
132       <sjtskX>-668050.419197776</sjtskX>
133       <sjtskY>-976823.636025029</sjtskY>
134     </sjtskPointCoordinates>
135   </endPoint>
136   <linearWithinLinearGNElement>
137     <sectionId>126853</sectionId>
138     <directionRelativeOnLinearSection>opposite</
↪ directionRelativeOnLinearSection>
139     <orderOfSection>0</orderOfSection>
140     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
141       <percentageDistanceAlong>9.224</percentageDistanceAlong>
142     </fromPoint>

```

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```

143         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
144             <percentageDistanceAlong>0</percentageDistanceAlong>
145         </toPoint>
146     </linearWithinLinearGNElement>
147     <linearWithinLinearGNElement>
148         <sectionId>126599</sectionId>
149         <directionRelativeOnLinearSection>opposite</
150     <directionRelativeOnLinearSection>
151         <orderOfSection>1</orderOfSection>
152         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
153             <percentageDistanceAlong>100</percentageDistanceAlong>
154         </fromPoint>
155         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
156             <percentageDistanceAlong>0</percentageDistanceAlong>
157         </toPoint>
158     </linearWithinLinearGNElement>
159     <linearWithinLinearGNElement>
160         <sectionId>1461717</sectionId>
161         <directionRelativeOnLinearSection>opposite</
162     <directionRelativeOnLinearSection>
163         <orderOfSection>2</orderOfSection>
164         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
165             <percentageDistanceAlong>100</percentageDistanceAlong>
166         </fromPoint>
167         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
168             <percentageDistanceAlong>0</percentageDistanceAlong>
169         </toPoint>
170     </linearWithinLinearGNElement>
171     <linearWithinLinearGNElement>
172         <sectionId>1461719</sectionId>
173         <directionRelativeOnLinearSection>opposite</
174     <directionRelativeOnLinearSection>
175         <orderOfSection>3</orderOfSection>
176         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
177             <percentageDistanceAlong>100</percentageDistanceAlong>
178         </fromPoint>
179         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
180             <percentageDistanceAlong>0</percentageDistanceAlong>
181         </toPoint>
182     </linearWithinLinearGNElement>
183     <linearWithinLinearGNElement>
184         <sectionId>2901326</sectionId>
185         <directionRelativeOnLinearSection>opposite</
186     <directionRelativeOnLinearSection>
187         <orderOfSection>4</orderOfSection>
188         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
189             <percentageDistanceAlong>100</percentageDistanceAlong>
190         </fromPoint>
191         <toPoint xsi:type="PercentageDistanceAlongLinearElement">
192             <percentageDistanceAlong>14.795</percentageDistanceAlong>
193         </toPoint>
194     </linearWithinLinearGNElement>
195 </globalNetworkLinear>

```

4.4 Location description using OpenLR extension

The OpenLR method in the *openlrExtendedLinear* element, uses a routing algorithm to describe the location and thus allows on-the-fly localization.

The method uses the reference points *openlrLocationReferencePoint* and *openlrLastLocationReferencePoint* containing coordinates, road direction, and a road type. These reference points are used as the start and end to find the road which is further refined using offsets to the actual extent of the event. In case 2 points are not enough the method allows to use intermediate points.

Note: To describe a **point location** (even the inside of an intersection or tunnel, i.e. a short segment) the *openlrExtendedLinear* method is also used, when offsets refer to a segment with zero length.

```

87 <openlrExtendedLinear>
88   <firstDirection>
89     <openlrLocationReferencePoint>
90       <openlrCoordinate>
91         <latitude>50.765282088427412</latitude>
92         <longitude>15.337521544012871</longitude>
93       </openlrCoordinate>
94       <openlrLineAttributes>
95         <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
96         <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
97         <openlrBearing>112</openlrBearing>
98       </openlrLineAttributes>
99       <openlrPathAttributes>
100         <openlrLowestFRCToNextLRPoint>FRC1</openlrLowestFRCToNextLRPoint>
101         <openlrDistanceToNextLRPoint>1240</openlrDistanceToNextLRPoint>
102       </openlrPathAttributes>
103     </openlrLocationReferencePoint>
104     <openlrLastLocationReferencePoint>
105       <openlrCoordinate>
106         <latitude>50.763208691718496</latitude>
107         <longitude>15.347566415800618</longitude>
108       </openlrCoordinate>
109       <openlrLineAttributes>
110         <openlrFunctionalRoadClass>FRC1</openlrFunctionalRoadClass>
111         <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
112         <openlrBearing>283</openlrBearing>
113       </openlrLineAttributes>
114     </openlrLastLocationReferencePoint>
115     <openlrOffsets>
116       <openlrPositiveOffset>120</openlrPositiveOffset>
117       <openlrNegativeOffset>418</openlrNegativeOffset>
118     </openlrOffsets>
119   </firstDirection>
120 </openlrExtendedLinear>

```

APPENDIX A: MESSAGE EXAMPLES

5.1 accident.xml - truck accident with a road closure

Publication contains exactly one message about a truck accident with a road closure.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
↪instance" xmlns="http://datex2.eu/schema/2/2_0" xsi:schemaLocation="http://datex2.
↪eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="SituationPublication" lang="cs">
    <feedDescription>
      <values>
        <value lang="cs">x-source:cz-ndic_d2-common-v1.2</value>
      </values>
    </feedDescription>
    <feedType>x-format:cz-ndic_d2-common-v1.2</feedType>
    <publicationTime>2023-08-30T09:18:24+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <situation id="ff808181-83c8-5229-0186-cfcf07786763" version="2">
      <situationVersionTime>2023-03-11T09:37:09+01:00</situationVersionTime>
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>urgent</urgency>
      </headerInformation>
      <!-- situation record 1 -->
      <situationRecord version="2" xsi:type="RoadOrCarriagewayOrLaneManagement" id=
↪"ff808181-83c8-5229-0186-cfcf07786763_RoadOrCarriagewayOrLaneManagement">
        <situationRecordCreationTime>2023-03-11T09:37:09+01:00</
↪situationRecordCreationTime>
        <situationRecordVersionTime>2023-03-11T09:37:09+01:00</
↪situationRecordVersionTime>
        <probabilityOfOccurrence>certain</probabilityOfOccurrence>
        <source>
          <sourceIdentification>PCR</sourceIdentification>
        </source>
      </situationRecord>
    </situation>
  </payloadPublication>
</d2LogicalModel>
```

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```

<validity>
  <validityStatus>definedByValidityTimeSpec</validityStatus>
  <validityTimeSpecification>
    <overallStartTime>2023-03-11T09:30:00+01:00</overallStartTime>
    <overallEndTime>2023-11-11T12:35:00+01:00</overallEndTime>
  </validityTimeSpecification>
</validity>
<impact>
  <capacityRemaining>0</capacityRemaining>
</impact>
<cause xsi:type="NonManagedCause">
  <causeType>earlierAccident</causeType>
</cause>
<generalPublicComment>
  <comment>
    <values>
      <value lang="cs">Od 11.3.2023 09:30 do 11.11.2023 12:35; na silnici 450_
↪ u obce Staré Město okres Bruntál; vozovka uzavřena v obou směrech; nehoda_
↪ nákladního vozidla; Objízdná trasa vede po místních komunikacích.</value>
    </values>
  </comment>
</generalPublicComment>
<urlLink>
  <urlLinkAddress>dopravniinfo.cz/te/ff808181-53b7-4de8-0154-d27e38fb58ea</
↪ urlLinkAddress>
  <urlLinkType>html</urlLinkType>
</urlLink>
<groupOfLocations xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>positive</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>11896</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>11895</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
  <linearExtension>
    <openlrExtendedLinear>
      <firstDirection>
        <openlrLocationReferencePoint>
          <openlrCoordinate>
            <latitude>49.996120448045858</latitude>
            <longitude>17.449393414128387</longitude>
          </openlrCoordinate>
          <openlrLineAttributes>
            <openlrFunctionalRoadClass>FRC2</openlrFunctionalRoadClass>
          </openlrLineAttributes>
        </openlrLocationReferencePoint>
      </firstDirection>
    </openlrExtendedLinear>
  </linearExtension>
</groupOfLocations>

```

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```

        <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
        <openlrBearing>321</openlrBearing>
      </openlrLineAttributes>
      <openlrPathAttributes>
        <openlrLowestFRCToNextLRPoint>FRC2</openlrLowestFRCToNextLRPoint>
        <openlrDistanceToNextLRPoint>642</openlrDistanceToNextLRPoint>
      </openlrPathAttributes>
    </openlrLocationReferencePoint>
    <openlrLastLocationReferencePoint>
      <openlrCoordinate>
        <latitude>50.0006509650643</latitude>
        <longitude>17.444126048234285</longitude>
      </openlrCoordinate>
      <openlrLineAttributes>
        <openlrFunctionalRoadClass>FRC2</openlrFunctionalRoadClass>
        <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
        <openlrBearing>123</openlrBearing>
      </openlrLineAttributes>
    </openlrLastLocationReferencePoint>
    <openlrOffsets>
      <openlrPositiveOffset>0</openlrPositiveOffset>
      <openlrNegativeOffset>0</openlrNegativeOffset>
    </openlrOffsets>
  </firstDirection>
</openlrExtendedLinear>
<globalNetworkLinear>
  <networkVersion>22.12</networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <!-- dummy coordinates -->
  <startPoint>
    <sjtskPointCoordinates>
      <sjtskX>-528652.81170224119</sjtskX>
      <sjtskY>-1078139.7963582373</sjtskY>
    </sjtskPointCoordinates>
  </startPoint>
  <endPoint>
    <sjtskPointCoordinates>
      <sjtskX>-528652.81170224119</sjtskX>
      <sjtskY>-1078139.7963582373</sjtskY>
    </sjtskPointCoordinates>
  </endPoint>
  <linearWithinLinearGNElement>
    <sectionId>1620017</sectionId>
    <directionRelativeOnLinearSection>aligned</
→directionRelativeOnLinearSection>
    <orderOfSection>0</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>100</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
</globalNetworkLinear>
</linearExtension>
</groupOfLocations>

```

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```

    <complianceOption>mandatory</complianceOption>
    <applicableForTrafficDirection>bothWays</applicableForTrafficDirection>
    <roadOrCarriagewayOrLaneManagementType>roadClosed</
↪roadOrCarriagewayOrLaneManagementType>
  </situationRecord>
  <!-- situation record 2 -->
  <situationRecord version="2" xsi:type="Accident" id="ff808181-83c8-5229-0186-
↪cfcf07786763_Accident">
    <situationRecordCreationTime>2023-03-11T09:37:09+01:00</
↪situationRecordCreationTime>
    <situationRecordVersionTime>2023-03-11T09:37:09+01:00</
↪situationRecordVersionTime>
    <probabilityOfOccurrence>certain</probabilityOfOccurrence>
    <source>
      <sourceIdentification>PCR</sourceIdentification>
    </source>
    <validity>
      <validityStatus>definedByValidityTimeSpec</validityStatus>
      <validityTimeSpecification>
        <overallStartTime>2023-03-11T09:30:00+01:00</overallStartTime>
        <overallEndTime>2023-11-11T12:35:00+01:00</overallEndTime>
      </validityTimeSpecification>
    </validity>
    <impact>
      <capacityRemaining>0</capacityRemaining>
    </impact>
    <generalPublicComment>
      <comment>
        <values>
          <value lang="cs">Od 11.3.2023 09:30 do 11.11.2023 12:35; na silnici 450_
↪u obce Staré Město okres Bruntál; vozovka uzavřena v obou směrech; nehoda_
↪nákladního vozidla; Objízdná trasa vede po místních komunikacích.</value>
        </values>
      </comment>
    </generalPublicComment>
    <urlLink>
      <urlLinkAddress>dopravniinfo.cz/te/ff808181-53b7-4de8-0154-d27e38fb58ea</
↪urlLinkAddress>
      <urlLinkType>html</urlLinkType>
    </urlLink>
    <groupOfLocations xsi:type="Linear">
      <alertCLinear xsi:type="AlertCMethod2Linear">
        <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
        <alertCLocationTableNumber>25</alertCLocationTableNumber>
        <alertCLocationTableVersion>9.0</alertCLocationTableVersion>
        <alertCDirection>
          <alertCDirectionCoded>positive</alertCDirectionCoded>
        </alertCDirection>
        <alertCMethod2PrimaryPointLocation>
          <alertCLocation>
            <specificLocation>11896</specificLocation>
          </alertCLocation>
        </alertCMethod2PrimaryPointLocation>
        <alertCMethod2SecondaryPointLocation>
          <alertCLocation>
            <specificLocation>11895</specificLocation>
          </alertCLocation>
        </alertCMethod2SecondaryPointLocation>
      </alertCLinear>
    </groupOfLocations>
  </situationRecord>

```

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```

    </alertCLocation>
  </alertCMethod2SecondaryPointLocation>
</alertCLinear>
<linearExtension>
  <openlrExtendedLinear>
    <firstDirection>
      <openlrLocationReferencePoint>
        <openlrCoordinate>
          <latitude>49.996120448045858</latitude>
          <longitude>17.449393414128387</longitude>
        </openlrCoordinate>
        <openlrLineAttributes>
          <openlrFunctionalRoadClass>FRC2</openlrFunctionalRoadClass>
          <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
          <openlrBearing>321</openlrBearing>
        </openlrLineAttributes>
        <openlrPathAttributes>
          <openlrLowestFRCToNextLRPoint>FRC2</openlrLowestFRCToNextLRPoint>
          <openlrDistanceToNextLRPoint>642</openlrDistanceToNextLRPoint>
        </openlrPathAttributes>
      </openlrLocationReferencePoint>
      <openlrLastLocationReferencePoint>
        <openlrCoordinate>
          <latitude>50.0006509650643</latitude>
          <longitude>17.444126048234285</longitude>
        </openlrCoordinate>
        <openlrLineAttributes>
          <openlrFunctionalRoadClass>FRC2</openlrFunctionalRoadClass>
          <openlrFormOfWay>singleCarriageway</openlrFormOfWay>
          <openlrBearing>123</openlrBearing>
        </openlrLineAttributes>
      </openlrLastLocationReferencePoint>
      <openlrOffsets>
        <openlrPositiveOffset>211</openlrPositiveOffset>
        <openlrNegativeOffset>431</openlrNegativeOffset>
      </openlrOffsets>
    </firstDirection>
  </openlrExtendedLinear>
</globalNetworkLinear>
<networkVersion>22.12</networkVersion>
<linearGeometryType>continuous</linearGeometryType>
<startPoint>
  <sjtskPointCoordinates>
    <sjtskX>-528652.81170224119</sjtskX>
    <sjtskY>-1078139.7963582373</sjtskY>
  </sjtskPointCoordinates>
</startPoint>
<endPoint>
  <sjtskPointCoordinates>
    <sjtskX>-528652.81170224119</sjtskX>
    <sjtskY>-1078139.7963582373</sjtskY>
  </sjtskPointCoordinates>
</endPoint>
<linearWithinLinearGNElement>
  <sectionId>1620017</sectionId>
  <directionRelativeOnLinearSection>aligned</

```

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```

<directionRelativeOnLinearSection>
  <orderOfSection>0</orderOfSection>
  <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>45.544</percentageDistanceAlong>
  </fromPoint>
  <toPoint xsi:type="PercentageDistanceAlongLinearElement">
    <percentageDistanceAlong>45.544</percentageDistanceAlong>
  </toPoint>
</linearWithinLinearGNElement>
</globalNetworkLinear>
</linearExtension>
</groupOfLocations>
  <!-- safety related message indication -->
<situationRecordExtension>
  <situationRecordExtendedApproved>
    <safetyRelatedMessage>true</safetyRelatedMessage>
  </situationRecordExtendedApproved>
</situationRecordExtension>
  <!-- accident cause -->
<accidentCause>limitedVisibility</accidentCause>
<accidentType>accidentInvolvingHeavyLorries</accidentType>
<groupOfVehiclesInvolved>
  <vehicleCharacteristics>
    <vehicleType>lorry</vehicleType>
  </vehicleCharacteristics>
</groupOfVehiclesInvolved>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *situation*, (in our case there is one).

Attribute *xsi:type="SituationPublication"* in element *payloadPublication* determines, that this envelope is of type Situation publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of one specific situation

The *situationRecord* element occurs twice in the *situation* element in this case. Once for the description of the road closure and the second time for the description of the accident that led to the closure.

Element *situationVersionTime* sets the time of the publication of this version of the situation.

Element *headerInformation* describes how to handle the information described.

The *urlLink* element contains a link to a website where there is more information about the situation.

Event description (what happened)

The first element *situationRecord* of type *xsi:type="Accident"* contains the description of the cause, i.e. the accident involving the truck *accidentInvolvingHeavyLorries*. The second element *situationRecord* of type *xsi:type="RoadOrCarriagewayOrLaneManagement"* contains a description of the impact, i.e. the complete closure of the road *roadClosed*.

The element *probabilityOfOccurrence* with the value *certain* specifies that the event has definitely happened.

The *groupOfVehiclesInvolved* element partially repeats the previous information by specifying *lorry*, i.e. a truck, as the characteristic of the vehicle involved in the accident.

Time validity

Estimated time validity is given in a structure *validity* by values of validity interval start *overallStartTime* and validity interval end *overallEndTime*.

Event location

The location of the event is described directly in the *groupOfLocations* element by the nested description of the *xsi:type="Linear"* type.

In the example 3 location descriptions are used:

- tmc location using location tables in the *alertCLinear* element of type *AlertCMethod2Linear*,
- description using references to Global Network segment id's in the *globalNetworkLinear* extension, and
- description using the OpenLR routing method in the *openlrExtendedLinear* extension.

The location of all published events is of type *Linear*, even in the case of point events of type accident, in which case the start and end points of the event are the same. Thus, the *startPoint* and *endPoint* in the *globalNetworkLinear* are the same.

Additional content

The document states time of its first version in *situationRecordCreationTime* and time of current version in *situationRecordVersionTime* for each *situationRecord*.

Element *source* lists the the creator of this information, who is responsible for its accuracy.

The *generalPublicComment* element contains a description of the situation, usually only in czech language, generated manually by the operator.

5.2 brokendownvehicles.xml - broken down lorry

Publication contains information about one broken down lorry at the D7 exit link.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns="http://datex2.eu/schema/2/2_0" xsi:schemaLocation="http://datex2.
eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>DIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="SituationPublication" lang="cs">
    <feedDescription>
      <values>
        <value lang="cs">x-source:cz-ndic_d2-common-v1.2</value>
      </values>
    </feedDescription>
    <feedType>x-format:cz-ndic_d2-common-v1.2</feedType>
    <publicationTime>2023-08-30T09:18:24+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <situation id="ff808181-877b-aada-018a-ef646b376b08" version="2">
      <situationVersionTime>2023-10-02T09:58:05+02:00</situationVersionTime>
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>urgent</urgency>
      </headerInformation>
      <situationRecord version="2" xsi:type="AbnormalTraffic" id="ff808181-877b-aada-
018a-ef646b376b08_AbnormalTraffic">
        <situationRecordCreationTime>2023-10-02T09:58:06+02:00</
situationRecordCreationTime>
        <situationRecordVersionTime>2023-10-02T09:58:05+02:00</
situationRecordVersionTime>
        <probabilityOfOccurrence>certain</probabilityOfOccurrence>
        <source>
          <sourceIdentification>PCR</sourceIdentification>
          <sourceName>
            <values>
              <value lang="cs"/>
            </values>
          </sourceName>
        </source>
        <validity>
          <validityStatus>definedByValidityTimeSpec</validityStatus>
          <validityTimeSpecification>
            <overallStartTime>2023-10-02T09:55:00+02:00</overallStartTime>
            <overallEndTime>2023-10-02T11:00:00+02:00</overallEndTime>
          </validityTimeSpecification>
        </validity>
        <cause xsi:type="NonManagedCause">
          <causeType>rubberNecking</causeType>
        </cause>
      </situationRecord>
    </situation>
  </payloadPublication>
</d2LogicalModel>
```

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```

</cause>
<generalPublicComment>
  <comment>
    <values>
      <value lang="cs">Od 2.10.2023 09:55 do 11:00; v ulici Tupolevova v obci
↪ Praha okres území Hlavního města Prahy; 2 havarovaná vozidla, pomalý provoz; bez
↪ zranění, blokují jeden JP</value>
    </values>
  </comment>
</generalPublicComment>
<groupOfLocations xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>10.1</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>positive</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>18231</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>10388</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
  <linearExtension>
    <openlrExtendedLinear>
      <firstDirection>
        <openlrLocationReferencePoint>
          <openlrCoordinate>
            <latitude>50.136887007413563</latitude>
            <longitude>14.504980597632043</longitude>
          </openlrCoordinate>
          <openlrLineAttributes>
            <openlrFunctionalRoadClass>FRC4</openlrFunctionalRoadClass>
            <openlrFormOfWay>multipleCarriageway</openlrFormOfWay>
            <openlrBearing>350</openlrBearing>
          </openlrLineAttributes>
          <openlrPathAttributes>
            <openlrLowestFRCToNextLRPoint>FRC4</openlrLowestFRCToNextLRPoint>
            <openlrDistanceToNextLRPoint>165</openlrDistanceToNextLRPoint>
          </openlrPathAttributes>
        </openlrLocationReferencePoint>
        <openlrLastLocationReferencePoint>
          <openlrCoordinate>
            <latitude>50.138363834263082</latitude>
            <longitude>14.504772075868614</longitude>
          </openlrCoordinate>
          <openlrLineAttributes>
            <openlrFunctionalRoadClass>FRC4</openlrFunctionalRoadClass>
            <openlrFormOfWay>multipleCarriageway</openlrFormOfWay>
            <openlrBearing>180</openlrBearing>

```

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```

        </openlrLineAttributes>
        </openlrLastLocationReferencePoint>
        <openlrOffsets>
            <openlrPositiveOffset>16</openlrPositiveOffset>
            <openlrNegativeOffset>4</openlrNegativeOffset>
        </openlrOffsets>
        </firstDirection>
    </openlrExtendedLinear>
    <globalNetworkLinear>
        <networkVersion>23.06</networkVersion>
        <linearGeometryType>continuous</linearGeometryType>
        <startPoint>
            <sjtskPointCoordinates>
                <!-- dummy coordinates -->
                <sjtskX>-736101.8975</sjtskX>
                <sjtskY>-1038259.9033</sjtskY>
            </sjtskPointCoordinates>
        </startPoint>
        <endPoint>
            <sjtskPointCoordinates>
                <sjtskX>-736123.68590948975</sjtskX>
                <sjtskY>-1038219.1256349033</sjtskY>
            </sjtskPointCoordinates>
        </endPoint>
        <linearWithinLinearGNElement>
            <sectionId>710586</sectionId>
            <directionRelativeOnLinearSection>aligned</
→ directionRelativeOnLinearSection>
            <orderOfSection>0</orderOfSection>
            <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
                <percentageDistanceAlong>0.156</percentageDistanceAlong>
            </fromPoint>
            <toPoint xsi:type="PercentageDistanceAlongLinearElement">
                <percentageDistanceAlong>97.576</percentageDistanceAlong>
            </toPoint>
        </linearWithinLinearGNElement>
    </globalNetworkLinear>
</linearExtension>
</groupOfLocations>
<abnormalTrafficType>slowTraffic</abnormalTrafficType>
<queueLength>160</queueLength>
</situationRecord>
→ <situationRecord version="2" xsi:type="VehicleObstruction" id="ff808181-877b-
aada-018a-ef646b376b08_VehicleObstruction">
    <situationRecordCreationTime>2023-10-02T09:58:06+02:00</
→ situationRecordCreationTime>
    <situationRecordVersionTime>2023-10-02T09:58:05+02:00</
→ situationRecordVersionTime>
    <probabilityOfOccurrence>certain</probabilityOfOccurrence>
    <source>
        <sourceIdentification>PCR</sourceIdentification>
        <sourceName>
            <values>
                <value lang="cs"/>
            </values>
        </sourceName>

```

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```

</source>
<validity>
  <validityStatus>definedByValidityTimeSpec</validityStatus>
  <validityTimeSpecification>
    <overallStartTime>2023-10-02T09:55:00+02:00</overallStartTime>
    <overallEndTime>2023-10-02T11:00:00+02:00</overallEndTime>
  </validityTimeSpecification>
</validity>
<generalPublicComment>
  <comment>
    <values>
      <value lang="cs">Od 2.10.2023 09:55 do 11:00; v ulici Tupolevova v obci
↳ Praha okres území Hlavního města Prahy; 2 havarovaná vozidla, pomalý provoz; bez
↳ zranění, blokují jeden JP</value>
    </values>
  </comment>
</generalPublicComment>
<groupOfLocations xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>10.1</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>positive</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>18231</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>10388</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
  <linearExtension>
    <openlrExtendedLinear>
      <firstDirection>
        <openlrLocationReferencePoint>
          <openlrCoordinate>
            <latitude>50.136887007413563</latitude>
            <longitude>14.504980597632043</longitude>
          </openlrCoordinate>
          <openlrLineAttributes>
            <openlrFunctionalRoadClass>FRC4</openlrFunctionalRoadClass>
            <openlrFormOfWay>multipleCarriageway</openlrFormOfWay>
            <openlrBearing>350</openlrBearing>
          </openlrLineAttributes>
          <openlrPathAttributes>
            <openlrLowestFRCToNextLRPoint>FRC4</openlrLowestFRCToNextLRPoint>
            <openlrDistanceToNextLRPoint>165</openlrDistanceToNextLRPoint>
          </openlrPathAttributes>
        </openlrLocationReferencePoint>
        <openlrLastLocationReferencePoint>
          <openlrCoordinate>

```

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```

        <latitude>50.138363834263082</latitude>
        <longitude>14.504772075868614</longitude>
    </openlrCoordinate>
    <openlrLineAttributes>
        <openlrFunctionalRoadClass>FRC4</openlrFunctionalRoadClass>
        <openlrFormOfWay>multipleCarriageway</openlrFormOfWay>
        <openlrBearing>180</openlrBearing>
    </openlrLineAttributes>
    </openlrLastLocationReferencePoint>
    <openlrOffsets>
        <openlrPositiveOffset>161</openlrPositiveOffset>
        <openlrNegativeOffset>4</openlrNegativeOffset>
    </openlrOffsets>
    </firstDirection>
</openlrExtendedLinear>
<globalNetworkLinear>
    <networkVersion>23.06</networkVersion>
    <linearGeometryType>continuous</linearGeometryType>
    <startPoint>
        <sjtskPointCoordinates>
            <sjtskX>-736123.68590948975</sjtskX>
            <sjtskY>-1038219.1256349033</sjtskY>
        </sjtskPointCoordinates>
    </startPoint>
    <endPoint>
        <sjtskPointCoordinates>
            <sjtskX>-736123.68590948975</sjtskX>
            <sjtskY>-1038219.1256349033</sjtskY>
        </sjtskPointCoordinates>
    </endPoint>
    <linearWithinLinearGNElement>
        <sectionId>710586</sectionId>
        <directionRelativeOnLinearSection>aligned</
->directionRelativeOnLinearSection>
        <orderOfSection>0</orderOfSection>
        <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>94.576</percentageDistanceAlong>
        </fromPoint>
        <toPoint xsi:type="PercentageDistanceAlongLinearElement">
            <percentageDistanceAlong>97.576</percentageDistanceAlong>
        </toPoint>
    </linearWithinLinearGNElement>
</globalNetworkLinear>
</linearExtension>
</groupOfLocations>
<numberOfObstructions>2</numberOfObstructions>
<vehicleObstructionType>brokenDownVehicle</vehicleObstructionType>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

5.2.1 Publication envelope, time validity and location

Publication envelope, time validity and location is using identical structures as previous example.

The only difference is in description of one concrete situation in element *situationRecord*.

Event description (what happened)

The *situationRecord* element of the *xsi:type="VehicleObstruction"* type contains the description of the cause, i.e. the broken down vehicle *brokenDownVehicle* and further, in the *impact* element it states in *delays* that the result of the delay of 259200 s.

The *probabilityOfOccurrence* element specifies by the value *certain* that the event definitely happened.

Time validity

As in previous example.

Event location

As in previous example.

Additional content

As in previous example.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:element name="globalNetworkLinear" type="D2LogicalModel:GlobalNetworkLinear"
  ↳ minOccurs="0" />
18      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
19    </xs:sequence>
20  </xs:complexType>
21  <xs:complexType name="_PointExtensionType">
22    <xs:sequence>
23      <xs:element name="pointWithinLinearGNElement" type=
  ↳ "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
24      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint"
  ↳ minOccurs="0" />
25      <xs:element name="pointByMultiCoordinates" type=
  ↳ "D2LogicalModel:PointByMultiCoordinates" minOccurs="0" />
26      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
27    </xs:sequence>
28  </xs:complexType>

```

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```

29 <xs:complexType name="_SituationRecordExtensionType">
30   <xs:sequence>
31     <xs:element name="situationRecordExtendedApproved" type=
32     ↪ "D2LogicalModel:SituationRecordExtendedApproved" minOccurs="0" />
33     <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
34     ↪ "unbounded" />
35   </xs:sequence>
36 </xs:complexType>
37 <xs:complexType name="AbnormalTraffic">
38   <xs:complexContent>
39     <xs:extension base="D2LogicalModel:TrafficElement">
40       <xs:sequence>
41         <xs:element name="abnormalTrafficType" type=
42         ↪ "D2LogicalModel:AbnormalTrafficTypeEnum" minOccurs="0" maxOccurs="1" />
43         <xs:element name="queueLength" type=
44         ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
45         <xs:element name="relativeTrafficFlow" type=
46         ↪ "D2LogicalModel:RelativeTrafficFlowEnum" minOccurs="0" maxOccurs="1" />
47         <xs:element name="trafficTrendType" type=
48         ↪ "D2LogicalModel:TrafficTrendTypeEnum" minOccurs="0" maxOccurs="1" />
49         <xs:element name="abnormalTrafficExtension" type="D2LogicalModel:_
50         ↪ ExtensionType" minOccurs="0" />
51       </xs:sequence>
52     </xs:extension>
53   </xs:complexContent>
54 </xs:complexType>
55 <xs:simpleType name="AbnormalTrafficTypeEnum">
56   <xs:restriction base="xs:string">
57     <xs:enumeration value="stationaryTraffic" />
58     <xs:enumeration value="queuingTraffic" />
59     <xs:enumeration value="slowTraffic" />
60     <xs:enumeration value="heavyTraffic" />
61     <xs:enumeration value="unspecifiedAbnormalTraffic" />
62     <xs:enumeration value="other" />
63   </xs:restriction>
64 </xs:simpleType>
65 <xs:complexType name="Accident">
66   <xs:complexContent>
67     <xs:extension base="D2LogicalModel:TrafficElement">
68       <xs:sequence>
69         <xs:element name="accidentCause" type="D2LogicalModel:AccidentCauseEnum"
70         ↪ minOccurs="0" maxOccurs="1" />
71         <xs:element name="accidentType" type="D2LogicalModel:AccidentTypeEnum"
72         ↪ minOccurs="1" maxOccurs="unbounded" />
73         <xs:element name="totalNumberOfVehiclesInvolved" type=
74         ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
75         <xs:element name="vehicleInvolved" type="D2LogicalModel:Vehicle" minOccurs=
76         ↪ "0" maxOccurs="unbounded" />
77         <xs:element name="groupOfVehiclesInvolved" type=
78         ↪ "D2LogicalModel:GroupOfVehiclesInvolved" minOccurs="0" maxOccurs="unbounded" />
79         <xs:element name="groupOfPeopleInvolved" type=
80         ↪ "D2LogicalModel:GroupOfPeopleInvolved" minOccurs="0" maxOccurs="unbounded" />
81         <xs:element name="accidentExtension" type="D2LogicalModel:_ExtensionType"
82         ↪ minOccurs="0" />
83       </xs:sequence>
84     </xs:extension>

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71     </xs:complexContent>
72 </xs:complexType>
73 <xs:simpleType name="AccidentCauseEnum">
74     <xs:restriction base="xs:string">
75         <xs:enumeration value="avoidanceOfObstacles" />
76         <xs:enumeration value="driverDistraction" />
77         <xs:enumeration value="driverDrugAbuse" />
78         <xs:enumeration value="driverIllness" />
79         <xs:enumeration value="exceedingSpeedsLimits" />
80         <xs:enumeration value="excessAlcohol" />
81         <xs:enumeration value="excessiveDriverTiredness" />
82         <xs:enumeration value="impermissibleManoeuvre" />
83         <xs:enumeration value="limitedVisibility" />
84         <xs:enumeration value="notKeepingASafeDistance" />
85         <xs:enumeration value="onTheWrongSideOfTheRoad" />
86         <xs:enumeration value="pedestrianInRoad" />
87         <xs:enumeration value="poorLaneAdherence" />
88         <xs:enumeration value="poorMergeEntryOrExitJudgement" />
89         <xs:enumeration value="poorRoadSurfaceCondition" />
90         <xs:enumeration value="poorSurfaceAdherence" />
91         <xs:enumeration value="undisclosed" />
92         <xs:enumeration value="unknown" />
93         <xs:enumeration value="vehicleFailure" />
94         <xs:enumeration value="other" />
95     </xs:restriction>
96 </xs:simpleType>
97 <xs:simpleType name="AccidentTypeEnum">
98     <xs:restriction base="xs:string">
99         <xs:enumeration value="accident" />
100        <xs:enumeration value="accidentInvolvingBicycles" />
101        <xs:enumeration value="accidentInvolvingBuses" />
102        <xs:enumeration value="accidentInvolvingHazardousMaterials" />
103        <xs:enumeration value="accidentInvolvingHeavyLorries" />
104        <xs:enumeration value="accidentInvolvingMassTransitVehicle" />
105        <xs:enumeration value="accidentInvolvingMopeds" />
106        <xs:enumeration value="accidentInvolvingMotorcycles" />
107        <xs:enumeration value="accidentInvolvingRadioactiveMaterial" />
108        <xs:enumeration value="accidentInvolvingTrain" />
109        <xs:enumeration value="chemicalSpillageAccident" />
110        <xs:enumeration value="collision" />
111        <xs:enumeration value="collisionWithAnimal" />
112        <xs:enumeration value="collisionWithObstruction" />
113        <xs:enumeration value="collisionWithPerson" />
114        <xs:enumeration value="earlierAccident" />
115        <xs:enumeration value="fuelSpillageAccident" />
116        <xs:enumeration value="headOnCollision" />
117        <xs:enumeration value="headOnOrSideCollision" />
118        <xs:enumeration value="jackknifedArticulatedLorry" />
119        <xs:enumeration value="jackknifedCaravan" />
120        <xs:enumeration value="jackknifedTrailer" />
121        <xs:enumeration value="multipleVehicleCollision" />
122        <xs:enumeration value="multivehicleAccident" />
123        <xs:enumeration value="oilSpillageAccident" />
124        <xs:enumeration value="overturnedHeavyLorry" />
125        <xs:enumeration value="overturnedTrailer" />
126        <xs:enumeration value="overturnedVehicle" />

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127     <xs:enumeration value="rearCollision" />
128     <xs:enumeration value="secondaryAccident" />
129     <xs:enumeration value="seriousAccident" />
130     <xs:enumeration value="sideCollision" />
131     <xs:enumeration value="vehicleOffRoad" />
132     <xs:enumeration value="vehicleSpunAround" />
133     <xs:enumeration value="other" />
134 </xs:restriction>
135 </xs:simpleType>
136 <xs:complexType name="Activity" abstract="true">
137     <xs:complexContent>
138         <xs:extension base="D2LogicalModel:TrafficElement">
139             <xs:sequence>
140                 <xs:element name="activityExtension" type="D2LogicalModel:_ExtensionType"
141 ↪ minOccurs="0" />
142             </xs:sequence>
143         </xs:extension>
144     </xs:complexContent>
145 </xs:complexType>
146 <xs:complexType name="AffectedCarriagewayAndLanes">
147     <xs:sequence>
148         <xs:element name="carriageway" type="D2LogicalModel:CarriagewayEnum" minOccurs=
149 ↪ "1" maxOccurs="1" />
150         <xs:element name="lane" type="D2LogicalModel:LaneEnum" minOccurs="0" maxOccurs=
151 ↪ "unbounded" />
152         <xs:element name="footpath" type="D2LogicalModel:Boolean" minOccurs="0"
153 ↪ maxOccurs="1" />
154         <xs:element name="lengthAffected" type="D2LogicalModel:MetresAsFloat" minOccurs=
155 ↪ "0" maxOccurs="1" />
156         <xs:element name="affectedCarriagewayAndLanesExtension" type="D2LogicalModel:_
157 ↪ ExtensionType" minOccurs="0" />
158     </xs:sequence>
159 </xs:complexType>
160 <xs:complexType name="AlertCDirection">
161     <xs:sequence>
162         <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum
163 ↪ " minOccurs="1" maxOccurs="1" />
164         <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString
165 ↪ " minOccurs="0" maxOccurs="1" />
166         <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
167 ↪ "0" maxOccurs="1" />
168         <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType
169 ↪ " minOccurs="0" />
170     </xs:sequence>
171 </xs:complexType>
172 <xs:simpleType name="AlertCDirectionEnum">
    <xs:restriction base="xs:string">
        <xs:enumeration value="both" />
        <xs:enumeration value="negative" />
        <xs:enumeration value="positive" />
        <xs:enumeration value="unknown" />
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="AlertCLinear" abstract="true">
    <xs:sequence>
        <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"

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173   <minOccurs="1" maxOccurs="1" />
      <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
174   <minOccurs="1" maxOccurs="1" />
      <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
175   <minOccurs="1" maxOccurs="1" />
      <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
176   <minOccurs="0" />
      </xs:sequence>
177   </xs:complexType>
178   <xs:complexType name="AlertCLinearByCode">
179     <xs:complexContent>
180       <xs:extension base="D2LogicalModel:AlertCLinear">
181         <xs:sequence>
182           <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
183           <xs:element name="locationCodeForLinearLocation" type=
184             "D2LogicalModel:AlertCLocation" />
185           <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
186             ExtensionType" minOccurs="0" />
187         </xs:sequence>
188       </xs:extension>
189     </xs:complexContent>
190   </xs:complexType>
191   <xs:complexType name="AlertCLocation">
192     <xs:sequence>
193       <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
194       <minOccurs="0" maxOccurs="1" />
195       <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
196       <minOccurs="1" maxOccurs="1" />
197       <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
198       <minOccurs="0" />
199     </xs:sequence>
200   </xs:complexType>
201   <xs:simpleType name="AlertCLocationCode">
202     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
203   </xs:simpleType>
204   <xs:complexType name="AlertCMethod2Linear">
205     <xs:complexContent>
206       <xs:extension base="D2LogicalModel:AlertCLinear">
207         <xs:sequence>
208           <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
209           <xs:element name="alertCMethod2PrimaryPointLocation" type=
210             "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
211           <xs:element name="alertCMethod2SecondaryPointLocation" type=
212             "D2LogicalModel:AlertCMethod2SecondaryPointLocation" />
213           <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
214             ExtensionType" minOccurs="0" />
215         </xs:sequence>
216       </xs:extension>
217     </xs:complexContent>
218   </xs:complexType>
219   <xs:complexType name="AlertCMethod2Point">
220     <xs:complexContent>
221       <xs:extension base="D2LogicalModel:AlertCPoint">
222         <xs:sequence>
223           <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
224           <xs:element name="alertCMethod2PrimaryPointLocation" type=

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217 <!-- "D2LogicalModel:AlertCMethod2PrimaryPointLocation" -->
    <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
218 <!-- ExtensionType" minOccurs="0" -->
    </xs:sequence>
219 </xs:extension>
220 </xs:complexContent>
221 </xs:complexType>
222 <xs:complexType name="AlertCMethod2PrimaryPointLocation">
223 <xs:sequence>
224 <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" -->
225 <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=
226 <!-- "D2LogicalModel:_ExtensionType" minOccurs="0" -->
    </xs:sequence>
227 </xs:complexType>
228 <xs:complexType name="AlertCMethod2SecondaryPointLocation">
229 <xs:sequence>
230 <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" -->
231 <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
232 <!-- "D2LogicalModel:_ExtensionType" minOccurs="0" -->
    </xs:sequence>
233 </xs:complexType>
234 <xs:complexType name="AlertCPoint" abstract="true">
235 <xs:sequence>
236 <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
237 <!-- minOccurs="1" maxOccurs="1" -->
    <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
238 <!-- minOccurs="1" maxOccurs="1" -->
    <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
239 <!-- minOccurs="1" maxOccurs="1" -->
    <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType"
240 <!-- minOccurs="0" -->
    </xs:sequence>
241 </xs:complexType>
242 <xs:simpleType name="AngleInDegrees">
243 <xs:restriction base="D2LogicalModel:NonNegativeInteger" -->
244 </xs:simpleType>
245 <xs:complexType name="AnimalPresenceObstruction">
246 <xs:complexContent>
247 <xs:extension base="D2LogicalModel:Obstruction">
248 <xs:sequence>
249 <xs:element name="alive" type="D2LogicalModel:Boolean" minOccurs="0"
250 <!-- maxOccurs="1" -->
    <xs:element name="animalPresenceType" type=
251 <!-- "D2LogicalModel:AnimalPresenceTypeEnum" minOccurs="1" maxOccurs="1" -->
    <xs:element name="animalPresenceObstructionExtension" type="D2LogicalModel:_
252 <!-- ExtensionType" minOccurs="0" -->
    </xs:sequence>
253 </xs:extension>
254 </xs:complexContent>
255 </xs:complexType>
256 <xs:simpleType name="AnimalPresenceTypeEnum">
257 <xs:restriction base="xs:string">
258 <xs:enumeration value="animalsOnTheRoad" -->
259 <xs:enumeration value="herdOfAnimalsOnTheRoad" -->
260 <xs:enumeration value="largeAnimalsOnTheRoad" -->
261 </xs:restriction>

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262 </xs:simpleType>
263 <xs:complexType name="ApplicationRateValue">
264   <xs:complexContent>
265     <xs:extension base="D2LogicalModel:DataValue">
266       <xs:sequence>
267         <xs:element name="applicationRate" type=
268 ↪ "D2LogicalModel:IntensityKilogramsPerSquareMetre" minOccurs="1" maxOccurs="1" />
269         <xs:element name="applicationRateValueExtension" type="D2LogicalModel:_
270 ↪ ExtensionType" minOccurs="0" />
271       </xs:sequence>
272     </xs:extension>
273   </xs:complexContent>
274 </xs:complexType>
275 <xs:complexType name="Area">
276   <xs:complexContent>
277     <xs:extension base="D2LogicalModel:Location">
278       <xs:sequence>
279         <xs:element name="areaExtension" type="D2LogicalModel:_ExtensionType"
280 ↪ minOccurs="0" />
281       </xs:sequence>
282     </xs:extension>
283   </xs:complexContent>
284 </xs:complexType>
285 <xs:complexType name="AreaDestination">
286   <xs:complexContent>
287     <xs:extension base="D2LogicalModel:Destination">
288       <xs:sequence>
289         <xs:element name="area" type="D2LogicalModel:Area" />
290         <xs:element name="areaDestinationExtension" type="D2LogicalModel:_
291 ↪ ExtensionType" minOccurs="0" />
292       </xs:sequence>
293     </xs:extension>
294   </xs:complexContent>
295 </xs:complexType>
296 <xs:complexType name="AuthorityOperation">
297   <xs:complexContent>
298     <xs:extension base="D2LogicalModel:Activity">
299       <xs:sequence>
300         <xs:element name="authorityOperationType" type=
301 ↪ "D2LogicalModel:AuthorityOperationTypeEnum" minOccurs="1" maxOccurs="1" />
302         <xs:element name="authorityOperationExtension" type="D2LogicalModel:_
303 ↪ ExtensionType" minOccurs="0" />
304       </xs:sequence>
305     </xs:extension>
306   </xs:complexContent>
307 </xs:complexType>
308 <xs:simpleType name="AuthorityOperationTypeEnum">
309   <xs:restriction base="xs:string">
310     <xs:enumeration value="accidentInvestigationWork" />
311     <xs:enumeration value="bombSquadInAction" />
    <xs:enumeration value="civilEmergency" />
    <xs:enumeration value="customsOperation" />
    <xs:enumeration value="juridicalReconstruction" />
    <xs:enumeration value="policeCheckPoint" />
    <xs:enumeration value="policeInvestigation" />
    <xs:enumeration value="roadOperatorCheckPoint" />

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312     <xs:enumeration value="survey" />
313     <xs:enumeration value="transportOfVip" />
314     <xs:enumeration value="undefinedAuthorityActivity" />
315     <xs:enumeration value="vehicleInspectionCheckPoint" />
316     <xs:enumeration value="vehicleWeighing" />
317     <xs:enumeration value="weighInMotion" />
318     <xs:enumeration value="other" />
319 </xs:restriction>
320 </xs:simpleType>
321 <xs:complexType name="AxleFlowValue">
322   <xs:complexContent>
323     <xs:extension base="D2LogicalModel:DataValue">
324       <xs:sequence>
325         <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour"
326           ↪ minOccurs="1" maxOccurs="1" />
327         <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_
328           ↪ ExtensionType" minOccurs="0" />
329       </xs:sequence>
330     </xs:extension>
331   </xs:complexContent>
332 </xs:complexType>
333 <xs:complexType name="AxleSpacing">
334   <xs:sequence>
335     <xs:element name="axleSpacing" type="D2LogicalModel:MetresAsFloat" minOccurs="1
336       ↪ " maxOccurs="1" />
337     <xs:element name="axleSpacingSequenceIdentifier" type=
338       ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
339     <xs:element name="axleSpacingExtension" type="D2LogicalModel:_ExtensionType"
340       ↪ minOccurs="0" />
341   </xs:sequence>
342 </xs:complexType>
343 <xs:simpleType name="AxlesPerHour">
344   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
345 </xs:simpleType>
346 <xs:complexType name="AxleWeight">
347   <xs:sequence>
348     <xs:element name="axlePositionIdentifier" type=
349       ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
350     <xs:element name="axleWeight" type="D2LogicalModel:Tonnes" minOccurs="0"
351       ↪ maxOccurs="1" />
352     <xs:element name="maximumPermittedAxleWeight" type="D2LogicalModel:Tonnes"
353       ↪ minOccurs="0" maxOccurs="1" />
354     <xs:element name="axleWeightExtension" type="D2LogicalModel:_ExtensionType"
355       ↪ minOccurs="0" />
356   </xs:sequence>
357 </xs:complexType>
358 <xs:simpleType name="Boolean">
359   <xs:restriction base="xs:boolean" />
360 </xs:simpleType>
361 <xs:simpleType name="CarParkConfigurationEnum">
362   <xs:restriction base="xs:string">
363     <xs:enumeration value="multiStorey" />
364     <xs:enumeration value="parkAndRide" />
365     <xs:enumeration value="singleLevel" />
366     <xs:enumeration value="underground" />
367   </xs:restriction>

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359 </xs:simpleType>
360 <xs:complexType name="CarParks">
361   <xs:complexContent>
362     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
363       <xs:sequence>
364         <xs:element name="carParkConfiguration" type=
365 ↪ "D2LogicalModel:CarParkConfigurationEnum" minOccurs="0" maxOccurs="1" />
366         <xs:element name="carParkIdentity" type="D2LogicalModel:String" minOccurs="1
367 ↪ " maxOccurs="1" />
368         <xs:element name="carParkOccupancy" type="D2LogicalModel:Percentage"
369 ↪ minOccurs="0" maxOccurs="1" />
370         <xs:element name="carParkStatus" type="D2LogicalModel:CarParkStatusEnum"
371 ↪ minOccurs="0" maxOccurs="1" />
372         <xs:element name="numberOfVacantParkingSpaces" type=
373 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
374         <xs:element name="occupiedSpaces" type="D2LogicalModel:NonNegativeInteger"
375 ↪ minOccurs="0" maxOccurs="1" />
376         <xs:element name="totalCapacity" type="D2LogicalModel:NonNegativeInteger"
377 ↪ minOccurs="0" maxOccurs="1" />
378         <xs:element name="carParksExtension" type="D2LogicalModel:_ExtensionType"
379 ↪ minOccurs="0" />
380       </xs:sequence>
381     </xs:extension>
382   </xs:complexContent>
383 </xs:complexType>
384 <xs:simpleType name="CarParkStatusEnum">
385   <xs:restriction base="xs:string">
386     <xs:enumeration value="carParkClosed" />
387     <xs:enumeration value="allCarParksFull" />
388     <xs:enumeration value="carParkFacilityFaulty" />
389     <xs:enumeration value="carParkFull" />
390     <xs:enumeration value="carParkStatusUnknown" />
391     <xs:enumeration value="enoughSpacesAvailable" />
392     <xs:enumeration value="multiStoryCarParksFull" />
393     <xs:enumeration value="noMoreParkingSpacesAvailable" />
394     <xs:enumeration value="noParkAndRideInformation" />
395     <xs:enumeration value="noParkingAllowed" />
396     <xs:enumeration value="noParkingInformationAvailable" />
397     <xs:enumeration value="normalParkingRestrictionsLifted" />
398     <xs:enumeration value="onlyAFewSpacesAvailable" />
399     <xs:enumeration value="parkAndRideServiceNotOperating" />
400     <xs:enumeration value="parkAndRideServiceOperating" />
401     <xs:enumeration value="specialParkingRestrictionsInForce" />
402   </xs:restriction>
403 </xs:simpleType>
404 <xs:simpleType name="CarriagewayEnum">
405   <xs:restriction base="xs:string">
406     <xs:enumeration value="connectingCarriageway" />
407     <xs:enumeration value="entrySlipRoad" />
408     <xs:enumeration value="exitSlipRoad" />
409     <xs:enumeration value="flyover" />
410     <xs:enumeration value="leftHandFeederRoad" />
411     <xs:enumeration value="leftHandParallelCarriageway" />
412     <xs:enumeration value="mainCarriageway" />
413     <xs:enumeration value="oppositeCarriageway" />
414     <xs:enumeration value="parallelCarriageway" />

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407     <xs:enumeration value="rightHandFeederRoad" />
408     <xs:enumeration value="rightHandParallelCarriageway" />
409     <xs:enumeration value="roundabout" />
410     <xs:enumeration value="serviceRoad" />
411     <xs:enumeration value="slipRoads" />
412     <xs:enumeration value="underpass" />
413 </xs:restriction>
414 </xs:simpleType>
415 <xs:complexType name="Cause" abstract="true">
416     <xs:sequence>
417         <xs:element name="causeExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
418     </xs:sequence>
419 </xs:complexType>
420 <xs:simpleType name="CauseTypeEnum">
421     <xs:restriction base="xs:string">
422         <xs:enumeration value="accident" />
423         <xs:enumeration value="congestion" />
424         <xs:enumeration value="earlierAccident" />
425         <xs:enumeration value="earlierEvent" />
426         <xs:enumeration value="earlierIncident" />
427         <xs:enumeration value="equipmentFailure" />
428         <xs:enumeration value="excessiveHeat" />
429         <xs:enumeration value="frost" />
430         <xs:enumeration value="holidayTraffic" />
431         <xs:enumeration value="infrastructureFailure" />
432         <xs:enumeration value="largeNumbersOfVisitors" />
433         <xs:enumeration value="obstruction" />
434         <xs:enumeration value="pollutionAlert" />
435         <xs:enumeration value="poorWeather" />
436         <xs:enumeration value="problemsAtBorderPost" />
437         <xs:enumeration value="problemsAtCustomPost" />
438         <xs:enumeration value="problemsOnLocalRoads" />
439         <xs:enumeration value="radioactiveLeakAlert" />
440         <xs:enumeration value="roadsideEvent" />
441         <xs:enumeration value="rubberNecking" />
442         <xs:enumeration value="securityIncident" />
443         <xs:enumeration value="shearWeightOfTraffic" />
444         <xs:enumeration value="technicalProblems" />
445         <xs:enumeration value="terrorism" />
446         <xs:enumeration value="toxicCloudAlert" />
447         <xs:enumeration value="vandalism" />
448         <xs:enumeration value="other" />
449     </xs:restriction>
450 </xs:simpleType>
451 <xs:complexType name="Comment">
452     <xs:sequence>
453         <xs:element name="comment" type="D2LogicalModel:MultilingualString" minOccurs="1"
↪ maxOccurs="1" />
454         <xs:element name="commentDateTime" type="D2LogicalModel:DateTime" minOccurs="0"
↪ maxOccurs="1" />
455         <xs:element name="commentType" type="D2LogicalModel:CommentTypeEnum" minOccurs=
↪ "0" maxOccurs="1" />
456         <xs:element name="commentExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
457     </xs:sequence>

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458 </xs:complexType>
459 <xs:simpleType name="CommentTypeEnum">
460   <xs:restriction base="xs:string">
461     <xs:enumeration value="abnormalLoadMovementNote" />
462     <xs:enumeration value="dataProcessingNote" />
463     <xs:enumeration value="description" />
464     <xs:enumeration value="internalNote" />
465     <xs:enumeration value="locationDescriptor" />
466     <xs:enumeration value="warning" />
467     <xs:enumeration value="other" />
468   </xs:restriction>
469 </xs:simpleType>
470 <xs:simpleType name="ComparisonOperatorEnum">
471   <xs:restriction base="xs:string">
472     <xs:enumeration value="equalTo" />
473     <xs:enumeration value="greaterThan" />
474     <xs:enumeration value="greaterThanOrEqualTo" />
475     <xs:enumeration value="lessThan" />
476     <xs:enumeration value="lessThanOrEqualTo" />
477   </xs:restriction>
478 </xs:simpleType>
479 <xs:simpleType name="ComplianceOptionEnum">
480   <xs:restriction base="xs:string">
481     <xs:enumeration value="advisory" />
482     <xs:enumeration value="mandatory" />
483   </xs:restriction>
484 </xs:simpleType>
485 <xs:simpleType name="ComputationMethodEnum">
486   <xs:restriction base="xs:string">
487     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples" /
488     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod" />
489     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod" />
490     <xs:enumeration value="medianOfSamplesInATimePeriod" />
491     <xs:enumeration value="movingAverageOfSamples" />
492   </xs:restriction>
493 </xs:simpleType>
494 <xs:simpleType name="ConcentrationKilogramsPerCubicMetre">
495   <xs:restriction base="D2LogicalModel:Float" />
496 </xs:simpleType>
497 <xs:complexType name="ConcentrationOfVehiclesValue">
498   <xs:complexContent>
499     <xs:extension base="D2LogicalModel:DataValue">
500       <xs:sequence>
501         <xs:element name="concentrationOfVehicles" type=
502         <xs:element name="concentrationOfVehiclesValueExtension" type=
503         <xs:sequence>
504       </xs:sequence>
505     </xs:extension>
506   </xs:complexContent>
507 </xs:complexType>
508 <xs:simpleType name="ConcentrationVehiclesPerKilometre">
509   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
510 </xs:simpleType>
511 <xs:complexType name="Conditions">

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511     <xs:complexContent>
512       <xs:extension base="D2LogicalModel:TrafficElement">
513         <xs:sequence>
514           <xs:element name="drivingConditionType" type=
515             ↪ "D2LogicalModel:DrivingConditionTypeEnum" minOccurs="0" maxOccurs="1" />
516           <xs:element name="conditionsExtension" type="D2LogicalModel:_ExtensionType" ↪
517             ↪ minOccurs="0" />
518         </xs:sequence>
519       </xs:extension>
520     </xs:complexContent>
521   </xs:complexType>
522   <xs:simpleType name="ConfidentialityValueEnum">
523     <xs:restriction base="xs:string">
524       <xs:enumeration value="internalUse" />
525       <xs:enumeration value="noRestriction" />
526       <xs:enumeration value="restrictedToAuthorities" />
527       <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
528       <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
529       <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
530     </xs:restriction>
531   </xs:simpleType>
532   <xs:complexType name="ConstructionWorks">
533     <xs:complexContent>
534       <xs:extension base="D2LogicalModel:Roadworks">
535         <xs:sequence>
536           <xs:element name="constructionWorkType" type=
537             ↪ "D2LogicalModel:ConstructionWorkTypeEnum" minOccurs="0" maxOccurs="1" />
538           <xs:element name="constructionWorksExtension" type="D2LogicalModel:_
539             ↪ ExtensionType" minOccurs="0" />
540         </xs:sequence>
541       </xs:extension>
542     </xs:complexContent>
543   </xs:complexType>
544   <xs:simpleType name="ConstructionWorkTypeEnum">
545     <xs:restriction base="xs:string">
546       <xs:enumeration value="blastingWork" />
547       <xs:enumeration value="constructionWork" />
548       <xs:enumeration value="demolitionWork" />
549       <xs:enumeration value="roadImprovementOrUpgrading" />
550       <xs:enumeration value="roadWideningWork" />
551     </xs:restriction>
552   </xs:simpleType>
553   <xs:simpleType name="CountryEnum">
554     <xs:restriction base="xs:string">
555       <xs:enumeration value="at" />
556       <xs:enumeration value="be" />
557       <xs:enumeration value="bg" />
558       <xs:enumeration value="ch" />
559       <xs:enumeration value="cs" />
560       <xs:enumeration value="cy" />
561       <xs:enumeration value="cz" />
562       <xs:enumeration value="de" />
563       <xs:enumeration value="dk" />
564       <xs:enumeration value="ee" />
565       <xs:enumeration value="es" />
566       <xs:enumeration value="fi" />

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563     <xs:enumeration value="fo" />
564     <xs:enumeration value="fr" />
565     <xs:enumeration value="gb" />
566     <xs:enumeration value="gg" />
567     <xs:enumeration value="gi" />
568     <xs:enumeration value="gr" />
569     <xs:enumeration value="hr" />
570     <xs:enumeration value="hu" />
571     <xs:enumeration value="ie" />
572     <xs:enumeration value="im" />
573     <xs:enumeration value="is" />
574     <xs:enumeration value="it" />
575     <xs:enumeration value="je" />
576     <xs:enumeration value="li" />
577     <xs:enumeration value="lt" />
578     <xs:enumeration value="lu" />
579     <xs:enumeration value="lv" />
580     <xs:enumeration value="ma" />
581     <xs:enumeration value="mc" />
582     <xs:enumeration value="mk" />
583     <xs:enumeration value="mt" />
584     <xs:enumeration value="nl" />
585     <xs:enumeration value="no" />
586     <xs:enumeration value="pl" />
587     <xs:enumeration value="pt" />
588     <xs:enumeration value="ro" />
589     <xs:enumeration value="se" />
590     <xs:enumeration value="si" />
591     <xs:enumeration value="sk" />
592     <xs:enumeration value="sm" />
593     <xs:enumeration value="tr" />
594     <xs:enumeration value="va" />
595     <xs:enumeration value="other" />
596 </xs:restriction>
597 </xs:simpleType>
598 <xs:simpleType name="CubicMetres">
599     <xs:restriction base="D2LogicalModel:Float" />
600 </xs:simpleType>
601 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
602     <xs:unique name="_d2LogicalModelSituationRecordConstraint">
603         <xs:selector xpath="."/D2LogicalModel:situationRecord" />
604         <xs:field xpath="@id" />
605         <xs:field xpath="@version" />
606     </xs:unique>
607     <xs:unique name="_d2LogicalModelSituationConstraint">
608         <xs:selector xpath="."/D2LogicalModel:situation" />
609         <xs:field xpath="@id" />
610         <xs:field xpath="@version" />
611     </xs:unique>
612 </xs:element>
613 <xs:complexType name="D2LogicalModel">
614     <xs:sequence>
615         <xs:element name="exchange" type="D2LogicalModel:Exchange" />
616         <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
617         ↪minOccurs="0" />
        <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"

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618     </xs:sequence>
619     <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
620 </xs:complexType>
621 <xs:simpleType name="DangerousGoodsRegulationsEnum">
622     <xs:restriction base="xs:string">
623         <xs:enumeration value="adr" />
624         <xs:enumeration value="iataIcao" />
625         <xs:enumeration value="imoImdg" />
626         <xs:enumeration value="railroadDangerousGoodsBook" />
627     </xs:restriction>
628 </xs:simpleType>
629 <xs:complexType name="DataValue" abstract="true">
630     <xs:sequence>
631         <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0"
632         <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString"
633         <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType"
634     </xs:sequence>
635     <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional" />
636     <xs:attribute name="computationalMethod" type=
637     "D2LogicalModel:ComputationMethodEnum" use="optional" />
638     <xs:attribute name="numberOfIncompleteInputs" type=
639     "D2LogicalModel:NonNegativeInteger" use="optional" />
640     <xs:attribute name="numberOfInputValuesUsed" type=
641     "D2LogicalModel:NonNegativeInteger" use="optional" />
642     <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional" />
643     <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional"
644     />
645     <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
646     " use="optional" />
647 </xs:complexType>
648 <xs:simpleType name="DateTime">
649     <xs:restriction base="xs:dateTime" />
650 </xs:simpleType>
651 <xs:complexType name="DateTimeValue">
652     <xs:complexContent>
653         <xs:extension base="D2LogicalModel:DataValue">
654             <xs:sequence>
655                 <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1"
656                 <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_
657                 ExtensionType" minOccurs="0" />
658             </xs:sequence>
659         </xs:extension>
660     </xs:complexContent>
661 </xs:complexType>
662 <xs:complexType name="Delays">
663     <xs:sequence>
664         <xs:element name="delaysType" type="D2LogicalModel:DelaysTypeEnum" minOccurs="0
665         " maxOccurs="1" />
666         <xs:element name="delayTimeValue" type="D2LogicalModel:Seconds" minOccurs="0"
667         maxOccurs="1" />
668         <xs:element name="delaysExtension" type="D2LogicalModel:_ExtensionType"

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661   <minOccurs="0" />
662 </xs:sequence>
663 </xs:complexType>
664 <xs:simpleType name="DelaysTypeEnum">
665   <xs:restriction base="xs:string">
666     <xs:enumeration value="delays" />
667     <xs:enumeration value="delaysOfUncertainDuration" />
668     <xs:enumeration value="longDelays" />
669     <xs:enumeration value="veryLongDelays" />
670   </xs:restriction>
671 </xs:simpleType>
672 <xs:complexType name="Destination" abstract="true">
673   <xs:sequence>
674     <xs:element name="destinationExtension" type="D2LogicalModel:_ExtensionType"
675       <minOccurs="0" />
676   </xs:sequence>
677 </xs:complexType>
678 <xs:complexType name="DirectionBearingValue">
679   <xs:complexContent>
680     <xs:extension base="D2LogicalModel:DataValue">
681       <xs:sequence>
682         <xs:element name="directionBearing" type="D2LogicalModel:AngleInDegrees"
683           <minOccurs="1" maxOccurs="1" />
684         <xs:element name="directionBearingValueExtension" type="D2LogicalModel:_
685           ExtensionType" minOccurs="0" />
686       </xs:sequence>
687     </xs:extension>
688   </xs:complexContent>
689 </xs:complexType>
690 <xs:simpleType name="DirectionCompassEnum">
691   <xs:restriction base="xs:string">
692     <xs:enumeration value="east" />
693     <xs:enumeration value="eastNorthEast" />
694     <xs:enumeration value="eastSouthEast" />
695     <xs:enumeration value="north" />
696     <xs:enumeration value="northEast" />
697     <xs:enumeration value="northNorthEast" />
698     <xs:enumeration value="northNorthWest" />
699     <xs:enumeration value="northWest" />
700     <xs:enumeration value="south" />
701     <xs:enumeration value="southEast" />
702     <xs:enumeration value="southSouthEast" />
703     <xs:enumeration value="southSouthWest" />
704     <xs:enumeration value="southWest" />
705     <xs:enumeration value="west" />
706     <xs:enumeration value="westNorthWest" />
707     <xs:enumeration value="westSouthWest" />
708   </xs:restriction>
709 </xs:simpleType>
710 <xs:complexType name="DirectionCompassValue">
711   <xs:complexContent>
712     <xs:extension base="D2LogicalModel:DataValue">
713       <xs:sequence>
714         <xs:element name="directionCompass" type=
715           "D2LogicalModel:DirectionCompassEnum" minOccurs="1" maxOccurs="1" />
716         <xs:element name="directionCompassValueExtension" type="D2LogicalModel:_

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712     <xs:sequence>
713     </xs:sequence>
714   </xs:complexType>
715 </xs:complexType>
716 <xs:simpleType name="DirectionEnum">
717   <xs:restriction base="xs:string">
718     <xs:enumeration value="allDirections" />
719     <xs:enumeration value="bothWays" />
720     <xs:enumeration value="clockwise" />
721     <xs:enumeration value="anticlockwise" />
722     <xs:enumeration value="innerRing" />
723     <xs:enumeration value="outerRing" />
724     <xs:enumeration value="northBound" />
725     <xs:enumeration value="northEastBound" />
726     <xs:enumeration value="eastBound" />
727     <xs:enumeration value="southEastBound" />
728     <xs:enumeration value="southBound" />
729     <xs:enumeration value="southWestBound" />
730     <xs:enumeration value="westBound" />
731     <xs:enumeration value="northWestBound" />
732     <xs:enumeration value="inboundTowardsTown" />
733     <xs:enumeration value="outboundFromTown" />
734     <xs:enumeration value="unknown" />
735     <xs:enumeration value="opposite" />
736     <xs:enumeration value="other" />
737   </xs:restriction>
738 </xs:simpleType>
739 <xs:complexType name="DistanceAlongLinearElement" abstract="true">
740   <xs:sequence>
741     <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
742     <xs:sequence>
743   </xs:complexType>
744 <xs:complexType name="DistanceFromLinearElementReferent">
745   <xs:complexContent>
746     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
747       <xs:sequence>
748         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat"
749         <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
750         <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=
751         <xs:element name="distanceFromLinearElementReferentExtension" type=
752         <xs:sequence>
753       </xs:extension>
754     </xs:complexContent>
755 </xs:complexType>
756 <xs:complexType name="DistanceFromLinearElementStart">
757   <xs:complexContent>
758     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
759       <xs:sequence>
760         <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat"
761         <xs:element name="distanceFromLinearElementStartExtension" type=

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762     </xs:sequence>
763   </xs:extension>
764 </xs:complexContent>
765 </xs:complexType>
766 <xs:complexType name="DisturbanceActivity">
767   <xs:complexContent>
768     <xs:extension base="D2LogicalModel:Activity">
769       <xs:sequence>
770         <xs:element name="disturbanceActivityType" type=
771           <xs:element name="disturbanceActivityExtension" type="D2LogicalModel:_
772             <xs:sequence>
773           </xs:extension>
774         </xs:complexContent>
775       </xs:complexType>
776     <xs:simpleType name="DisturbanceActivityTypeEnum">
777       <xs:restriction base="xs:string">
778         <xs:enumeration value="airRaid" />
779         <xs:enumeration value="altercationOfVehicleOccupants" />
780         <xs:enumeration value="assault" />
781         <xs:enumeration value="assetDestruction" />
782         <xs:enumeration value="attack" />
783         <xs:enumeration value="attackOnVehicle" />
784         <xs:enumeration value="blockadeOrBarrier" />
785         <xs:enumeration value="bombAlert" />
786         <xs:enumeration value="crowd" />
787         <xs:enumeration value="demonstration" />
788         <xs:enumeration value="evacuation" />
789         <xs:enumeration value="filterBlockade" />
790         <xs:enumeration value="goSlowOperation" />
791         <xs:enumeration value="gunfireOnRoadway" />
792         <xs:enumeration value="illVehicleOccupants" />
793         <xs:enumeration value="march" />
794         <xs:enumeration value="publicDisturbance" />
795         <xs:enumeration value="radioactiveLeakAlert" />
796         <xs:enumeration value="riot" />
797         <xs:enumeration value="sabotage" />
798         <xs:enumeration value="securityAlert" />
799         <xs:enumeration value="securityIncident" />
800         <xs:enumeration value="sightseersObstructingAccess" />
801         <xs:enumeration value="strike" />
802         <xs:enumeration value="terroristIncident" />
803         <xs:enumeration value="theft" />
804         <xs:enumeration value="toxicCloudAlert" />
805         <xs:enumeration value="unspecifiedAlert" />
806         <xs:enumeration value="other" />
807       </xs:restriction>
808     </xs:simpleType>
809   <xs:simpleType name="DrivingConditionTypeEnum">
810     <xs:restriction base="xs:string">
811       <xs:enumeration value="impossible" />
812       <xs:enumeration value="hazardous" />
813       <xs:enumeration value="normal" />
814       <xs:enumeration value="passableWithCare" />

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815     <xs:enumeration value="unknown" />
816     <xs:enumeration value="veryHazardous" />
817     <xs:enumeration value="winterConditions" />
818     <xs:enumeration value="other" />
819   </xs:restriction>
820 </xs:simpleType>
821 <xs:complexType name="ElaboratedDataFault">
822   <xs:complexContent>
823     <xs:extension base="D2LogicalModel:Fault">
824       <xs:sequence>
825         <xs:element name="elaboratedDataFault" type=
826 ↪ "D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1" />
827         <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
828 ↪ ExtensionType" minOccurs="0" />
829       </xs:sequence>
830     </xs:extension>
831   </xs:complexContent>
832 </xs:complexType>
833 <xs:simpleType name="ElaboratedDataFaultEnum">
834   <xs:restriction base="xs:string">
835     <xs:enumeration value="intermittentDataValues" />
836     <xs:enumeration value="noDataValuesAvailable" />
837     <xs:enumeration value="spuriousUnreliableDataValues" />
838     <xs:enumeration value="unspecifiedOrUnknownFault" />
839     <xs:enumeration value="other" />
840   </xs:restriction>
841 </xs:simpleType>
842 <xs:complexType name="EnvironmentalObstruction">
843   <xs:complexContent>
844     <xs:extension base="D2LogicalModel:Obstruction">
845       <xs:sequence>
846         <xs:element name="depth" type="D2LogicalModel:MetresAsFloat" minOccurs="0"
847 ↪ maxOccurs="1" />
848         <xs:element name="environmentalObstructionType" type=
849 ↪ "D2LogicalModel:EnvironmentalObstructionTypeEnum" minOccurs="1" maxOccurs="1" />
850         <xs:element name="environmentalObstructionExtension" type="D2LogicalModel:_
851 ↪ ExtensionType" minOccurs="0" />
852       </xs:sequence>
853     </xs:extension>
854   </xs:complexContent>
855 </xs:complexType>
856 <xs:simpleType name="EnvironmentalObstructionTypeEnum">
857   <xs:restriction base="xs:string">
858     <xs:enumeration value="avalanches" />
859     <xs:enumeration value="earthquakeDamage" />
860     <xs:enumeration value="fallenTrees" />
861     <xs:enumeration value="fallingIce" />
862     <xs:enumeration value="fallingLightIceOrSnow" />
863     <xs:enumeration value="flashFloods" />
864     <xs:enumeration value="flooding" />
865     <xs:enumeration value="forestFire" />
866     <xs:enumeration value="grassFire" />
867     <xs:enumeration value="landslips" />
868     <xs:enumeration value="mudSlide" />
869     <xs:enumeration value="sewerOverflow" />
870     <xs:enumeration value="rockfalls" />

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866     <xs:enumeration value="seriousFire" />
867     <xs:enumeration value="smokeOrFumes" />
868     <xs:enumeration value="stormDamage" />
869     <xs:enumeration value="subsidence" />
870     <xs:enumeration value="other" />
871 </xs:restriction>
872 </xs:simpleType>
873 <xs:complexType name="EquipmentOrSystemFault">
874     <xs:complexContent>
875         <xs:extension base="D2LogicalModel:TrafficElement">
876             <xs:sequence>
877                 <xs:element name="equipmentOrSystemFaultType" type=
878 ↪ "D2LogicalModel:EquipmentOrSystemFaultTypeEnum" minOccurs="1" maxOccurs="1" />
879                 <xs:element name="faultyEquipmentOrSystemType" type=
880 ↪ "D2LogicalModel:EquipmentOrSystemTypeEnum" minOccurs="1" maxOccurs="1" />
881                 <xs:element name="equipmentOrSystemFaultExtension" type="D2LogicalModel:_
882 ↪ ExtensionType" minOccurs="0" />
883             </xs:sequence>
884         </xs:extension>
885     </xs:complexContent>
886 </xs:complexType>
887 <xs:simpleType name="EquipmentOrSystemFaultTypeEnum">
888     <xs:restriction base="xs:string">
889         <xs:enumeration value="notWorking" />
890         <xs:enumeration value="outOfService" />
891         <xs:enumeration value="workingIncorrectly" />
892         <xs:enumeration value="workingIntermittently" />
893     </xs:restriction>
894 </xs:simpleType>
895 <xs:simpleType name="EquipmentOrSystemTypeEnum">
896     <xs:restriction base="xs:string">
897         <xs:enumeration value="anprCameras" />
898         <xs:enumeration value="automatedTollSystem" />
899         <xs:enumeration value="cctvCameras" />
900         <xs:enumeration value="emergencyRoadsideTelephones" />
901         <xs:enumeration value="galleryLights" />
902         <xs:enumeration value="laneControlSigns" />
903         <xs:enumeration value="levelCrossing" />
904         <xs:enumeration value="matrixSigns" />
905         <xs:enumeration value="rampControls" />
906         <xs:enumeration value="roadsideCommunicationsSystem" />
907         <xs:enumeration value="roadsidePowerSystem" />
908         <xs:enumeration value="speedControlSigns" />
909         <xs:enumeration value="streetLighting" />
910         <xs:enumeration value="temporaryTrafficLights" />
911         <xs:enumeration value="tollGates" />
912         <xs:enumeration value="trafficLightSets" />
913         <xs:enumeration value="trafficSignals" />
914         <xs:enumeration value="tunnelLights" />
915         <xs:enumeration value="tunnelVentilation" />
916         <xs:enumeration value="variableMessageSigns" />
917         <xs:enumeration value="other" />
918     </xs:restriction>
919 </xs:simpleType>
920 <xs:complexType name="Exchange">
921     <xs:sequence>

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919     <xs:element name="supplierIdentification" type=
↪ "D2LogicalModel:InternationalIdentifier" />
920     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />
921     </xs:sequence>
922 </xs:complexType>
923 <xs:complexType name="Fault">
924     <xs:sequence>
925         <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0" ↪
↪ maxOccurs="1" />
926         <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0" ↪
↪ maxOccurs="1" />
927         <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0"
↪ " maxOccurs="1" />
928         <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↪ "1" maxOccurs="1" />
929         <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum" ↪
↪ minOccurs="0" maxOccurs="1" />
930         <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType" ↪
↪ minOccurs="0" />
931     </xs:sequence>
932 </xs:complexType>
933 <xs:simpleType name="FaultSeverityEnum">
934     <xs:restriction base="xs:string">
935         <xs:enumeration value="low" />
936         <xs:enumeration value="medium" />
937         <xs:enumeration value="high" />
938         <xs:enumeration value="unknown" />
939     </xs:restriction>
940 </xs:simpleType>
941 <xs:simpleType name="Float">
942     <xs:restriction base="xs:float" />
943 </xs:simpleType>
944 <xs:complexType name="FloatingPointMetreDistanceValue">
945     <xs:complexContent>
946         <xs:extension base="D2LogicalModel:DataValue">
947             <xs:sequence>
948                 <xs:element name="floatingPointMetreDistance" type=
↪ "D2LogicalModel:MetresAsFloat" minOccurs="1" maxOccurs="1" />
949                 <xs:element name="floatingPointMetreDistanceValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
950             </xs:sequence>
951         </xs:extension>
952     </xs:complexContent>
953 </xs:complexType>
954 <xs:complexType name="GeneralInstructionOrMessageToRoadUsers">
955     <xs:complexContent>
956         <xs:extension base="D2LogicalModel:NetworkManagement">
957             <xs:sequence>
958                 <xs:element name="generalInstructionToRoadUsersType" type=
↪ "D2LogicalModel:GeneralInstructionToRoadUsersTypeEnum" minOccurs="0" maxOccurs="1" /
↪ >
959                 <xs:element name="generalMessageToRoadUsers" type=
↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
960                 <xs:element name="generalInstructionOrMessageToRoadUsersExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />

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961     </xs:sequence>
962   </xs:extension>
963 </xs:complexContent>
964 </xs:complexType>
965 <xs:simpleType name="GeneralInstructionToRoadUsersTypeEnum">
966   <xs:restriction base="xs:string">
967     <xs:enumeration value="allowEmergencyVehiclesToPass" />
968     <xs:enumeration value="approachWithCare" />
969     <xs:enumeration value="avoidTheArea" />
970     <xs:enumeration value="closeAllWindowsTurnOffHeaterAndVents" />
971     <xs:enumeration value="crossJunctionWithCare" />
972     <xs:enumeration value="doNotAllowUnnecessaryGaps" />
973     <xs:enumeration value="doNotLeaveYourVehicle" />
974     <xs:enumeration value="doNotThrowOutAnyBurningObjects" />
975     <xs:enumeration value="doNotUseNavigationSystems" />
976     <xs:enumeration value="driveCarefully" />
977     <xs:enumeration value="driveWithExtremeCaution" />
978     <xs:enumeration value="flashYourLights" />
979     <xs:enumeration value="followTheVehicleInFrontSmoothly" />
980     <xs:enumeration value="increaseNormalFollowingDistance" />
981     <xs:enumeration value="inEmergencyWaitForPatrolService" />
982     <xs:enumeration value="keepYourDistance" />
983     <xs:enumeration value="leaveYourVehicleProceedToNextSafePlace" />
984     <xs:enumeration value="noNakedFlames" />
985     <xs:enumeration value="noOvertaking" />
986     <xs:enumeration value="noSmoking" />
987     <xs:enumeration value="noStopping" />
988     <xs:enumeration value="noUTurns" />
989     <xs:enumeration value="observeAmberAlert" />
990     <xs:enumeration value="observeSignals" />
991     <xs:enumeration value="observeSigns" />
992     <xs:enumeration value="onlyTravelIfAbsolutelyNecessary" />
993     <xs:enumeration value="overtakeWithCare" />
994     <xs:enumeration value="pullOverToTheEdgeOfTheRoadway" />
995     <xs:enumeration value="stopAtNextSafePlace" />
996     <xs:enumeration value="stopAtNextServiceArea" />
997     <xs:enumeration value="switchOffEngine" />
998     <xs:enumeration value="switchOffMobilePhonesAndTwoWayRadios" />
999     <xs:enumeration value="testYourBrakes" />
1000    <xs:enumeration value="useBusService" />

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1001    <xs:enumeration value="useFogLights" />
1002    <xs:enumeration value="useHazardWarningLights" />
1003    <xs:enumeration value="useHeadlights" />
1004    <xs:enumeration value="useRailService" />
1005    <xs:enumeration value="useTramService" />
1006    <xs:enumeration value="useUndergroundService" />
1007    <xs:enumeration value="waitForEscortVehicle" />
1008    <xs:enumeration value="other" />
1009  </xs:restriction>
1010 </xs:simpleType>
1011 <xs:complexType name="GeneralNetworkManagement">
1012   <xs:complexContent>
1013     <xs:extension base="D2LogicalModel:NetworkManagement">
1014       <xs:sequence>
1015         <xs:element name="generalNetworkManagementType" type=

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1016 <↪ "D2LogicalModel:GeneralNetworkManagementTypeEnum" minOccurs="1" maxOccurs="1" />
    <xs:element name="trafficManuallyDirectedBy" type=
1017 <↪ "D2LogicalModel:PersonCategoryEnum" minOccurs="0" maxOccurs="1" />
    <xs:element name="generalNetworkManagementExtension" type="D2LogicalModel:_
1018 <↪ ExtensionType" minOccurs="0" />
    </xs:sequence>
1019 </xs:extension>
1020 </xs:complexContent>
1021 </xs:complexType>
1022 <xs:simpleType name="GeneralNetworkManagementTypeEnum">
1023 <xs:restriction base="xs:string">
1024 <xs:enumeration value="bridgeSwingInOperation" />
1025 <xs:enumeration value="convoyService" />
1026 <xs:enumeration value="obstacleSignalling" />
1027 <xs:enumeration value="rampMeteringInOperation" />
1028 <xs:enumeration value="temporaryTrafficLights" />
1029 <xs:enumeration value="tollGatesOpen" />
1030 <xs:enumeration value="trafficBeingManuallyDirected" />
1031 <xs:enumeration value="trafficHeld" />
1032 <xs:enumeration value="other" />
1033 </xs:restriction>
1034 </xs:simpleType>
1035 <xs:complexType name="GeneralObstruction">
1036 <xs:complexContent>
1037 <xs:extension base="D2LogicalModel:Obstruction">
1038 <xs:sequence>
1039 <xs:element name="obstructionType" type="D2LogicalModel:ObstructionTypeEnum
1040 <↪ " minOccurs="1" maxOccurs="unbounded" />
    <xs:element name="groupOfPeopleInvolved" type=
1041 <↪ "D2LogicalModel:GroupOfPeopleInvolved" minOccurs="0" maxOccurs="unbounded" />
    <xs:element name="generalObstructionExtension" type="D2LogicalModel:_
1042 <↪ ExtensionType" minOccurs="0" />
    </xs:sequence>
1043 </xs:extension>
1044 </xs:complexContent>
1045 </xs:complexType>
1046 <xs:complexType name="GenericSituationRecord" abstract="true">
1047 <xs:complexContent>
1048 <xs:extension base="D2LogicalModel:SituationRecord">
1049 <xs:sequence>
1050 <xs:element name="genericSituationRecordName" type="D2LogicalModel:String"
1051 <↪ minOccurs="1" maxOccurs="1" />
    <xs:element name="genericSituationRecordExtension" type="D2LogicalModel:_
1052 <↪ ExtensionType" minOccurs="0" />
    </xs:sequence>
1053 </xs:extension>
1054 </xs:complexContent>
1055 </xs:complexType>
1056 <xs:simpleType name="GeometryTypeEnum">
1057 <xs:restriction base="xs:string" />
1058 </xs:simpleType>
1059 <xs:complexType name="GlobalNetworkLinear">
1060 <xs:sequence>
1061 <xs:element name="networkVersion" type="D2LogicalModel:String" minOccurs="1"
1062 <↪ maxOccurs="1" />
    <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum"

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1063   minOccurs="1" maxOccurs="1" />
1064     <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates"
1065     minOccurs="0" />
1066     <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
1067     minOccurs="0" />
1068     <xs:element name="linearWithinLinearGNElement" type=
1069     "D2LogicalModel:LinearWithinLinearGNElement" maxOccurs="unbounded" />
1070   </xs:sequence>
1071 </xs:complexType>
1072 <xs:complexType name="GrossWeightCharacteristic">
1073   <xs:sequence>
1074     <xs:element name="comparisonOperator" type=
1075     "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1076     <xs:element name="grossVehicleWeight" type="D2LogicalModel:Tonnes" minOccurs="1"
1077     maxOccurs="1" />
1078     <xs:element name="grossWeightCharacteristicExtension" type="D2LogicalModel:_
1079     ExtensionType" minOccurs="0" />
1080   </xs:sequence>
1081 </xs:complexType>
1082 <xs:complexType name="GroupOfLocations" abstract="true">
1083   <xs:sequence>
1084     <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType"
1085     minOccurs="0" />
1086   </xs:sequence>
1087 </xs:complexType>
1088 <xs:complexType name="GroupOfPeopleInvolved">
1089   <xs:sequence>
1090     <xs:element name="numberOfPeople" type="D2LogicalModel:NonNegativeInteger"
1091     minOccurs="0" maxOccurs="1" />
1092     <xs:element name="injuryStatus" type="D2LogicalModel:InjuryStatusTypeEnum"
1093     minOccurs="0" maxOccurs="1" />
1094     <xs:element name="involvementRole" type="D2LogicalModel:InvolvementRolesEnum"
1095     minOccurs="0" maxOccurs="1" />
1096     <xs:element name="categoryOfPeopleInvolved" type=
1097     "D2LogicalModel:PersonCategoryEnum" minOccurs="0" maxOccurs="1" />
1098     <xs:element name="groupOfPeopleInvolvedExtension" type="D2LogicalModel:_
1099     ExtensionType" minOccurs="0" />
1100   </xs:sequence>
1101 </xs:complexType>
1102 <xs:complexType name="GroupOfVehiclesInvolved">
1103   <xs:sequence>
1104     <xs:element name="numberOfVehicles" type="D2LogicalModel:NonNegativeInteger"
1105     minOccurs="0" maxOccurs="1" />
1106     <xs:element name="vehicleStatus" type="D2LogicalModel:VehicleStatusEnum"
1107     minOccurs="0" maxOccurs="1" />
1108     <xs:element name="vehicleCharacteristics" type=
1109     "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
1110     <xs:element name="groupOfVehiclesInvolvedExtension" type="D2LogicalModel:_
1111     ExtensionType" minOccurs="0" />
1112   </xs:sequence>
1113 </xs:complexType>
1114 <xs:complexType name="HazardousMaterials">
1115   <xs:sequence>
1116     <xs:element name="chemicalName" type="D2LogicalModel:MultilingualString"
1117     minOccurs="1" maxOccurs="1" />
1118     <xs:element name="dangerousGoodsFlashPoint" type=

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1101 <xs:element name="dangerousGoodsRegulations" type=
1102 <xs:element name="hazardCodeIdentification" type="D2LogicalModel:String"
1103 minOccurs="0" maxOccurs="1" />
1104 <xs:element name="hazardCodeVersionNumber" type=
1105 <xs:element name="hazardSubstanceItemPageNumber" type="D2LogicalModel:String"
1106 minOccurs="0" maxOccurs="1" />
1107 <xs:element name="tremCardNumber" type="D2LogicalModel:String" minOccurs="0"
1108 maxOccurs="1" />
1109 <xs:element name="undgNumber" type="D2LogicalModel:String" minOccurs="0"
1110 maxOccurs="1" />
1111 <xs:element name="volumeOfDangerousGoods" type="D2LogicalModel:CubicMetres"
1112 minOccurs="0" maxOccurs="1" />
1113 <xs:element name="weightOfDangerousGoods" type="D2LogicalModel:Tonnes"
1114 minOccurs="0" maxOccurs="1" />
1115 <xs:element name="hazardousMaterialsExtension" type="D2LogicalModel:_
1116 ExtensionType" minOccurs="0" />
1117 </xs:sequence>
1118 </xs:complexType>
1119 <xs:complexType name="HeaderInformation">
1120 <xs:sequence>
1121 <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
1122 " minOccurs="1" maxOccurs="1" />
1123 <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
1124 " minOccurs="1" maxOccurs="1" />
1125 <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
1126 maxOccurs="1" />
1127 <xs:element name="headerInformationExtension" type="D2LogicalModel:_
1128 ExtensionType" minOccurs="0" />
1129 </xs:sequence>
1130 </xs:complexType>
1131 <xs:complexType name="HeaviestAxleWeightCharacteristic">
1132 <xs:sequence>
1133 <xs:element name="comparisonOperator" type=
1134 "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1135 <xs:element name="heaviestAxleWeight" type="D2LogicalModel:Tonnes" minOccurs="1"
1136 maxOccurs="1" />
1137 <xs:element name="heaviestAxleWeightCharacteristicExtension" type=
1138 "D2LogicalModel:_ExtensionType" minOccurs="0" />
1139 </xs:sequence>
1140 </xs:complexType>
1141 <xs:complexType name="HeightCharacteristic">
1142 <xs:sequence>
1143 <xs:element name="comparisonOperator" type=
1144 "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1145 <xs:element name="vehicleHeight" type="D2LogicalModel:MetresAsFloat" minOccurs=
1146 "1" maxOccurs="1" />
1147 <xs:element name="heightCharacteristicExtension" type="D2LogicalModel:_
1148 ExtensionType" minOccurs="0" />
1149 </xs:sequence>
1150 </xs:complexType>
1151 <xs:simpleType name="HeightGradeEnum">
1152 <xs:restriction base="xs:string">
1153 <xs:enumeration value="aboveGrade" />

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1137     <xs:enumeration value="atGrade" />
1138     <xs:enumeration value="belowGrade" />
1139   </xs:restriction>
1140 </xs:simpleType>
1141 <xs:complexType name="Impact">
1142   <xs:sequence>
1143     <xs:element name="capacityRemaining" type="D2LogicalModel:Percentage" minOccurs=
1144     ↪ "0" maxOccurs="1" />
1145     <xs:element name="numberOfLanesRestricted" type=
1146     ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1147     <xs:element name="numberOfOperationalLanes" type=
1148     ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1149     <xs:element name="originalNumberOfLanes" type="D2LogicalModel:NonNegativeInteger
1150     ↪ " minOccurs="0" maxOccurs="1" />
1151     <xs:element name="trafficConstrictionType" type=
1152     ↪ "D2LogicalModel:TrafficConstrictionTypeEnum" minOccurs="0" maxOccurs="1" />
1153     <xs:element name="delays" type="D2LogicalModel:Delays" minOccurs="0" />
1154     <xs:element name="impactExtension" type="D2LogicalModel:_ExtensionType"
1155     ↪ minOccurs="0" />
1156   </xs:sequence>
1157 </xs:complexType>
1158 <xs:simpleType name="InformationStatusEnum">
1159   <xs:restriction base="xs:string">
1160     <xs:enumeration value="real" />
1161     <xs:enumeration value="securityExercise" />
1162     <xs:enumeration value="technicalExercise" />
1163     <xs:enumeration value="test" />
1164   </xs:restriction>
1165 </xs:simpleType>
1166 <xs:complexType name="InfrastructureDamageObstruction">
1167   <xs:complexContent>
1168     <xs:extension base="D2LogicalModel:Obstruction">
1169       <xs:sequence>
1170         <xs:element name="infrastructureDamageType" type=
1171         ↪ "D2LogicalModel:InfrastructureDamageTypeEnum" minOccurs="1" maxOccurs="1" />
1172         <xs:element name="infrastructureDamageObstructionExtension" type=
1173         ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1174       </xs:sequence>
1175     </xs:extension>
1176   </xs:complexContent>
1177 </xs:complexType>
1178 <xs:simpleType name="InfrastructureDamageTypeEnum">
1179   <xs:restriction base="xs:string">
1180     <xs:enumeration value="burstPipe" />
1181     <xs:enumeration value="burstWaterMain" />
1182     <xs:enumeration value="collapsedSewer" />
1183     <xs:enumeration value="damagedBridge" />
1184     <xs:enumeration value="damagedCrashBarrier" />
1185     <xs:enumeration value="damagedFlyover" />
1186     <xs:enumeration value="damagedGallery" />
1187     <xs:enumeration value="damagedGantry" />
1188     <xs:enumeration value="damagedRoadSurface" />
1189     <xs:enumeration value="damagedTunnel" />
1190     <xs:enumeration value="damagedViaduct" />
1191     <xs:enumeration value="fallenPowerCables" />
1192     <xs:enumeration value="gasLeak" />

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1185     <xs:enumeration value="weakBridge" />
1186     <xs:enumeration value="other" />
1187   </xs:restriction>
1188 </xs:simpleType>
1189 <xs:simpleType name="InjuryStatusTypeEnum">
1190   <xs:restriction base="xs:string">
1191     <xs:enumeration value="dead" />
1192     <xs:enumeration value="injured" />
1193     <xs:enumeration value="seriouslyInjured" />
1194     <xs:enumeration value="slightlyInjured" />
1195     <xs:enumeration value="uninjured" />
1196     <xs:enumeration value="unknown" />
1197   </xs:restriction>
1198 </xs:simpleType>
1199 <xs:simpleType name="Integer">
1200   <xs:restriction base="xs:integer" />
1201 </xs:simpleType>
1202 <xs:complexType name="IntegerMetreDistanceValue">
1203   <xs:complexContent>
1204     <xs:extension base="D2LogicalModel:DataValue">
1205       <xs:sequence>
1206         <xs:element name="integerMetreDistance" type=
1207 ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1" />
1208         <xs:element name="integerMetreDistanceValueExtension" type="D2LogicalModel:_
1209 ↪ ExtensionType" minOccurs="0" />
1210       </xs:sequence>
1211     </xs:extension>
1212   </xs:complexContent>
1213 </xs:complexType>
1214 <xs:simpleType name="IntensityKilogramsPerSquareMetre">
1215   <xs:restriction base="D2LogicalModel:Float" />
1216 </xs:simpleType>
1217 <xs:simpleType name="IntensityMillimetresPerHour">
1218   <xs:restriction base="D2LogicalModel:Float" />
1219 </xs:simpleType>
1220 <xs:complexType name="InternationalIdentifier">
1221   <xs:sequence>
1222     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
1223 ↪ maxOccurs="1" />
1224     <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
1225 ↪ maxOccurs="1" />
1226     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
1227 ↪ ExtensionType" minOccurs="0" />
1228   </xs:sequence>
1229 </xs:complexType>
1230 <xs:simpleType name="InvolvementRolesEnum">
1231   <xs:restriction base="xs:string">
1232     <xs:enumeration value="cyclist" />
1233     <xs:enumeration value="pedestrian" />
1234     <xs:enumeration value="unknown" />
1235     <xs:enumeration value="vehicleDriver" />
1236     <xs:enumeration value="vehicleOccupant" />
1237     <xs:enumeration value="vehiclePassenger" />
1238     <xs:enumeration value="witness" />
1239   </xs:restriction>
1240 </xs:simpleType>

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1236 <xs:complexType name="KilogramsConcentrationValue">
1237   <xs:complexContent>
1238     <xs:extension base="D2LogicalModel:DataValue">
1239       <xs:sequence>
1240         <xs:element name="kilogramsConcentration" type=
1241 ↪ "D2LogicalModel:ConcentrationKilogramsPerCubicMetre" minOccurs="1" maxOccurs="1" />
1242         <xs:element name="kilogramsConcentrationValueExtension" type=
1243 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1244       </xs:sequence>
1245     </xs:extension>
1246   </xs:complexContent>
1247 </xs:complexType>
1248 <xs:simpleType name="KilometresPerHour">
1249   <xs:restriction base="D2LogicalModel:Float" />
1250 </xs:simpleType>
1251 <xs:simpleType name="LaneEnum">
1252   <xs:restriction base="xs:string">
1253     <xs:enumeration value="allLanesCompleteCarriageway" />
1254     <xs:enumeration value="busLane" />
1255     <xs:enumeration value="busStop" />
1256     <xs:enumeration value="carPoolLane" />
1257     <xs:enumeration value="centralReservation" />
1258     <xs:enumeration value="crawlerLane" />
1259     <xs:enumeration value="emergencyLane" />
1260     <xs:enumeration value="escapeLane" />
1261     <xs:enumeration value="expressLane" />
1262     <xs:enumeration value="hardShoulder" />
1263     <xs:enumeration value="heavyVehicleLane" />
1264     <xs:enumeration value="lane1" />
1265     <xs:enumeration value="lane2" />
1266     <xs:enumeration value="lane3" />
1267     <xs:enumeration value="lane4" />
1268     <xs:enumeration value="lane5" />
1269     <xs:enumeration value="lane6" />
1270     <xs:enumeration value="lane7" />
1271     <xs:enumeration value="lane8" />
1272     <xs:enumeration value="lane9" />
1273     <xs:enumeration value="layBy" />
1274     <xs:enumeration value="leftHandTurningLane" />
1275     <xs:enumeration value="leftLane" />
1276     <xs:enumeration value="localTrafficLane" />
1277     <xs:enumeration value="middleLane" />
1278     <xs:enumeration value="opposingLanes" />
1279     <xs:enumeration value="overtakingLane" />
1280     <xs:enumeration value="rightHandTurningLane" />
1281     <xs:enumeration value="rightLane" />
1282     <xs:enumeration value="rushHourLane" />
1283     <xs:enumeration value="setDownArea" />
1284     <xs:enumeration value="slowVehicleLane" />
1285     <xs:enumeration value="throughTrafficLane" />
1286     <xs:enumeration value="tidalFlowLane" />
1287     <xs:enumeration value="turningLane" />
1288     <xs:enumeration value="verge" />
1289   </xs:restriction>
1290 </xs:simpleType>
1291 <xs:simpleType name="Language">

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1290     <xs:restriction base="xs:language" />
1291   </xs:simpleType>
1292   <xs:complexType name="LengthCharacteristic">
1293     <xs:sequence>
1294       <xs:element name="comparisonOperator" type=
1295 ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1296       <xs:element name="vehicleLength" type="D2LogicalModel:MetresAsFloat" minOccurs=
1297 ↪ "1" maxOccurs="1" />
1298       <xs:element name="lengthCharacteristicExtension" type="D2LogicalModel:_
1299 ↪ ExtensionType" minOccurs="0" />
1300     </xs:sequence>
1301   </xs:complexType>
1302   <xs:complexType name="Linear">
1303     <xs:complexContent>
1304       <xs:extension base="D2LogicalModel:NetworkLocation">
1305         <xs:sequence>
1306           <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
1307 ↪ minOccurs="0" />
1308           <xs:element name="linearWithinLinearElement" type=
1309 ↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
1310           <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType
1311 ↪ " minOccurs="0" />
1312         </xs:sequence>
1313       </xs:extension>
1314     </xs:complexContent>
1315   </xs:complexType>
1316   <xs:complexType name="LinearElement">
1317     <xs:sequence>
1318       <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
1319 ↪ "0" maxOccurs="1" />
1320       <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
1321 ↪ maxOccurs="1" />
1322       <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
1323 ↪ minOccurs="0" maxOccurs="1" />
1324       <xs:element name="linearElementReferenceModelVersion" type=
1325 ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
1326       <xs:element name="linearElementNature" type=
1327 ↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
1328       <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
1329 ↪ minOccurs="0" />
1330     </xs:sequence>
1331   </xs:complexType>
1332   <xs:complexType name="LinearElementByCode">
1333     <xs:complexContent>
1334       <xs:extension base="D2LogicalModel:LinearElement">
1335         <xs:sequence>
1336           <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"
1337 ↪ minOccurs="1" maxOccurs="1" />
1338           <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
1339 ↪ ExtensionType" minOccurs="0" />
1340         </xs:sequence>
1341       </xs:extension>
1342     </xs:complexContent>
1343   </xs:complexType>
1344   <xs:complexType name="LinearElementByPoints">
1345     <xs:complexContent>

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1332     <xs:extension base="D2LogicalModel:LinearElement">
1333         <xs:sequence>
1334             <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent" />
1335             <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
1336             <IntermediatePointOnLinearElement" minOccurs="0" maxOccurs="unbounded" />
1337             <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
1338             <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
1339             <ExtensionType" minOccurs="0" />
1340         </xs:sequence>
1341     </xs:extension>
1342 </xs:complexType>
1343 <xs:simpleType name="LinearElementNatureEnum">
1344     <xs:restriction base="xs:string">
1345         <xs:enumeration value="road" />
1346         <xs:enumeration value="roadSection" />
1347         <xs:enumeration value="slipRoad" />
1348         <xs:enumeration value="other" />
1349     </xs:restriction>
1350 </xs:simpleType>
1351 <xs:simpleType name="LinearReferencingDirectionEnum">
1352     <xs:restriction base="xs:string">
1353         <xs:enumeration value="both" />
1354         <xs:enumeration value="opposite" />
1355         <xs:enumeration value="aligned" />
1356         <xs:enumeration value="unknown" />
1357     </xs:restriction>
1358 </xs:simpleType>
1359 <xs:complexType name="LinearWithinLinearElement">
1360     <xs:sequence>
1361         <xs:element name="administrativeAreaOfLinearSection" type=
1362         <"D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1363         <xs:element name="directionBoundOnLinearSection" type=
1364         <"D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="1" />
1365         <xs:element name="directionRelativeOnLinearSection" type=
1366         <"D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
1367         <xs:element name="heightGradeOfLinearSection" type=
1368         <"D2LogicalModel:HeightGradeEnum" minOccurs="0" maxOccurs="1" />
1369         <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
1370         <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1371         <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1372         <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
1373         <ExtensionType" minOccurs="0" />
1374     </xs:sequence>
1375 </xs:complexType>
1376 <xs:complexType name="LinearWithinLinearGNElement">
1377     <xs:sequence>
1378         <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
1379         <"0" maxOccurs="1" />
1380         <xs:element name="directionRelativeOnLinearSection" type=
1381         <"D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
1382         <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger" />
1383         <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
1384         <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />

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1377     <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1378   </xs:sequence>
1379 </xs:complexType>
1380 <xs:simpleType name="LoadTypeEnum">
1381   <xs:restriction base="xs:string">
1382     <xs:enumeration value="abnormalLoad" />
1383     <xs:enumeration value="ammunition" />
1384     <xs:enumeration value="chemicals" />
1385     <xs:enumeration value="combustibleMaterials" />
1386     <xs:enumeration value="corrosiveMaterials" />
1387     <xs:enumeration value="debris" />
1388     <xs:enumeration value="empty" />
1389     <xs:enumeration value="explosiveMaterials" />
1390     <xs:enumeration value="extraHighLoad" />
1391     <xs:enumeration value="extraLongLoad" />
1392     <xs:enumeration value="extraWideLoad" />
1393     <xs:enumeration value="fuel" />
1394     <xs:enumeration value="glass" />
1395     <xs:enumeration value="goods" />
1396     <xs:enumeration value="hazardousMaterials" />
1397     <xs:enumeration value="liquid" />
1398     <xs:enumeration value="livestock" />
1399     <xs:enumeration value="materials" />
1400     <xs:enumeration value="materialsDangerousForPeople" />
1401     <xs:enumeration value="materialsDangerousForTheEnvironment" />
1402     <xs:enumeration value="materialsDangerousForWater" />
1403     <xs:enumeration value="oil" />
1404     <xs:enumeration value="ordinary" />
1405     <xs:enumeration value="perishableProducts" />
1406     <xs:enumeration value="petrol" />
1407     <xs:enumeration value="pharmaceuticalMaterials" />
1408     <xs:enumeration value="radioactiveMaterials" />
1409     <xs:enumeration value="refuse" />
1410     <xs:enumeration value="toxicMaterials" />
1411     <xs:enumeration value="vehicles" />
1412     <xs:enumeration value="other" />
1413   </xs:restriction>
1414 </xs:simpleType>
1415 <xs:complexType name="Location" abstract="true">
1416   <xs:complexContent>
1417     <xs:extension base="D2LogicalModel:GroupOfLocations">
1418       <xs:sequence>
1419         <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType" ↪
↪minOccurs="0" />
1420       </xs:sequence>
1421     </xs:extension>
1422   </xs:complexContent>
1423 </xs:complexType>
1424 <xs:simpleType name="LocationDescriptorEnum">
1425   <xs:restriction base="xs:string">
1426     <xs:enumeration value="aroundABendInRoad" />
1427     <xs:enumeration value="atMotorwayInterchange" />
1428     <xs:enumeration value="atRestArea" />
1429     <xs:enumeration value="atServiceArea" />
1430     <xs:enumeration value="atTollPlaza" />

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1431     <xs:enumeration value="atTunnelEntryOrExit" />
1432     <xs:enumeration value="inbound" />
1433     <xs:enumeration value="inGallery" />
1434     <xs:enumeration value="inTheCentre" />
1435     <xs:enumeration value="inTheOppositeDirection" />
1436     <xs:enumeration value="inTunnel" />
1437     <xs:enumeration value="onBorder" />
1438     <xs:enumeration value="onBridge" />
1439     <xs:enumeration value="onConnector" />
1440     <xs:enumeration value="onElevatedSection" />
1441     <xs:enumeration value="onFlyover" />
1442     <xs:enumeration value="onIceRoad" />
1443     <xs:enumeration value="onLevelCrossing" />
1444     <xs:enumeration value="onLinkRoad" />
1445     <xs:enumeration value="onPass" />
1446     <xs:enumeration value="onRoundabout" />
1447     <xs:enumeration value="onTheLeft" />
1448     <xs:enumeration value="onTheRight" />
1449     <xs:enumeration value="onTheRoadway" />
1450     <xs:enumeration value="onUndergroundSection" />
1451     <xs:enumeration value="onUnderpass" />
1452     <xs:enumeration value="outbound" />
1453     <xs:enumeration value="overCrestOfHill" />
1454     <xs:enumeration value="withinJunction" />
1455 </xs:restriction>
1456 </xs:simpleType>
1457 <xs:simpleType name="MaintenanceVehicleActionsEnum">
1458     <xs:restriction base="xs:string">
1459         <xs:enumeration value="maintenanceVehiclesMergingIntoTrafficFlow" />
1460         <xs:enumeration value="saltAndGritSpreading" />
1461         <xs:enumeration value="slowMoving" />
1462         <xs:enumeration value="snowClearing" />
1463         <xs:enumeration value="stoppingToServiceEquipments" />
1464     </xs:restriction>
1465 </xs:simpleType>
1466 <xs:complexType name="MaintenanceVehicles">
1467     <xs:sequence>
1468         <xs:element name="numberOfMaintenanceVehicles" type=
1469         ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1470         <xs:element name="maintenanceVehicleActions" type=
1471         ↪ "D2LogicalModel:MaintenanceVehicleActionsEnum" minOccurs="0" maxOccurs="unbounded" /
1472         ↪ >
1473         <xs:element name="maintenanceVehiclesExtension" type="D2LogicalModel:_
1474         ↪ ExtensionType" minOccurs="0" />
1475     </xs:sequence>
1476 </xs:complexType>
1477 <xs:complexType name="MaintenanceWorks">
1478     <xs:complexContent>
1479         <xs:extension base="D2LogicalModel:Roadworks">
1480             <xs:sequence>
1481                 <xs:element name="roadMaintenanceType" type=
1482                 ↪ "D2LogicalModel:RoadMaintenanceTypeEnum" minOccurs="1" maxOccurs="unbounded" />
1483                 <xs:element name="maintenanceWorksExtension" type="D2LogicalModel:_
1484                 ↪ ExtensionType" minOccurs="0" />
1485             </xs:sequence>
1486         </xs:extension>

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1481     </xs:complexContent>
1482 </xs:complexType>
1483 <xs:complexType name="MeasurementEquipmentFault">
1484     <xs:complexContent>
1485         <xs:extension base="D2LogicalModel:Fault">
1486             <xs:sequence>
1487                 <xs:element name="measurementEquipmentFault" type=
1488 ↪ "D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1" />
1489                 <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
1490 ↪ ExtensionType" minOccurs="0" />
1491             </xs:sequence>
1492         </xs:extension>
1493     </xs:complexContent>
1494 </xs:complexType>
1495 <xs:simpleType name="MeasurementEquipmentFaultEnum">
1496     <xs:restriction base="xs:string">
1497         <xs:enumeration value="intermittentDataValues" />
1498         <xs:enumeration value="noDataValuesAvailable" />
1499         <xs:enumeration value="spuriousUnreliableDataValues" />
1500         <xs:enumeration value="unspecifiedOrUnknownFault" />
1501         <xs:enumeration value="other" />
1502     </xs:restriction>
1503 </xs:simpleType>
1504 <xs:simpleType name="MetresAsFloat">
1505     <xs:restriction base="D2LogicalModel:Float" />
1506 </xs:simpleType>
1507 <xs:simpleType name="MetresAsNonNegativeInteger">
1508     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1509 </xs:simpleType>
1510 <xs:complexType name="Mobility">
1511     <xs:sequence>
1512         <xs:element name="mobilityType" type="D2LogicalModel:MobilityEnum" minOccurs="1
1513 ↪ " maxOccurs="1" />
1514         <xs:element name="mobilityExtension" type="D2LogicalModel:_ExtensionType"
1515 ↪ minOccurs="0" />
1516     </xs:sequence>
1517 </xs:complexType>
1518 <xs:simpleType name="MobilityEnum">
1519     <xs:restriction base="xs:string">
1520         <xs:enumeration value="mobile" />
1521         <xs:enumeration value="stationary" />
1522         <xs:enumeration value="unknown" />
1523     </xs:restriction>
1524 </xs:simpleType>
1525 <xs:complexType name="MultilingualString">
1526     <xs:sequence>
1527         <xs:element name="values">
1528             <xs:complexType>
1529                 <xs:sequence>
1530                     <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
1531 ↪ maxOccurs="unbounded" />
1532                 </xs:sequence>
1533             </xs:complexType>
1534         </xs:element>
1535     </xs:sequence>
1536 </xs:complexType>

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1532 <xs:complexType name="MultilingualStringValue">
1533   <xs:simpleContent>
1534     <xs:extension base="D2LogicalModel:MultilingualStringValue" type="D2LogicalModel:MultilingualStringValue" />
1535     <xs:attribute name="lang" type="xs:language" />
1536   </xs:simpleContent>
1537 </xs:complexType>
1538 <xs:simpleType name="MultilingualStringValue" type="xs:string">
1539   <xs:restriction base="xs:string">
1540     <xs:maxLength value="1024" />
1541   </xs:restriction>
1542 </xs:simpleType>
1543 <xs:complexType name="NetworkLocation" abstract="true">
1544   <xs:complexContent>
1545     <xs:extension base="D2LogicalModel:Location">
1546       <xs:sequence>
1547         <xs:element name="supplementaryPositionalDescription" type="D2LogicalModel:SupplementaryPositionalDescription" minOccurs="0" />
1548         <xs:element name="destination" type="D2LogicalModel:Destination" minOccurs="0" />
1549         <xs:element name="networkLocationExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
1550       </xs:sequence>
1551     </xs:extension>
1552   </xs:complexContent>
1553 </xs:complexType>
1554 <xs:complexType name="NetworkManagement" abstract="true">
1555   <xs:complexContent>
1556     <xs:extension base="D2LogicalModel:OperatorAction">
1557       <xs:sequence>
1558         <xs:element name="complianceOption" type="D2LogicalModel:ComplianceOptionEnum" minOccurs="1" maxOccurs="1" />
1559         <xs:element name="applicableForTrafficDirection" type="D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="unbounded" />
1560         <xs:element name="applicableForTrafficType" type="D2LogicalModel:TrafficTypeEnum" minOccurs="0" maxOccurs="unbounded" />
1561         <xs:element name="placesAtWhichApplicable" type="D2LogicalModel:PlacesEnum" minOccurs="0" maxOccurs="unbounded" />
1562         <xs:element name="forVehiclesWithCharacteristicsOf" type="D2LogicalModel:VehicleCharacteristics" minOccurs="0" maxOccurs="unbounded" />
1563         <xs:element name="networkManagementExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
1564       </xs:sequence>
1565     </xs:extension>
1566   </xs:complexContent>
1567 </xs:complexType>
1568 <xs:complexType name="NonManagedCause">
1569   <xs:complexContent>
1570     <xs:extension base="D2LogicalModel:Cause">
1571       <xs:sequence>
1572         <xs:element name="causeType" type="D2LogicalModel:CauseTypeEnum" minOccurs="0" maxOccurs="1" />
1573         <xs:element name="nonManagedCauseExtension" type="D2LogicalModel:_ExtensionType" minOccurs="0" />
1574       </xs:sequence>
1575     </xs:extension>
1576   </xs:complexContent>

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1577     </xs:complexContent>
1578 </xs:complexType>
1579 <xs:simpleType name="NonNegativeInteger">
1580   <xs:restriction base="xs:nonNegativeInteger" />
1581 </xs:simpleType>
1582 <xs:complexType name="NonRoadEventInformation" abstract="true">
1583   <xs:complexContent>
1584     <xs:extension base="D2LogicalModel:SituationRecord">
1585       <xs:sequence>
1586         <xs:element name="nonRoadEventInformationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1587       </xs:sequence>
1588     </xs:extension>
1589   </xs:complexContent>
1590 </xs:complexType>
1591 <xs:complexType name="NonWeatherRelatedRoadConditions">
1592   <xs:complexContent>
1593     <xs:extension base="D2LogicalModel:RoadConditions">
1594       <xs:sequence>
1595         <xs:element name="nonWeatherRelatedRoadConditionType" type=
↪ "D2LogicalModel:NonWeatherRelatedRoadConditionTypeEnum" minOccurs="1" maxOccurs=
↪ "unbounded" />
1596         <xs:element name="nonWeatherRelatedRoadConditionsExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1597       </xs:sequence>
1598     </xs:extension>
1599   </xs:complexContent>
1600 </xs:complexType>
1601 <xs:simpleType name="NonWeatherRelatedRoadConditionTypeEnum">
1602   <xs:restriction base="xs:string">
1603     <xs:enumeration value="dieselOnRoad" />
1604     <xs:enumeration value="leavesOnRoad" />
1605     <xs:enumeration value="looseChippings" />
1606     <xs:enumeration value="looseSandOnRoad" />
1607     <xs:enumeration value="mudOnRoad" />
1608     <xs:enumeration value="oilOnRoad" />
1609     <xs:enumeration value="petrolOnRoad" />
1610     <xs:enumeration value="roadSurfaceInPoorCondition" />
1611     <xs:enumeration value="slipperyRoad" />
1612     <xs:enumeration value="other" />
1613   </xs:restriction>
1614 </xs:simpleType>
1615 <xs:complexType name="NumberOfAxlesCharacteristic">
1616   <xs:sequence>
1617     <xs:element name="comparisonOperator" type=
↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
1618     <xs:element name="numberOfAxles" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="1" maxOccurs="1" />
1619     <xs:element name="numberOfAxlesCharacteristicExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1620   </xs:sequence>
1621 </xs:complexType>
1622 <xs:complexType name="Obstruction" abstract="true">
1623   <xs:complexContent>
1624     <xs:extension base="D2LogicalModel:TrafficElement">
1625       <xs:sequence>

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1626     <xs:element name="numberOfObstructions" type=
1627     ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
1628     <xs:element name="mobilityOfObstruction" type="D2LogicalModel:Mobility"
1629     ↪ minOccurs="0" />
1630     <xs:element name="obstructionExtension" type="D2LogicalModel:_ExtensionType"
1631     ↪ minOccurs="0" />
1632   </xs:sequence>
1633 </xs:extension>
1634 </xs:complexContent>
1635 </xs:complexType>
1636 <xs:simpleType name="ObstructionTypeEnum">
1637   <xs:restriction base="xs:string">
1638     <xs:enumeration value="airCrash" />
1639     <xs:enumeration value="childrenOnRoadway" />
1640     <xs:enumeration value="clearanceWork" />
1641     <xs:enumeration value="craneOperating" />
1642     <xs:enumeration value="cyclistsOnRoadway" />
1643     <xs:enumeration value="debris" />
1644     <xs:enumeration value="explosion" />
1645     <xs:enumeration value="explosionHazard" />
1646     <xs:enumeration value="hazardsOnTheRoad" />
1647     <xs:enumeration value="highSpeedChase" />
1648     <xs:enumeration value="houseFire" />
1649     <xs:enumeration value="incident" />
1650     <xs:enumeration value="industrialAccident" />
1651     <xs:enumeration value="objectOnTheRoad" />
1652     <xs:enumeration value="objectsFallingFromMovingVehicle" />
1653     <xs:enumeration value="obstructionOnTheRoad" />
1654     <xs:enumeration value="peopleOnRoadway" />
1655     <xs:enumeration value="railCrash" />
1656     <xs:enumeration value="recklessDriver" />
1657     <xs:enumeration value="rescueAndRecoveryWork" />
1658     <xs:enumeration value="severeFrostDamagedRoadway" />
1659     <xs:enumeration value="shedLoad" />
1660     <xs:enumeration value="snowAndIceDebris" />
1661     <xs:enumeration value="spillageOccurringFromMovingVehicle" />
1662     <xs:enumeration value="spillageOnTheRoad" />
1663     <xs:enumeration value="unprotectedAccidentArea" />
1664     <xs:enumeration value="other" />
1665   </xs:restriction>
1666 </xs:simpleType>
1667 <xs:complexType name="OccupancyChangeValue">
1668   <xs:complexContent>
1669     <xs:extension base="D2LogicalModel:DataValue">
1670       <xs:sequence>
1671         <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
1672         ↪ "1" maxOccurs="1" />
1673         <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
1674         ↪ ExtensionType" minOccurs="0" />
1675       </xs:sequence>
1676     </xs:extension>
1677   </xs:complexContent>
1678 </xs:complexType>
1679 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
1680   <xs:sequence>
1681     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />

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1677     <xs:element name="openlrLineAttributes" type=
1678     ↪ "D2LogicalModel:OpenlrLineAttributes" />
1679     <xs:element name="openlrBaseLocationReferencePointExtension" type=
1680     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1681     </xs:sequence>
1682   </xs:complexType>
1683   <xs:complexType name="OpenlrBasePointLocation" abstract="true">
1684     <xs:sequence>
1685       <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum" ↪
1686       ↪ minOccurs="1" maxOccurs="1" />
1687       <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum"
1688       ↪ minOccurs="1" maxOccurs="1" />
1689       <xs:element name="openlrPositiveOffset" type=
1690       ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1691       <xs:element name="openlrLocationReferencePoint" type=
1692       ↪ "D2LogicalModel:OpenlrLocationReferencePoint" />
1693       <xs:element name="openlrLastLocationReferencePoint" type=
1694       ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1695       <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
1696       ↪ ExtensionType" minOccurs="0" />
1697     </xs:sequence>
1698   </xs:complexType>
1699   <xs:complexType name="OpenlrExtendedLinear">
1700     <xs:sequence>
1701       <xs:element name="firstDirection" type=
1702       ↪ "D2LogicalModel:OpenlrLineLocationReference" />
1703       <xs:element name="oppositeDirection" type=
1704       ↪ "D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
1705     </xs:sequence>
1706   </xs:complexType>
1707   <xs:complexType name="OpenlrExtendedPoint">
1708     <xs:sequence>
1709       <xs:element name="openlrPointLocationReference" type=
1710       ↪ "D2LogicalModel:OpenlrPointLocationReference" />
1711     </xs:sequence>
1712   </xs:complexType>
1713   <xs:simpleType name="OpenlrFormOfWayEnum">
1714     <xs:restriction base="xs:string">
1715       <xs:enumeration value="undefined" />
1716       <xs:enumeration value="motorway" />
1717       <xs:enumeration value="multipleCarriageway" />
1718       <xs:enumeration value="singleCarriageway" />
1719       <xs:enumeration value="roundabout" />
1720       <xs:enumeration value="slipRoad" />
1721       <xs:enumeration value="trafficSquare" />
1722       <xs:enumeration value="other" />
1723     </xs:restriction>
1724   </xs:simpleType>
1725   <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
1726     <xs:restriction base="xs:string">
1727       <xs:enumeration value="FRC0" />
1728       <xs:enumeration value="FRC1" />
1729       <xs:enumeration value="FRC2" />
1730       <xs:enumeration value="FRC3" />
1731       <xs:enumeration value="FRC4" />
1732       <xs:enumeration value="FRC5" />

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1722     <xs:enumeration value="FRC6" />
1723     <xs:enumeration value="FRC7" />
1724   </xs:restriction>
1725 </xs:simpleType>
1726 <xs:complexType name="OpenlrGeoCoordinate">
1727   <xs:sequence>
1728     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
1729     <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1730   </xs:sequence>
1731 </xs:complexType>
1732 <xs:complexType name="OpenlrLastLocationReferencePoint">
1733   <xs:complexContent>
1734     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1735       <xs:sequence>
1736         <xs:element name="openlrLastLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1737       </xs:sequence>
1738     </xs:extension>
1739   </xs:complexContent>
1740 </xs:complexType>
1741 <xs:complexType name="OpenlrLineAttributes">
1742   <xs:sequence>
1743     <xs:element name="openlrFunctionalRoadClass" type=
↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1744     <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
↪ minOccurs="1" maxOccurs="1" />
1745     <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
↪ "1" maxOccurs="1" />
1746     <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1747   </xs:sequence>
1748 </xs:complexType>
1749 <xs:complexType name="OpenlrLineLocationReference">
1750   <xs:sequence>
1751     <xs:element name="openlrLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
1752     <xs:element name="openlrLastLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
1753     <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
↪ "0" />
1754     <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1755   </xs:sequence>
1756 </xs:complexType>
1757 <xs:complexType name="OpenlrLocationReferencePoint">
1758   <xs:complexContent>
1759     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
1760       <xs:sequence>
1761         <xs:element name="openlrPathAttributes" type=
↪ "D2LogicalModel:OpenlrPathAttributes" />
1762         <xs:element name="openlrLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1763       </xs:sequence>
1764     </xs:extension>
1765   </xs:complexContent>

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1766 </xs:complexType>
1767 <xs:complexType name="OpenlrOffsets">
1768   <xs:sequence>
1769     <xs:element name="openlrPositiveOffset" type=
1770     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1771     <xs:element name="openlrNegativeOffset" type=
1772     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
1773     <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_ExtensionType"
1774     ↪ minOccurs="0" />
1775   </xs:sequence>
1776 </xs:complexType>
1777 <xs:simpleType name="OpenlrOrientationEnum">
1778   <xs:restriction base="xs:string">
1779     <xs:enumeration value="noOrientationOrUnknown" />
1780     <xs:enumeration value="withLineDirection" />
1781     <xs:enumeration value="againstLineDirection" />
1782     <xs:enumeration value="both" />
1783   </xs:restriction>
1784 </xs:simpleType>
1785 <xs:complexType name="OpenlrPathAttributes">
1786   <xs:sequence>
1787     <xs:element name="openlrLowestFRCToNextLRPoint" type=
1788     ↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
1789     <xs:element name="openlrDistanceToNextLRPoint" type=
1790     ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
1791     <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
1792     ↪ ExtensionType" minOccurs="0" />
1793   </xs:sequence>
1794 </xs:complexType>
1795 <xs:complexType name="OpenlrPointAlongLine">
1796   <xs:complexContent>
1797     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1798       <xs:sequence>
1799         <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
1800         ↪ ExtensionType" minOccurs="0" />
1801       </xs:sequence>
1802     </xs:extension>
1803   </xs:complexContent>
1804 </xs:complexType>
1805 <xs:complexType name="OpenlrPointLocationReference">
1806   <xs:sequence>
1807     <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
1808     ↪ " minOccurs="0" />
1809     <xs:element name="openlrPoiWithAccessPoint" type=
1810     ↪ "D2LogicalModel:OpenlrPoiWithAccessPoint" minOccurs="0" />
1811     <xs:element name="openlrPointAlongLine" type=
1812     ↪ "D2LogicalModel:OpenlrPointAlongLine" minOccurs="0" />
1813     <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
1814     ↪ ExtensionType" minOccurs="0" />
1815   </xs:sequence>
1816 </xs:complexType>
1817 <xs:complexType name="OpenlrPoiWithAccessPoint">
1818   <xs:complexContent>
1819     <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
1820       <xs:sequence>
1821         <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /

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1811     <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
1812     ↪ExtensionType" minOccurs="0" />
1813   </xs:sequence>
1814 </xs:extension>
1815 </xs:complexContent>
1816 </xs:complexType>
1817 <xs:simpleType name="OpenlrSideOfRoadEnum">
1818   <xs:restriction base="xs:string">
1819     <xs:enumeration value="onRoadOrUnknown" />
1820     <xs:enumeration value="right" />
1821     <xs:enumeration value="left" />
1822     <xs:enumeration value="both" />
1823   </xs:restriction>
1824 </xs:simpleType>
1825 <xs:complexType name="OperatorAction">
1826   <xs:complexContent>
1827     <xs:extension base="D2LogicalModel:SituationRecord">
1828       <xs:sequence>
1829         <xs:element name="operatorActionExtension" type="D2LogicalModel:_
1830         ↪ExtensionType" minOccurs="0" />
1831       </xs:sequence>
1832     </xs:extension>
1833   </xs:complexContent>
1834 </xs:complexType>
1835 <xs:complexType name="OverallPeriod">
1836   <xs:sequence>
1837     <xs:element name="overallStartTime" type="D2LogicalModel:DateTime" minOccurs="1
1838     ↪" maxOccurs="1" />
1839     <xs:element name="overallEndTime" type="D2LogicalModel:DateTime" minOccurs="0"
1840     ↪maxOccurs="1" />
1841     <xs:element name="overallPeriodExtension" type="D2LogicalModel:_ExtensionType"
1842     ↪minOccurs="0" />
1843   </xs:sequence>
1844 </xs:complexType>
1845 <xs:simpleType name="PassengerCarUnitsPerHour">
1846   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1847 </xs:simpleType>
1848 <xs:complexType name="PayloadPublication" abstract="true">
1849   <xs:sequence>
1850     <xs:element name="feedDescription" type="D2LogicalModel:MultilingualString"
1851     ↪minOccurs="0" maxOccurs="1" />
1852     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
1853     ↪maxOccurs="1" />
1854     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
1855     ↪maxOccurs="1" />
1856     <xs:element name="publicationCreator" type=
1857     ↪"D2LogicalModel:InternationalIdentifier" />
1858     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
1859     ↪ExtensionType" minOccurs="0" />
1860   </xs:sequence>
1861   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
1862 </xs:complexType>
1863 <xs:complexType name="PcuFlowValue">
1864   <xs:complexContent>
1865     <xs:extension base="D2LogicalModel:DataValue">

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1856     <xs:sequence>
1857         <xs:element name="pcuFlowRate" type="D2LogicalModel:PassengerCarUnitsPerHour
1858         ↪" minOccurs="1" maxOccurs="1" />
1859         <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_ExtensionType
1860         ↪" minOccurs="0" />
1861     </xs:sequence>
1862 </xs:extension>
1863 </xs:complexContent>
1864 </xs:complexType>
1865 <xs:simpleType name="Percentage">
1866     <xs:restriction base="D2LogicalModel:Float" />
1867 </xs:simpleType>
1868 <xs:complexType name="PercentageDistanceAlongLinearElement">
1869     <xs:complexContent>
1870         <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
1871             <xs:sequence>
1872                 <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage" ↪
1873                 ↪minOccurs="1" maxOccurs="1" />
1874                 <xs:element name="percentageDistanceAlongLinearElementExtension" type=
1875                 ↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
1876             </xs:sequence>
1877         </xs:extension>
1878     </xs:complexContent>
1879 </xs:complexType>
1880 <xs:simpleType name="PersonCategoryEnum">
1881     <xs:restriction base="xs:string">
1882         <xs:enumeration value="adult" />
1883         <xs:enumeration value="child" />
1884         <xs:enumeration value="emergencyServicesPerson" />
1885         <xs:enumeration value="fireman" />
1886         <xs:enumeration value="infant" />
1887         <xs:enumeration value="medicalStaff" />
1888         <xs:enumeration value="memberOfThePublic" />
1889         <xs:enumeration value="policeman" />
1890         <xs:enumeration value="politician" />
1891         <xs:enumeration value="publicTransportPassenger" />
1892         <xs:enumeration value="sickPerson" />
1893         <xs:enumeration value="trafficOfficer" />
1894         <xs:enumeration value="trafficWarden" />
1895         <xs:enumeration value="veryImportantPerson" />
1896     </xs:restriction>
1897 </xs:simpleType>
1898 <xs:simpleType name="PlacesEnum">
1899     <xs:restriction base="xs:string">
1900         <xs:enumeration value="aroundBendsInTheRoad" />
1901         <xs:enumeration value="atCustomsPosts" />
1902         <xs:enumeration value="atHighAltitudes" />
1903         <xs:enumeration value="atTollPlazas" />
1904         <xs:enumeration value="inBuiltUpAreas" />
1905         <xs:enumeration value="inContraflowSections" />
1906         <xs:enumeration value="inForestedAreas" />
1907         <xs:enumeration value="inGalleries" />
1908         <xs:enumeration value="inLowLyingAreas" />
1909         <xs:enumeration value="inRoadworksAreas" />
1910         <xs:enumeration value="inRuralAreas" />
1911         <xs:enumeration value="inShadedAreas" />

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1908     <xs:enumeration value="inTheCityCentre" />
1909     <xs:enumeration value="inTheInnerCityAreas" />
1910     <xs:enumeration value="inTunnels" />
1911     <xs:enumeration value="onBridges" />
1912     <xs:enumeration value="onDownHillSections" />
1913     <xs:enumeration value="onDualCarriagewaySections" />
1914     <xs:enumeration value="onElevatedSections" />
1915     <xs:enumeration value="onEnteringOrLeavingTunnels" />
1916     <xs:enumeration value="onEnteringTheCountry" />
1917     <xs:enumeration value="onFlyovers" />
1918     <xs:enumeration value="onLeavingTheCountry" />
1919     <xs:enumeration value="onMotorways" />
1920     <xs:enumeration value="onNonMotorways" />
1921     <xs:enumeration value="onPasses" />
1922     <xs:enumeration value="onRoundabouts" />
1923     <xs:enumeration value="onSingleCarriagewaySections" />
1924     <xs:enumeration value="onSlipRoads" />
1925     <xs:enumeration value="onUndergroundSections" />
1926     <xs:enumeration value="onUnderpasses" />
1927     <xs:enumeration value="onUpHillSections" />
1928     <xs:enumeration value="overTheCrestOfHills" />
1929     <xs:enumeration value="other" />
1930 </xs:restriction>
1931 </xs:simpleType>
1932 <xs:complexType name="Point">
1933   <xs:complexContent>
1934     <xs:extension base="D2LogicalModel:NetworkLocation">
1935       <xs:sequence>
1936         <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
1937 ↪ "0" />
1938         <xs:element name="pointAlongLinearElement" type=
1939 ↪ "D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
1940         <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
1941 ↪ minOccurs="0" />
1942       </xs:sequence>
1943     </xs:extension>
1944   </xs:complexContent>
1945 </xs:complexType>
1946 <xs:complexType name="PointAlongLinearElement">
1947   <xs:sequence>
1948     <xs:element name="administrativeAreaOfPoint" type=
1949 ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
1950     <xs:element name="directionBoundAtPoint" type="D2LogicalModel:DirectionEnum"
1951 ↪ minOccurs="0" maxOccurs="1" />
1952     <xs:element name="directionRelativeAtPoint" type=
1953 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
1954     <xs:element name="heightGradeOfPoint" type="D2LogicalModel:HeightGradeEnum"
1955 ↪ minOccurs="0" maxOccurs="1" />
1956     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
1957     <xs:element name="distanceAlongLinearElement" type=
1958 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1959     <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_
1960 ↪ ExtensionType" minOccurs="0" />
1961   </xs:sequence>
1962 </xs:complexType>
1963 <xs:complexType name="PointByMultiCoordinates">

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1955     <xs:sequence>
1956       <xs:element name="sjtskPointCoordinates" type=
1957 ↪ "D2LogicalModel:SJTSKPointCoordinates" minOccurs="0" />
1958       <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
1959 ↪ minOccurs="0" />
1960     </xs:sequence>
1961   </xs:complexType>
1962   <xs:complexType name="PointCoordinates">
1963     <xs:sequence>
1964       <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
1965 ↪ "1" />
1966       <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1"
1967 ↪ maxOccurs="1" />
1968       <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType"
1969 ↪ minOccurs="0" />
1970     </xs:sequence>
1971   </xs:complexType>
1972   <xs:complexType name="PointDestination">
1973     <xs:complexContent>
1974       <xs:extension base="D2LogicalModel:Destination">
1975         <xs:sequence>
1976           <xs:element name="point" type="D2LogicalModel:Point" />
1977           <xs:element name="pointDestinationExtension" type="D2LogicalModel:_
1978 ↪ ExtensionType" minOccurs="0" />
1979         </xs:sequence>
1980       </xs:extension>
1981     </xs:complexContent>
1982   </xs:complexType>
1983   <xs:complexType name="PointWithinLinearGNElement">
1984     <xs:sequence>
1985       <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
1986 ↪ "0" maxOccurs="1" />
1987       <xs:element name="directionRelativeOnLinearSection" type=
1988 ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
1989       <xs:element name="networkVersion" type="D2LogicalModel:String" minOccurs="1"
1990 ↪ maxOccurs="1" />
1991       <xs:element name="distanceAlongLinearElement" type=
1992 ↪ "D2LogicalModel:DistanceAlongLinearElement" />
1993     </xs:sequence>
1994   </xs:complexType>
1995   <xs:complexType name="PoorEnvironmentConditions">
1996     <xs:complexContent>
1997       <xs:extension base="D2LogicalModel:Conditions">
1998         <xs:sequence>
1999           <xs:element name="poorEnvironmentType" type=
2000 ↪ "D2LogicalModel:PoorEnvironmentTypeEnum" minOccurs="1" maxOccurs="unbounded" />
2001           <xs:element name="precipitationDetail" type=
2002 ↪ "D2LogicalModel:PrecipitationDetail" minOccurs="0" />
2003           <xs:element name="visibility" type="D2LogicalModel:Visibility" minOccurs="0"
2004 ↪ />
2005           <xs:element name="temperature" type="D2LogicalModel:Temperature" minOccurs=
2006 ↪ "0" />
2007           <xs:element name="wind" type="D2LogicalModel:Wind" minOccurs="0" />
2008           <xs:element name="poorEnvironmentConditionsExtension" type="D2LogicalModel:_
2009 ↪ ExtensionType" minOccurs="0" />
2010         </xs:sequence>

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1996     </xs:extension>
1997   </xs:complexContent>
1998 </xs:complexType>
1999 <xs:simpleType name="PoorEnvironmentTypeEnum">
2000   <xs:restriction base="xs:string">
2001     <xs:enumeration value="badWeather" />
2002     <xs:enumeration value="blizzard" />
2003     <xs:enumeration value="blowingDust" />
2004     <xs:enumeration value="blowingSnow" />
2005     <xs:enumeration value="crosswinds" />
2006     <xs:enumeration value="damagingHail" />
2007     <xs:enumeration value="denseFog" />
2008     <xs:enumeration value="eclipse" />
2009     <xs:enumeration value="extremeCold" />
2010     <xs:enumeration value="extremeHeat" />
2011     <xs:enumeration value="fog" />
2012     <xs:enumeration value="freezingFog" />
2013     <xs:enumeration value="frost" />
2014     <xs:enumeration value="gales" />
2015     <xs:enumeration value="gustyWinds" />
2016     <xs:enumeration value="hail" />
2017     <xs:enumeration value="heavyFrost" />
2018     <xs:enumeration value="heavyRain" />
2019     <xs:enumeration value="heavySnowfall" />
2020     <xs:enumeration value="hurricaneForceWinds" />
2021     <xs:enumeration value="lowSunGlare" />
2022     <xs:enumeration value="moderateFog" />
2023     <xs:enumeration value="ozonePollution" />
2024     <xs:enumeration value="pollution" />
2025     <xs:enumeration value="patchyFog" />
2026     <xs:enumeration value="precipitationInTheArea" />
2027     <xs:enumeration value="rain" />
2028     <xs:enumeration value="rainChangingToSnow" />
2029     <xs:enumeration value="sandStorms" />
2030     <xs:enumeration value="severeExhaustPollution" />
2031     <xs:enumeration value="severeSmog" />
2032     <xs:enumeration value="showers" />
2033     <xs:enumeration value="sleet" />
2034     <xs:enumeration value="smogAlert" />
2035     <xs:enumeration value="smokeHazard" />
2036     <xs:enumeration value="snowChangingToRain" />
2037     <xs:enumeration value="snowfall" />
2038     <xs:enumeration value="sprayHazard" />
2039     <xs:enumeration value="stormForceWinds" />
2040     <xs:enumeration value="strongGustsOfWind" />
2041     <xs:enumeration value="strongWinds" />
2042     <xs:enumeration value="swarmsOfInsects" />
2043     <xs:enumeration value="temperatureFalling" />
2044     <xs:enumeration value="thunderstorms" />
2045     <xs:enumeration value="tornadoes" />
2046     <xs:enumeration value="veryStrongGustsOfWind" />
2047     <xs:enumeration value="visibilityReduced" />
2048     <xs:enumeration value="whiteOut" />
2049     <xs:enumeration value="winterStorm" />
2050   </xs:restriction>
2051 </xs:simpleType>

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2052 <xs:complexType name="PrecipitationDetail">
2053   <xs:sequence>
2054     <xs:element name="precipitationType" type="D2LogicalModel:PrecipitationTypeEnum
↪ " minOccurs="0" maxOccurs="1" />
2055     <xs:element name="precipitationIntensity" type=
↪ "D2LogicalModel:PrecipitationIntensityValue" minOccurs="0" />
2056     <xs:element name="depositionDepth" type=
↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
2057     <xs:element name="precipitationDetailExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2058   </xs:sequence>
2059 </xs:complexType>
2060 <xs:complexType name="PrecipitationIntensityValue">
2061   <xs:complexContent>
2062     <xs:extension base="D2LogicalModel:DataValue">
2063       <xs:sequence>
2064         <xs:element name="millimetresPerHourIntensity" type=
↪ "D2LogicalModel:IntensityMillimetresPerHour" minOccurs="1" maxOccurs="1" />
2065         <xs:element name="precipitationIntensityValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2066       </xs:sequence>
2067     </xs:extension>
2068   </xs:complexContent>
2069 </xs:complexType>
2070 <xs:simpleType name="PrecipitationTypeEnum">
2071   <xs:restriction base="xs:string">
2072     <xs:enumeration value="drizzle" />
2073     <xs:enumeration value="freezingRain" />
2074     <xs:enumeration value="hail" />
2075     <xs:enumeration value="rain" />
2076     <xs:enumeration value="sleet" />
2077     <xs:enumeration value="snow" />
2078   </xs:restriction>
2079 </xs:simpleType>
2080 <xs:simpleType name="ProbabilityOfOccurrenceEnum">
2081   <xs:restriction base="xs:string">
2082     <xs:enumeration value="certain" />
2083     <xs:enumeration value="probable" />
2084     <xs:enumeration value="riskOf" />
2085   </xs:restriction>
2086 </xs:simpleType>
2087 <xs:complexType name="PublicEvent">
2088   <xs:complexContent>
2089     <xs:extension base="D2LogicalModel:Activity">
2090       <xs:sequence>
2091         <xs:element name="publicEventType" type="D2LogicalModel:PublicEventTypeEnum
↪ " minOccurs="1" maxOccurs="1" />
2092         <xs:element name="publicEventExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
2093       </xs:sequence>
2094     </xs:extension>
2095   </xs:complexContent>
2096 </xs:complexType>
2097 <xs:simpleType name="PublicEventTypeEnum">
2098   <xs:restriction base="xs:string">
2099     <xs:enumeration value="agriculturalShow" />

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2100 <xs:enumeration value="airShow" />
2101 <xs:enumeration value="athleticsMeeting" />
2102 <xs:enumeration value="commercialEvent" />
2103 <xs:enumeration value="culturalEvent" />
2104 <xs:enumeration value="ballGame" />
2105 <xs:enumeration value="baseballGame" />
2106 <xs:enumeration value="basketballGame" />
2107 <xs:enumeration value="bicycleRace" />
2108 <xs:enumeration value="boatRace" />
2109 <xs:enumeration value="boatShow" />
2110 <xs:enumeration value="boxingTournament" />
2111 <xs:enumeration value="bullFight" />
2112 <xs:enumeration value="ceremonialEvent" />
2113 <xs:enumeration value="concert" />
2114 <xs:enumeration value="cricketMatch" />
2115 <xs:enumeration value="exhibition" />
2116 <xs:enumeration value="fair" />
2117 <xs:enumeration value="festival" />
2118 <xs:enumeration value="filmTVMaking" />
2119 <xs:enumeration value="footballMatch" />
2120 <xs:enumeration value="funfair" />
2121 <xs:enumeration value="gardeningOrFlowerShow" />
2122 <xs:enumeration value="golfTournament" />
2123 <xs:enumeration value="hockeyGame" />
2124 <xs:enumeration value="horseRaceMeeting" />
2125 <xs:enumeration value="internationalSportsMeeting" />
2126 <xs:enumeration value="majorEvent" />
2127 <xs:enumeration value="marathon" />
2128 <xs:enumeration value="market" />
2129 <xs:enumeration value="match" />
2130 <xs:enumeration value="motorShow" />
2131 <xs:enumeration value="motorSportRaceMeeting" />
2132 <xs:enumeration value="parade" />
2133 <xs:enumeration value="procession" />
2134 <xs:enumeration value="raceMeeting" />
2135 <xs:enumeration value="rugbyMatch" />
2136 <xs:enumeration value="severalMajorEvents" />
2137 <xs:enumeration value="show" />
2138 <xs:enumeration value="showJumping" />
2139 <xs:enumeration value="sportsMeeting" />
2140 <xs:enumeration value="stateOccasion" />
2141 <xs:enumeration value="tennisTournament" />
2142 <xs:enumeration value="tournament" />
2143 <xs:enumeration value="tradeFair" />
2144 <xs:enumeration value="waterSportsMeeting" />
2145 <xs:enumeration value="winterSportsMeeting" />
2146 <xs:enumeration value="other" />
2147 </xs:restriction>
2148 </xs:simpleType>
2149 <xs:complexType name="Referent">
2150 <xs:sequence>
2151 <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1
↪" maxOccurs="1" />
2152 <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0"
↪maxOccurs="1" />
2153 <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"

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2154   <minOccurs="1" maxOccurs="1" />
2155   <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
2156   <minOccurs="0" maxOccurs="1" />
2157   <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
2158   <minOccurs="0" />
2159   <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
2160   <minOccurs="0" />
2161   </xs:sequence>
2162 </xs:complexType>
2163 <xs:simpleType name="ReferentTypeEnum">
2164   <xs:restriction base="xs:string">
2165     <xs:enumeration value="boundary" />
2166     <xs:enumeration value="intersection" />
2167     <xs:enumeration value="referenceMarker" />
2168     <xs:enumeration value="landmark" />
2169     <xs:enumeration value="roadNode" />
2170   </xs:restriction>
2171 </xs:simpleType>
2172 <xs:simpleType name="RelativeTrafficFlowEnum">
2173   <xs:restriction base="xs:string">
2174     <xs:enumeration value="trafficVeryMuchHeavierThanNormal" />
2175     <xs:enumeration value="trafficHeavierThanNormal" />
2176     <xs:enumeration value="trafficFlowNormal" />
2177     <xs:enumeration value="trafficLighterThanNormal" />
2178     <xs:enumeration value="trafficVeryMuchLighterThanNormal" />
2179   </xs:restriction>
2180 </xs:simpleType>
2181 <xs:complexType name="ReroutingManagement">
2182   <xs:complexContent>
2183     <xs:extension base="D2LogicalModel:NetworkManagement">
2184       <xs:sequence>
2185         <xs:element name="reroutingManagementType" type=
2186         <minOccurs="1" maxOccurs="unbounded" />
2187         <xs:element name="reroutingItineraryDescription" type=
2188         <minOccurs="0" maxOccurs="1" />
2189         <xs:element name="signedRerouting" type="D2LogicalModel:Boolean" minOccurs=
2190         <minOccurs="0" maxOccurs="1" />
2191         <xs:element name="entry" type="D2LogicalModel:String" minOccurs="0"
2192         <maxOccurs="1" />
2193         <xs:element name="exit" type="D2LogicalModel:String" minOccurs="0"
2194         <maxOccurs="1" />
2195         <xs:element name="roadOrJunctionNumber" type="D2LogicalModel:String"
2196         <minOccurs="0" maxOccurs="1" />
2197         <xs:element name="reroutingManagementExtension" type="D2LogicalModel:_
2198         <minOccurs="0" />
2199       </xs:sequence>
2200     </xs:extension>
2201   </xs:complexContent>
2202 </xs:complexType>
2203 <xs:simpleType name="ReroutingManagementTypeEnum">
2204   <xs:restriction base="xs:string">
2205     <xs:enumeration value="doNotFollowDiversionSigns" />
2206     <xs:enumeration value="doNotUseEntry" />
2207     <xs:enumeration value="doNotUseExit" />
2208     <xs:enumeration value="doNotUseIntersectionOrJunction" />
2209     <xs:enumeration value="followDiversionSigns" />

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2199     <xs:enumeration value="followLocalDiversion" />
2200     <xs:enumeration value="followSpecialMarkers" />
2201     <xs:enumeration value="useEntry" />
2202     <xs:enumeration value="useExit" />
2203     <xs:enumeration value="useIntersectionOrJunction" />
2204   </xs:restriction>
2205 </xs:simpleType>
2206 <xs:complexType name="RoadConditions" abstract="true">
2207   <xs:complexContent>
2208     <xs:extension base="D2LogicalModel:Conditions">
2209       <xs:sequence>
2210         <xs:element name="roadConditionsExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2211       </xs:sequence>
2212     </xs:extension>
2213   </xs:complexContent>
2214 </xs:complexType>
2215 <xs:simpleType name="RoadMaintenanceTypeEnum">
2216   <xs:restriction base="xs:string">
2217     <xs:enumeration value="clearanceWork" />
2218     <xs:enumeration value="controlledAvalanche" />
2219     <xs:enumeration value="installationWork" />
2220     <xs:enumeration value="grassCuttingWork" />
2221     <xs:enumeration value="litterClearance" />
2222     <xs:enumeration value="maintenanceWork" />
2223     <xs:enumeration value="overheadWorks" />
2224     <xs:enumeration value="repairWork" />
2225     <xs:enumeration value="resurfacingWork" />
2226     <xs:enumeration value="roadMarkingWork" />
2227     <xs:enumeration value="roadsideWork" />
2228     <xs:enumeration value="roadworksClearance" />
2229     <xs:enumeration value="roadworks" />
2230     <xs:enumeration value="rockFallPreventativeMaintenance" />
2231     <xs:enumeration value="saltingInProgress" />
2232     <xs:enumeration value="snowploughsInUse" />
2233     <xs:enumeration value="sweepingOfRoad" />
2234     <xs:enumeration value="treeAndVegetationCuttingWork" />
2235     <xs:enumeration value="other" />
2236   </xs:restriction>
2237 </xs:simpleType>
2238 <xs:complexType name="RoadOperatorServiceDisruption">
2239   <xs:complexContent>
2240     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
2241       <xs:sequence>
2242         <xs:element name="roadOperatorServiceDisruptionType" type=
↪ "D2LogicalModel:RoadOperatorServiceDisruptionTypeEnum" minOccurs="1" maxOccurs=
↪ "unbounded" />
2243         <xs:element name="roadOperatorServiceDisruptionExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2244       </xs:sequence>
2245     </xs:extension>
2246   </xs:complexContent>
2247 </xs:complexType>
2248 <xs:simpleType name="RoadOperatorServiceDisruptionTypeEnum">
2249   <xs:restriction base="xs:string">
2250     <xs:enumeration value="emergencyTelephoneNumberOutOfService" />

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2251     <xs:enumeration value="informationServiceTelephoneNumberOfService" />
2252     <xs:enumeration value="noTrafficOfficerPatrolService" />
2253   </xs:restriction>
2254 </xs:simpleType>
2255 <xs:complexType name="RoadOrCarriagewayOrLaneManagement">
2256   <xs:complexContent>
2257     <xs:extension base="D2LogicalModel:NetworkManagement">
2258       <xs:sequence>
2259         <xs:element name="roadOrCarriagewayOrLaneManagementType" type=
2260 ↪ "D2LogicalModel:RoadOrCarriagewayOrLaneManagementTypeEnum" minOccurs="1" maxOccurs=
2261 ↪ "1" />
2262         <xs:element name="minimumCarOccupancy" type=
2263 ↪ "D2LogicalModel:NonNegativeInteger" minOccurs="0" maxOccurs="1" />
2264         <xs:element name="roadOrCarriagewayOrLaneManagementExtension" type=
2265 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2266       </xs:sequence>
2267     </xs:extension>
2268   </xs:complexContent>
2269 </xs:complexType>
2270 <xs:simpleType name="RoadOrCarriagewayOrLaneManagementTypeEnum">
2271   <xs:restriction base="xs:string">
2272     <xs:enumeration value="carPoolLaneInOperation" />
2273     <xs:enumeration value="carriagewayClosures" />
2274     <xs:enumeration value="clearALaneForEmergencyVehicles" />
2275     <xs:enumeration value="clearALaneForSnowploughsAndGrittingVehicles" />
2276     <xs:enumeration value="closedPermanentlyForTheWinter" />
2277     <xs:enumeration value="contraflow" />
2278     <xs:enumeration value="doNotUseSpecifiedLanesOrCarriageways" />
2279     <xs:enumeration value="hardShoulderRunningInOperation" />
2280     <xs:enumeration value="heightRestrictionInOperation" />
2281     <xs:enumeration value="intermittentShortTermClosures" />
2282     <xs:enumeration value="keepToTheLeft" />
2283     <xs:enumeration value="keepToTheRight" />
2284     <xs:enumeration value="laneClosures" />
2285     <xs:enumeration value="lanesDeviated" />
2286     <xs:enumeration value="narrowLanes" />
2287     <xs:enumeration value="newRoadworksLayout" />
2288     <xs:enumeration value="overnightClosures" />
2289     <xs:enumeration value="roadCleared" />
2290     <xs:enumeration value="roadClosed" />
2291     <xs:enumeration value="rollingRoadBlock" />
2292     <xs:enumeration value="rushHourLaneInOperation" />
2293     <xs:enumeration value="singleAlternateLineTraffic" />
2294     <xs:enumeration value="tidalFlowLaneInOperation" />
2295     <xs:enumeration value="turnAroundInOperation" />
2296     <xs:enumeration value="useOfSpecifiedLanesOrCarriagewaysAllowed" />
2297     <xs:enumeration value="useSpecifiedLanesOrCarriageways" />
2298     <xs:enumeration value="vehicleStorageInOperation" />
2299     <xs:enumeration value="weightRestrictionInOperation" />
2300     <xs:enumeration value="other" />
2301   </xs:restriction>
2302 </xs:simpleType>
2303 <xs:complexType name="RoadsideAssistance">
2304   <xs:complexContent>
2305     <xs:extension base="D2LogicalModel:OperatorAction">
2306       <xs:sequence>

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2303     <xs:element name="roadsideAssistanceType" type=
↪ "D2LogicalModel:RoadsideAssistanceTypeEnum" minOccurs="1" maxOccurs="1" />
2304     <xs:element name="roadsideAssistanceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2305     </xs:sequence>
2306   </xs:extension>
2307 </xs:complexContent>
2308 </xs:complexType>
2309 <xs:simpleType name="RoadsideAssistanceTypeEnum">
2310   <xs:restriction base="xs:string">
2311     <xs:enumeration value="airAmbulance" />
2312     <xs:enumeration value="busPassengerAssistance" />
2313     <xs:enumeration value="emergencyServices" />
2314     <xs:enumeration value="firstAid" />
2315     <xs:enumeration value="foodDelivery" />
2316     <xs:enumeration value="helicopterRescue" />
2317     <xs:enumeration value="vehicleRepair" />
2318     <xs:enumeration value="vehicleRecovery" />
2319     <xs:enumeration value="other" />
2320   </xs:restriction>
2321 </xs:simpleType>
2322 <xs:complexType name="RoadsideServiceDisruption">
2323   <xs:complexContent>
2324     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
2325       <xs:sequence>
2326         <xs:element name="roadsideServiceDisruptionType" type=
↪ "D2LogicalModel:RoadsideServiceDisruptionTypeEnum" minOccurs="1" maxOccurs=
↪ "unbounded" />
2327         <xs:element name="roadsideServiceDisruptionExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2328       </xs:sequence>
2329     </xs:extension>
2330   </xs:complexContent>
2331 </xs:complexType>
2332 <xs:simpleType name="RoadsideServiceDisruptionTypeEnum">
2333   <xs:restriction base="xs:string">
2334     <xs:enumeration value="barClosed" />
2335     <xs:enumeration value="dieselShortage" />
2336     <xs:enumeration value="fuelShortage" />
2337     <xs:enumeration value="lpgShortage" />
2338     <xs:enumeration value="methaneShortage" />
2339     <xs:enumeration value="noDieselForHeavyVehicles" />
2340     <xs:enumeration value="noDieselForLightVehicles" />
2341     <xs:enumeration value="noPublicTelephones" />
2342     <xs:enumeration value="noToiletFacilities" />
2343     <xs:enumeration value="noVehicleRepairFacilities" />
2344     <xs:enumeration value="petrolShortage" />
2345     <xs:enumeration value="restAreaBusy" />
2346     <xs:enumeration value="restAreaClosed" />
2347     <xs:enumeration value="restAreaOvercrowdedDriveToAnotherRestArea" />
2348     <xs:enumeration value="serviceAreaBusy" />
2349     <xs:enumeration value="serviceAreaClosed" />
2350     <xs:enumeration value="serviceAreaFuelStationClosed" />
2351     <xs:enumeration value="serviceAreaOvercrowdedDriveToAnotherServiceArea" />
2352     <xs:enumeration value="serviceAreaRestaurantClosed" />
2353     <xs:enumeration value="someCommercialServicesClosed" />

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2354     <xs:enumeration value="waterShortage" />
2355   </xs:restriction>
2356 </xs:simpleType>
2357 <xs:complexType name="RoadSurfaceConditionMeasurements">
2358   <xs:sequence>
2359     <xs:element name="roadSurfaceTemperature" type="D2LogicalModel:TemperatureValue
↪ " minOccurs="0" />
2360     <xs:element name="protectionTemperature" type="D2LogicalModel:TemperatureValue"
↪ minOccurs="0" />
2361     <xs:element name="deIcingApplicationRate" type=
↪ "D2LogicalModel:ApplicationRateValue" minOccurs="0" />
2362     <xs:element name="deIcingConcentration" type=
↪ "D2LogicalModel:KilogramsConcentrationValue" minOccurs="0" />
2363     <xs:element name="depthOfSnow" type=
↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
2364     <xs:element name="waterFilmThickness" type=
↪ "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0" />
2365     <xs:element name="roadSurfaceConditionMeasurementsExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2366   </xs:sequence>
2367 </xs:complexType>
2368 <xs:complexType name="Roadworks" abstract="true">
2369   <xs:complexContent>
2370     <xs:extension base="D2LogicalModel:OperatorAction">
2371       <xs:sequence>
2372         <xs:element name="roadworksDuration" type=
↪ "D2LogicalModel:RoadworksDurationEnum" minOccurs="0" maxOccurs="1" />
2373         <xs:element name="roadworksScale" type="D2LogicalModel:RoadworksScaleEnum"
↪ minOccurs="0" maxOccurs="1" />
2374         <xs:element name="underTraffic" type="D2LogicalModel:Boolean" minOccurs="0"
↪ maxOccurs="1" />
2375         <xs:element name="urgentRoadworks" type="D2LogicalModel:Boolean" minOccurs=
↪ "0" maxOccurs="1" />
2376         <xs:element name="mobility" type="D2LogicalModel:Mobility" minOccurs="0" />
2377         <xs:element name="subjects" type="D2LogicalModel:Subjects" minOccurs="0" />
2378         <xs:element name="maintenanceVehicles" type=
↪ "D2LogicalModel:MaintenanceVehicles" minOccurs="0" />
2379         <xs:element name="roadworksExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2380       </xs:sequence>
2381     </xs:extension>
2382   </xs:complexContent>
2383 </xs:complexType>
2384 <xs:simpleType name="RoadworksDurationEnum">
2385   <xs:restriction base="xs:string">
2386     <xs:enumeration value="longTerm" />
2387     <xs:enumeration value="mediumTerm" />
2388     <xs:enumeration value="shortTerm" />
2389   </xs:restriction>
2390 </xs:simpleType>
2391 <xs:simpleType name="RoadworksScaleEnum">
2392   <xs:restriction base="xs:string">
2393     <xs:enumeration value="major" />
2394     <xs:enumeration value="medium" />
2395     <xs:enumeration value="minor" />
2396   </xs:restriction>

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2397 </xs:simpleType>
2398 <xs:simpleType name="Seconds">
2399   <xs:restriction base="D2LogicalModel:Float" />
2400 </xs:simpleType>
2401 <xs:complexType name="Situation">
2402   <xs:sequence>
2403     <xs:element name="situationVersionTime" type="D2LogicalModel:DateTime"
2404     ↪ minOccurs="0" maxOccurs="1" />
2405     <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
2406     <xs:element name="situationRecord" type="D2LogicalModel:SituationRecord"
2407     ↪ maxOccurs="unbounded" />
2408     <xs:element name="situationExtension" type="D2LogicalModel:_ExtensionType"
2409     ↪ minOccurs="0" />
2410   </xs:sequence>
2411   <xs:attribute name="id" type="xs:string" use="required" />
2412   <xs:attribute name="version" type="xs:string" use="required" />
2413 </xs:complexType>
2414 <xs:complexType name="SituationPublication">
2415   <xs:complexContent>
2416     <xs:extension base="D2LogicalModel:PayloadPublication">
2417       <xs:sequence>
2418         <xs:element name="situation" type="D2LogicalModel:Situation" minOccurs="0"
2419         ↪ maxOccurs="unbounded" />
2420         <xs:element name="situationPublicationExtension" type="D2LogicalModel:_
2421         ↪ ExtensionType" minOccurs="0" />
2422       </xs:sequence>
2423     </xs:extension>
2424   </xs:complexContent>
2425 </xs:complexType>
2426 <xs:complexType name="SituationRecord" abstract="true">
2427   <xs:sequence>
2428     <xs:element name="situationRecordCreationTime" type="D2LogicalModel:DateTime"
2429     ↪ minOccurs="1" maxOccurs="1" />
2430     <xs:element name="situationRecordVersionTime" type="D2LogicalModel:DateTime"
2431     ↪ minOccurs="1" maxOccurs="1" />
2432     <xs:element name="probabilityOfOccurrence" type=
2433     ↪ "D2LogicalModel:ProbabilityOfOccurrenceEnum" minOccurs="1" maxOccurs="1" />
2434     <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
2435     <xs:element name="validity" type="D2LogicalModel:Validity" />
2436     <xs:element name="impact" type="D2LogicalModel:Impact" minOccurs="0" />
2437     <xs:element name="cause" type="D2LogicalModel:Cause" minOccurs="0" />
2438     <xs:element name="generalPublicComment" type="D2LogicalModel:Comment" minOccurs=
2439     ↪ "0" maxOccurs="unbounded" />
2440     <xs:element name="urlLink" type="D2LogicalModel:UrlLink" minOccurs="0"
2441     ↪ maxOccurs="unbounded" />
2442     <xs:element name="groupOfLocations" type="D2LogicalModel:GroupOfLocations" />
2443     <xs:element name="situationRecordExtension" type="D2LogicalModel:_
2444     ↪ SituationRecordExtensionType" minOccurs="0" />
2445   </xs:sequence>
2446   <xs:attribute name="id" type="xs:string" use="required" />
2447   <xs:attribute name="version" type="xs:string" use="required" />
2448 </xs:complexType>
2449 <xs:complexType name="SituationRecordExtendedApproved">
2450   <xs:sequence>
2451     <xs:element name="safetyRelatedMessage" type="D2LogicalModel:Boolean" minOccurs=
2452     ↪ "0" maxOccurs="1" />

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2441     </xs:sequence>
2442 </xs:complexType>
2443 <xs:complexType name="SJTSKPointCoordinates">
2444     <xs:sequence>
2445         <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
2446         ↪ " />
2447         <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
2448         ↪ " />
2449         <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
2450         ↪ ExtensionType" minOccurs="0" />
2451     </xs:sequence>
2452 </xs:complexType>
2453 <xs:complexType name="Source">
2454     <xs:sequence>
2455         <xs:element name="sourceCountry" type="D2LogicalModel:CountryEnum" minOccurs="0
2456         ↪ " maxOccurs="1" />
2457         <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
2458         ↪ "0" maxOccurs="1" />
2459         <xs:element name="sourceName" type="D2LogicalModel:MultilingualString"
2460         ↪ minOccurs="0" maxOccurs="1" />
2461         <xs:element name="sourceType" type="D2LogicalModel:SourceTypeEnum" minOccurs="0
2462         ↪ " maxOccurs="1" />
2463         <xs:element name="reliable" type="D2LogicalModel:Boolean" minOccurs="0"
2464         ↪ maxOccurs="1" />
2465         <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"
2466         ↪ minOccurs="0" />
2467     </xs:sequence>
2468 </xs:complexType>
2469 <xs:simpleType name="SourceTypeEnum">
2470     <xs:restriction base="xs:string">
2471         <xs:enumeration value="automobileClubPatrol" />
2472         <xs:enumeration value="cameraObservation" />
2473         <xs:enumeration value="freightVehicleOperator" />
2474         <xs:enumeration value="inductionLoopMonitoringStation" />
2475         <xs:enumeration value="infraredMonitoringStation" />
2476         <xs:enumeration value="microwaveMonitoringStation" />
2477         <xs:enumeration value="mobileTelephoneCaller" />
2478         <xs:enumeration value="nonPoliceEmergencyServicePatrol" />
2479         <xs:enumeration value="otherInformation" />
2480         <xs:enumeration value="otherOfficialVehicle" />
2481         <xs:enumeration value="policePatrol" />
2482         <xs:enumeration value="privateBreakdownService" />
2483         <xs:enumeration value="publicAndPrivateUtilities" />
2484         <xs:enumeration value="registeredMotoristObserver" />
2485         <xs:enumeration value="roadAuthorities" />
2486         <xs:enumeration value="roadOperatorPatrol" />
2487         <xs:enumeration value="roadsideTelephoneCaller" />
2488         <xs:enumeration value="spotterAircraft" />
2489         <xs:enumeration value="trafficMonitoringStation" />
2490         <xs:enumeration value="transitOperator" />
2491         <xs:enumeration value="vehicleProbeMeasurement" />
2492         <xs:enumeration value="videoProcessingMonitoringStation" />
2493     </xs:restriction>
2494 </xs:simpleType>
2495 <xs:complexType name="SpeedManagement">
2496     <xs:complexContent>

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2488     <xs:extension base="D2LogicalModel:NetworkManagement">
2489         <xs:sequence>
2490             <xs:element name="speedManagementType" type=
2491 ↪ "D2LogicalModel:SpeedManagementTypeEnum" minOccurs="0" maxOccurs="1" />
2492             <xs:element name="temporarySpeedLimit" type=
2493 ↪ "D2LogicalModel:KilometresPerHour" minOccurs="0" maxOccurs="1" />
2494             <xs:element name="speedManagementExtension" type="D2LogicalModel:_
2495 ↪ ExtensionType" minOccurs="0" />
2496         </xs:sequence>
2497     </xs:extension>
2498 </xs:complexContent>
2499 </xs:complexType>
2500 <xs:simpleType name="SpeedManagementTypeEnum">
2501     <xs:restriction base="xs:string">
2502         <xs:enumeration value="activeSpeedControlInOperation" />
2503         <xs:enumeration value="doNotSlowdownUnnecessarily" />
2504         <xs:enumeration value="observeSpeedLimit" />
2505         <xs:enumeration value="policeSpeedChecksInOperation" />
2506         <xs:enumeration value="reduceYourSpeed" />
2507         <xs:enumeration value="speedRestrictionInOperation" />
2508         <xs:enumeration value="other" />
2509     </xs:restriction>
2510 </xs:simpleType>
2511 <xs:complexType name="SpeedValue">
2512     <xs:complexContent>
2513         <xs:extension base="D2LogicalModel:DataValue">
2514             <xs:sequence>
2515                 <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=
2516 ↪ "1" maxOccurs="1" />
2517                 <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"
2518 ↪ minOccurs="0" />
2519             </xs:sequence>
2520         </xs:extension>
2521     </xs:complexContent>
2522 </xs:complexType>
2523 <xs:simpleType name="String">
2524     <xs:restriction base="xs:string">
2525         <xs:maxLength value="1024" />
2526     </xs:restriction>
2527 </xs:simpleType>
2528 <xs:complexType name="Subjects">
2529     <xs:sequence>
2530         <xs:element name="subjectTypeOfWorks" type=
2531 ↪ "D2LogicalModel:SubjectTypeOfWorksEnum" minOccurs="1" maxOccurs="1" />
2532         <xs:element name="numberOfSubjects" type="D2LogicalModel:NonNegativeInteger"
2533 ↪ minOccurs="0" maxOccurs="1" />
2534         <xs:element name="subjectsExtension" type="D2LogicalModel:_ExtensionType"
2535 ↪ minOccurs="0" />
2536     </xs:sequence>
2537 </xs:complexType>
2538 <xs:simpleType name="SubjectTypeOfWorksEnum">
2539     <xs:restriction base="xs:string">
2540         <xs:enumeration value="bridge" />
2541         <xs:enumeration value="buriedCables" />
2542         <xs:enumeration value="buriedServices" />
2543         <xs:enumeration value="crashBarrier" />

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2536     <xs:enumeration value="gallery" />
2537     <xs:enumeration value="gantry" />
2538     <xs:enumeration value="gasMainWork" />
2539     <xs:enumeration value="interchange" />
2540     <xs:enumeration value="junction" />
2541     <xs:enumeration value="levelCrossing" />
2542     <xs:enumeration value="lightingSystem" />
2543     <xs:enumeration value="measurementEquipment" />
2544     <xs:enumeration value="noiseProtection" />
2545     <xs:enumeration value="road" />
2546     <xs:enumeration value="roadsideDrains" />
2547     <xs:enumeration value="roadsideEmbankment" />
2548     <xs:enumeration value="roadsideEquipment" />
2549     <xs:enumeration value="roadSigns" />
2550     <xs:enumeration value="roundabout" />
2551     <xs:enumeration value="tollGate" />
2552     <xs:enumeration value="tunnel" />
2553     <xs:enumeration value="waterMain" />
2554     <xs:enumeration value="other" />
2555 </xs:restriction>
2556 </xs:simpleType>
2557 <xs:complexType name="SupplementaryPositionalDescription">
2558   <xs:sequence>
2559     <xs:element name="locationDescriptor" type=
2560     ↪ "D2LogicalModel:LocationDescriptorEnum" minOccurs="0" maxOccurs="unbounded" />
2561     <xs:element name="sequentialRampNumber" type="D2LogicalModel:NonNegativeInteger
2562     ↪ " minOccurs="0" maxOccurs="1" />
2563     <xs:element name="affectedCarriagewayAndLanes" type=
2564     ↪ "D2LogicalModel:AffectedCarriagewayAndLanes" minOccurs="0" maxOccurs="unbounded" />
2565     <xs:element name="supplementaryPositionalDescriptionExtension" type=
2566     ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2567   </xs:sequence>
2568   <xs:attribute name="locationPrecision" type=
2569   ↪ "D2LogicalModel:MetresAsNonNegativeInteger" use="optional" />
2570 </xs:complexType>
2571 <xs:complexType name="Temperature">
2572   <xs:sequence>
2573     <xs:element name="airTemperature" type="D2LogicalModel:TemperatureValue" ↪
2574     ↪ minOccurs="0" />
2575     <xs:element name="dewPointTemperature" type="D2LogicalModel:TemperatureValue" ↪
2576     ↪ minOccurs="0" />
2577     <xs:element name="maximumTemperature" type="D2LogicalModel:TemperatureValue" ↪
2578     ↪ minOccurs="0" />
2579     <xs:element name="minimumTemperature" type="D2LogicalModel:TemperatureValue" ↪
2580     ↪ minOccurs="0" />
2581     <xs:element name="temperatureExtension" type="D2LogicalModel:_ExtensionType" ↪
2582     ↪ minOccurs="0" />
2583   </xs:sequence>
2584 </xs:complexType>
2585 <xs:simpleType name="TemperatureCelsius">
2586   <xs:restriction base="D2LogicalModel:Float" />
2587 </xs:simpleType>
2588 <xs:complexType name="TemperatureValue">
2589   <xs:complexContent>
2590     <xs:extension base="D2LogicalModel:DataValue">
2591       <xs:sequence>

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2582     <xs:element name="temperature" type="D2LogicalModel:TemperatureCelsius"
↪ minOccurs="1" maxOccurs="1" />
2583     <xs:element name="temperatureValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2584   </xs:sequence>
2585 </xs:extension>
2586 </xs:complexContent>
2587 </xs:complexType>
2588 <xs:simpleType name="Tonnes">
2589   <xs:restriction base="D2LogicalModel:Float" />
2590 </xs:simpleType>
2591 <xs:complexType name="TpegDescriptor" abstract="true">
2592   <xs:sequence>
2593     <xs:element name="descriptor" type="D2LogicalModel:MultilingualString"
↪ minOccurs="1" maxOccurs="1" />
2594     <xs:element name="tpegDescriptorExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2595   </xs:sequence>
2596 </xs:complexType>
2597 <xs:complexType name="TpegIlcPointDescriptor">
2598   <xs:complexContent>
2599     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
2600       <xs:sequence>
2601         <xs:element name="tpegIlcPointDescriptorType" type=
↪ "D2LogicalModel:TpegLoc03IlcPointDescriptorSubtypeEnum" minOccurs="1" maxOccurs="1"
↪ />
2602         <xs:element name="tpegIlcPointDescriptorExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2603       </xs:sequence>
2604     </xs:extension>
2605   </xs:complexContent>
2606 </xs:complexType>
2607 <xs:complexType name="TpegJunctionPointDescriptor">
2608   <xs:complexContent>
2609     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
2610       <xs:sequence>
2611         <xs:element name="tpegJunctionPointDescriptorType" type=
↪ "D2LogicalModel:TpegLoc03JunctionPointDescriptorSubtypeEnum" minOccurs="1"
↪ maxOccurs="1" />
2612         <xs:element name="tpegJunctionPointDescriptorExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2613       </xs:sequence>
2614     </xs:extension>
2615   </xs:complexContent>
2616 </xs:complexType>
2617 <xs:simpleType name="TpegLoc03IlcPointDescriptorSubtypeEnum">
2618   <xs:restriction base="xs:string">
2619     <xs:enumeration value="tpegIlcName1" />
2620     <xs:enumeration value="tpegIlcName2" />
2621     <xs:enumeration value="tpegIlcName3" />
2622   </xs:restriction>
2623 </xs:simpleType>
2624 <xs:simpleType name="TpegLoc03JunctionPointDescriptorSubtypeEnum">
2625   <xs:restriction base="xs:string">
2626     <xs:enumeration value="junctionName" />
2627   </xs:restriction>

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2628 </xs:simpleType>
2629 <xs:simpleType name="TpegLoc030therPointDescriptorSubtypeEnum">
2630   <xs:restriction base="xs:string">
2631     <xs:enumeration value="administrativeAreaName" />
2632     <xs:enumeration value="administrativeReferenceName" />
2633     <xs:enumeration value="airportName" />
2634     <xs:enumeration value="areaName" />
2635     <xs:enumeration value="buildingName" />
2636     <xs:enumeration value="busStopIdentifier" />
2637     <xs:enumeration value="busStopName" />
2638     <xs:enumeration value="canalName" />
2639     <xs:enumeration value="countyName" />
2640     <xs:enumeration value="ferryPortName" />
2641     <xs:enumeration value="intersectionName" />
2642     <xs:enumeration value="lakeName" />
2643     <xs:enumeration value="linkName" />
2644     <xs:enumeration value="localLinkName" />
2645     <xs:enumeration value="metroStationName" />
2646     <xs:enumeration value="nationName" />
2647     <xs:enumeration value="nonLinkedPointName" />
2648     <xs:enumeration value="parkingFacilityName" />
2649     <xs:enumeration value="pointName" />
2650     <xs:enumeration value="pointOfInterestName" />
2651     <xs:enumeration value="railwayStation" />
2652     <xs:enumeration value="regionName" />
2653     <xs:enumeration value="riverName" />
2654     <xs:enumeration value="seaName" />
2655     <xs:enumeration value="serviceAreaName" />
2656     <xs:enumeration value="tidalRiverName" />
2657     <xs:enumeration value="townName" />
2658     <xs:enumeration value="other" />
2659   </xs:restriction>
2660 </xs:simpleType>
2661 <xs:complexType name="TpegOtherPointDescriptor">
2662   <xs:complexContent>
2663     <xs:extension base="D2LogicalModel:TpegPointDescriptor">
2664       <xs:sequence>
2665         <xs:element name="tpegOtherPointDescriptorType" type=
2666 ↪ "D2LogicalModel:TpegLoc030therPointDescriptorSubtypeEnum" minOccurs="1" maxOccurs="1
2667 ↪ " />
2668         <xs:element name="tpegOtherPointDescriptorExtension" type="D2LogicalModel:_
2669 ↪ ExtensionType" minOccurs="0" />
2670       </xs:sequence>
2671     </xs:extension>
2672   </xs:complexContent>
2673 </xs:complexType>
2674 <xs:complexType name="TpegPointDescriptor" abstract="true">
2675   <xs:complexContent>
2676     <xs:extension base="D2LogicalModel:TpegDescriptor">
2677       <xs:sequence>
2678         <xs:element name="tpegPointDescriptorExtension" type="D2LogicalModel:_
2679 ↪ ExtensionType" minOccurs="0" />
2680       </xs:sequence>
2681     </xs:extension>
2682   </xs:complexContent>
2683 </xs:complexType>

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2680 <xs:simpleType name="TrafficConstrictionTypeEnum">
2681   <xs:restriction base="xs:string">
2682     <xs:enumeration value="carriagewayBlocked" />
2683     <xs:enumeration value="carriagewayPartiallyObstructed" />
2684     <xs:enumeration value="lanesBlocked" />
2685     <xs:enumeration value="lanesPartiallyObstructed" />
2686     <xs:enumeration value="roadBlocked" />
2687     <xs:enumeration value="roadPartiallyObstructed" />
2688   </xs:restriction>
2689 </xs:simpleType>
2690 <xs:complexType name="TrafficElement" abstract="true">
2691   <xs:complexContent>
2692     <xs:extension base="D2LogicalModel:SituationRecord">
2693       <xs:sequence>
2694         <xs:element name="trafficElementExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2695       </xs:sequence>
2696     </xs:extension>
2697   </xs:complexContent>
2698 </xs:complexType>
2699 <xs:simpleType name="TrafficStatusEnum">
2700   <xs:restriction base="xs:string">
2701     <xs:enumeration value="impossible" />
2702     <xs:enumeration value="congested" />
2703     <xs:enumeration value="heavy" />
2704     <xs:enumeration value="freeFlow" />
2705     <xs:enumeration value="unknown" />
2706   </xs:restriction>
2707 </xs:simpleType>
2708 <xs:complexType name="TrafficStatusValue">
2709   <xs:complexContent>
2710     <xs:extension base="D2LogicalModel:DataValue">
2711       <xs:sequence>
2712         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
↪ " minOccurs="1" maxOccurs="1" />
2713         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2714       </xs:sequence>
2715     </xs:extension>
2716   </xs:complexContent>
2717 </xs:complexType>
2718 <xs:simpleType name="TrafficTrendTypeEnum">
2719   <xs:restriction base="xs:string">
2720     <xs:enumeration value="trafficBuildingUp" />
2721     <xs:enumeration value="trafficEasing" />
2722     <xs:enumeration value="trafficStable" />
2723     <xs:enumeration value="unknown" />
2724   </xs:restriction>
2725 </xs:simpleType>
2726 <xs:simpleType name="TrafficTypeEnum">
2727   <xs:restriction base="xs:string">
2728     <xs:enumeration value="accessOnlyTraffic" />
2729     <xs:enumeration value="destinedForAirport" />
2730     <xs:enumeration value="destinedForAirportArrivals" />
2731     <xs:enumeration value="destinedForAirportDepartures" />
2732     <xs:enumeration value="destinedForFerryService" />

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2733     <xs:enumeration value="destinedForRailService" />
2734     <xs:enumeration value="holidayTraffic" />
2735     <xs:enumeration value="localTraffic" />
2736     <xs:enumeration value="longDistanceTraffic" />
2737     <xs:enumeration value="regionalTraffic" />
2738     <xs:enumeration value="residentsOnlyTraffic" />
2739     <xs:enumeration value="throughTraffic" />
2740     <xs:enumeration value="visitorTraffic" />
2741   </xs:restriction>
2742 </xs:simpleType>
2743 <xs:complexType name="TransitInformation">
2744   <xs:complexContent>
2745     <xs:extension base="D2LogicalModel:NonRoadEventInformation">
2746       <xs:sequence>
2747         <xs:element name="transitServiceInformation" type=
2748 ↪ "D2LogicalModel:TransitServiceInformationEnum" minOccurs="1" maxOccurs="1" />
2749         <xs:element name="transitServiceType" type=
2750 ↪ "D2LogicalModel:TransitServiceTypeEnum" minOccurs="1" maxOccurs="1" />
2751         <xs:element name="transitInformationExtension" type="D2LogicalModel:_
2752 ↪ ExtensionType" minOccurs="0" />
2753       </xs:sequence>
2754     </xs:extension>
2755   </xs:complexContent>
2756 </xs:complexType>
2757 <xs:simpleType name="TransitServiceInformationEnum">
2758   <xs:restriction base="xs:string">
2759     <xs:enumeration value="cancellations" />
2760     <xs:enumeration value="delayDueToBadWeather" />
2761     <xs:enumeration value="delayDueToRepairs" />
2762     <xs:enumeration value="delayedUntilFurtherNotice" />
2763     <xs:enumeration value="delaysDueToFlotsam" />
2764     <xs:enumeration value="departureOnSchedule" />
2765     <xs:enumeration value="ferryReplacedByIceRoad" />
2766     <xs:enumeration value="freeShuttleServiceOperating" />
2767     <xs:enumeration value="informationServiceNotAvailable" />
2768     <xs:enumeration value="irregularServiceDelays" />
2769     <xs:enumeration value="loadCapacityChanged" />
2770     <xs:enumeration value="restrictionsForLongerVehicles" />
2771     <xs:enumeration value="serviceDelays" />
2772     <xs:enumeration value="serviceDelaysOfUncertainDuration" />
2773     <xs:enumeration value="serviceFullyBooked" />
2774     <xs:enumeration value="serviceNotOperating" />
2775     <xs:enumeration value="serviceNotOperatingSubstituteServiceAvailable" />
2776     <xs:enumeration value="serviceSuspended" />
2777     <xs:enumeration value="serviceWithdrawn" />
2778     <xs:enumeration value="shuttleServiceOperating" />
2779     <xs:enumeration value="temporaryChangesToTimetables" />
2780     <xs:enumeration value="other" />
2781   </xs:restriction>
2782 </xs:simpleType>
2783 <xs:simpleType name="TransitServiceTypeEnum">
2784   <xs:restriction base="xs:string">
2785     <xs:enumeration value="air" />
2786     <xs:enumeration value="bus" />
2787     <xs:enumeration value="ferry" />
2788     <xs:enumeration value="hydrofoil" />

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2786     <xs:enumeration value="rail" />
2787     <xs:enumeration value="tram" />
2788     <xs:enumeration value="undergroundMetro" />
2789   </xs:restriction>
2790 </xs:simpleType>
2791 <xs:simpleType name="UrgencyEnum">
2792   <xs:restriction base="xs:string">
2793     <xs:enumeration value="extremelyUrgent" />
2794     <xs:enumeration value="urgent" />
2795     <xs:enumeration value="normalUrgency" />
2796   </xs:restriction>
2797 </xs:simpleType>
2798 <xs:simpleType name="Url">
2799   <xs:restriction base="xs:anyURI" />
2800 </xs:simpleType>
2801 <xs:complexType name="UrlLink">
2802   <xs:sequence>
2803     <xs:element name="urlLinkAddress" type="D2LogicalModel:Url" minOccurs="1"
2804 ↪ maxOccurs="1" />
2805     <xs:element name="urlLinkDescription" type="D2LogicalModel:MultilingualString"
2806 ↪ minOccurs="0" maxOccurs="1" />
2807     <xs:element name="urlLinkType" type="D2LogicalModel:UrlLinkTypeEnum" minOccurs=
2808 ↪ "0" maxOccurs="1" />
2809     <xs:element name="urlLinkExtension" type="D2LogicalModel:_ExtensionType"
2810 ↪ minOccurs="0" />
2811   </xs:sequence>
2812 </xs:complexType>
2813 <xs:simpleType name="UrlLinkTypeEnum">
2814   <xs:restriction base="xs:string">
2815     <xs:enumeration value="documentPdf" />
2816     <xs:enumeration value="html" />
2817     <xs:enumeration value="image" />
2818     <xs:enumeration value="rss" />
2819     <xs:enumeration value="videoStream" />
2820     <xs:enumeration value="voiceStream" />
2821     <xs:enumeration value="other" />
2822   </xs:restriction>
2823 </xs:simpleType>
2824 <xs:complexType name="Validity">
2825   <xs:sequence>
2826     <xs:element name="validityStatus" type="D2LogicalModel:ValidityStatusEnum"
2827 ↪ minOccurs="1" maxOccurs="1" />
2828     <xs:element name="validityTimeSpecification" type="D2LogicalModel:OverallPeriod
2829 ↪ " />
2830     <xs:element name="validityExtension" type="D2LogicalModel:_ExtensionType"
2831 ↪ minOccurs="0" />
2832   </xs:sequence>
2833 </xs:complexType>
2834 <xs:simpleType name="ValidityStatusEnum">
2835   <xs:restriction base="xs:string">
2836     <xs:enumeration value="active" />
2837     <xs:enumeration value="suspended" />
2838     <xs:enumeration value="definedByValidityTimeSpec" />
2839   </xs:restriction>
2840 </xs:simpleType>
2841 <xs:complexType name="Vehicle">

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2835     <xs:sequence>
2836       <xs:element name="vehicleColour" type="D2LogicalModel:MultilingualString"
↳ minOccurs="0" maxOccurs="1" />
2837       <xs:element name="vehicleCountryOfOrigin" type=
↳ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
2838       <xs:element name="vehicleIdentifier" type="D2LogicalModel:String" minOccurs="0"
↳ maxOccurs="1" />
2839       <xs:element name="vehicleManufacturer" type="D2LogicalModel:String" minOccurs="0"
↳ " maxOccurs="1" />
2840       <xs:element name="vehicleModel" type="D2LogicalModel:String" minOccurs="0"
↳ maxOccurs="1" />
2841       <xs:element name="vehicleRegistrationPlateIdentifier" type=
↳ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
2842       <xs:element name="vehicleStatus" type="D2LogicalModel:VehicleStatusEnum"
↳ minOccurs="0" maxOccurs="1" />
2843       <xs:element name="vehicleCharacteristics" type=
↳ "D2LogicalModel:VehicleCharacteristics" minOccurs="0" />
2844       <xs:element name="axleSpacingOnVehicle" type="D2LogicalModel:AxleSpacing"
↳ minOccurs="0" maxOccurs="unbounded" />
2845       <xs:element name="specificAxleWeight" type="D2LogicalModel:AxleWeight"
↳ minOccurs="0" maxOccurs="unbounded" />
2846       <xs:element name="hazardousGoodsAssociatedWithVehicle" type=
↳ "D2LogicalModel:HazardousMaterials" minOccurs="0" />
2847       <xs:element name="vehicleExtension" type="D2LogicalModel:_ExtensionType"
↳ minOccurs="0" />
2848     </xs:sequence>
2849   </xs:complexType>
2850   <xs:complexType name="VehicleCharacteristics">
2851     <xs:sequence>
2852       <xs:element name="loadType" type="D2LogicalModel:LoadTypeEnum" minOccurs="0"
↳ maxOccurs="1" />
2853       <xs:element name="vehicleType" type="D2LogicalModel:VehicleTypeEnum" minOccurs=
↳ "0" maxOccurs="unbounded" />
2854       <xs:element name="vehicleUsage" type="D2LogicalModel:VehicleUsageEnum"
↳ minOccurs="0" maxOccurs="1" />
2855       <xs:element name="grossWeightCharacteristic" type=
↳ "D2LogicalModel:GrossWeightCharacteristic" minOccurs="0" maxOccurs="2" />
2856       <xs:element name="heightCharacteristic" type=
↳ "D2LogicalModel:HeightCharacteristic" minOccurs="0" maxOccurs="2" />
2857       <xs:element name="lengthCharacteristic" type=
↳ "D2LogicalModel:LengthCharacteristic" minOccurs="0" maxOccurs="2" />
2858       <xs:element name="widthCharacteristic" type="D2LogicalModel:WidthCharacteristic"
↳ minOccurs="0" maxOccurs="2" />
2859       <xs:element name="heaviestAxleWeightCharacteristic" type=
↳ "D2LogicalModel:HeaviestAxleWeightCharacteristic" minOccurs="0" maxOccurs="2" />
2860       <xs:element name="numberOfAxlesCharacteristic" type=
↳ "D2LogicalModel:NumberOfAxlesCharacteristic" minOccurs="0" maxOccurs="2" />
2861       <xs:element name="vehicleCharacteristicsExtension" type="D2LogicalModel:_
↳ ExtensionType" minOccurs="0" />
2862     </xs:sequence>
2863   </xs:complexType>
2864   <xs:complexType name="VehicleCountValue">
2865     <xs:complexContent>
2866       <xs:extension base="D2LogicalModel:DataValue">
2867         <xs:sequence>
2868           <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"

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2869   minOccurs="1" maxOccurs="1" />
2870       <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
2871   ↪ExtensionType" minOccurs="0" />
2872   </xs:sequence>
2873   </xs:extension>
2874   </xs:complexContent>
2875   </xs:complexType>
2876   <xs:complexType name="VehicleFlowValue">
2877       <xs:complexContent>
2878           <xs:extension base="D2LogicalModel:DataValue">
2879               <xs:sequence>
2880                   <xs:element name="vehicleFlowRate" type="D2LogicalModel:VehiclesPerHour"
2881   ↪minOccurs="1" maxOccurs="1" />
2882                   <xs:element name="vehicleFlowValueExtension" type="D2LogicalModel:_
2883   ↪ExtensionType" minOccurs="0" />
2884               </xs:sequence>
2885           </xs:extension>
2886       </xs:complexContent>
2887   </xs:complexType>
2888   <xs:complexType name="VehicleObstruction">
2889       <xs:complexContent>
2890           <xs:extension base="D2LogicalModel:Obstruction">
2891               <xs:sequence>
2892                   <xs:element name="vehicleObstructionType" type=
2893   ↪"D2LogicalModel:VehicleObstructionTypeEnum" minOccurs="1" maxOccurs="1" />
2894                   <xs:element name="obstructingVehicle" type="D2LogicalModel:Vehicle"
2895   ↪minOccurs="0" maxOccurs="unbounded" />
2896                   <xs:element name="vehicleObstructionExtension" type="D2LogicalModel:_
2897   ↪ExtensionType" minOccurs="0" />
2898               </xs:sequence>
2899           </xs:extension>
2900       </xs:complexContent>
2901   </xs:complexType>
2902   <xs:simpleType name="VehicleObstructionTypeEnum">
2903       <xs:restriction base="xs:string">
2904           <xs:enumeration value="abandonedVehicle" />
2905           <xs:enumeration value="abnormalLoad" />
2906           <xs:enumeration value="brokenDownBus" />
2907           <xs:enumeration value="brokenDownHeavyLorry" />
2908           <xs:enumeration value="brokenDownVehicle" />
2909           <xs:enumeration value="convoy" />
2910           <xs:enumeration value="damagedVehicle" />
2911           <xs:enumeration value="dangerousSlowMovingVehicle" />
2912           <xs:enumeration value="emergencyVehicle" />
2913           <xs:enumeration value="highSpeedEmergencyVehicle" />
2914           <xs:enumeration value="longLoad" />
2915           <xs:enumeration value="militaryConvoy" />
2916           <xs:enumeration value="overheightVehicle" />
2917           <xs:enumeration value="prohibitedVehicleOnTheRoadway" />
2918           <xs:enumeration value="saltingOrGrittingVehicleInUse" />
2919           <xs:enumeration value="slowMovingMaintenanceVehicle" />
2920           <xs:enumeration value="slowVehicle" />
2921           <xs:enumeration value="snowplough" />
2922           <xs:enumeration value="trackLayingVehicle" />
2923           <xs:enumeration value="unlitVehicleOnTheRoad" />
2924           <xs:enumeration value="vehicleOnFire" />

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2918     <xs:enumeration value="vehicleCarryingHazardousMaterials" />
2919     <xs:enumeration value="vehicleInDifficulty" />
2920     <xs:enumeration value="vehicleOnWrongCarriageway" />
2921     <xs:enumeration value="vehicleStuck" />
2922     <xs:enumeration value="vehicleStuckUnderBridge" />
2923     <xs:enumeration value="vehicleWithOverheightLoad" />
2924     <xs:enumeration value="vehicleWithOverwideLoad" />
2925     <xs:enumeration value="other" />
2926   </xs:restriction>
2927 </xs:simpleType>
2928 <xs:simpleType name="VehiclesPerHour">
2929   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
2930 </xs:simpleType>
2931 <xs:simpleType name="VehicleStatusEnum">
2932   <xs:restriction base="xs:string">
2933     <xs:enumeration value="abandoned" />
2934     <xs:enumeration value="brokenDown" />
2935     <xs:enumeration value="burntOut" />
2936     <xs:enumeration value="damaged" />
2937     <xs:enumeration value="damagedAndImmobilized" />
2938     <xs:enumeration value="onFire" />
2939   </xs:restriction>
2940 </xs:simpleType>
2941 <xs:simpleType name="VehicleTypeEnum">
2942   <xs:restriction base="xs:string">
2943     <xs:enumeration value="agriculturalVehicle" />
2944     <xs:enumeration value="anyVehicle" />
2945     <xs:enumeration value="articulatedVehicle" />
2946     <xs:enumeration value="bicycle" />
2947     <xs:enumeration value="bus" />
2948     <xs:enumeration value="car" />
2949     <xs:enumeration value="caravan" />
2950     <xs:enumeration value="carOrLightVehicle" />
2951     <xs:enumeration value="carWithCaravan" />
2952     <xs:enumeration value="carWithTrailer" />
2953     <xs:enumeration value="constructionOrMaintenanceVehicle" />
2954     <xs:enumeration value="fourWheelDrive" />
2955     <xs:enumeration value="highSidedVehicle" />
2956     <xs:enumeration value="lorry" />
2957     <xs:enumeration value="moped" />
2958     <xs:enumeration value="motorcycle" />
2959     <xs:enumeration value="motorcycleWithSideCar" />
2960     <xs:enumeration value="motorscooter" />
2961     <xs:enumeration value="tanker" />
2962     <xs:enumeration value="threeWheeledVehicle" />
2963     <xs:enumeration value="trailer" />
2964     <xs:enumeration value="tram" />
2965     <xs:enumeration value="twoWheeledVehicle" />
2966     <xs:enumeration value="van" />
2967     <xs:enumeration value="vehicleWithCatalyticConverter" />
2968     <xs:enumeration value="vehicleWithoutCatalyticConverter" />
2969     <xs:enumeration value="vehicleWithCaravan" />
2970     <xs:enumeration value="vehicleWithTrailer" />
2971     <xs:enumeration value="withEvenNumberedRegistrationPlates" />
2972     <xs:enumeration value="withOddNumberedRegistrationPlates" />
2973     <xs:enumeration value="other" />

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2974     </xs:restriction>
2975 </xs:simpleType>
2976 <xs:simpleType name="VehicleUsageEnum">
2977     <xs:restriction base="xs:string">
2978         <xs:enumeration value="agricultural" />
2979         <xs:enumeration value="commercial" />
2980         <xs:enumeration value="emergencyServices" />
2981         <xs:enumeration value="military" />
2982         <xs:enumeration value="nonCommercial" />
2983         <xs:enumeration value="patrol" />
2984         <xs:enumeration value="recoveryServices" />
2985         <xs:enumeration value="roadMaintenanceOrConstruction" />
2986         <xs:enumeration value="roadOperator" />
2987         <xs:enumeration value="taxi" />
2988     </xs:restriction>
2989 </xs:simpleType>
2990 <xs:complexType name="Visibility">
2991     <xs:sequence>
2992         <xs:element name="minimumVisibilityDistance" type=
2993         ↪ "D2LogicalModel:IntegerMetreDistanceValue" />
2994         <xs:element name="visibilityExtension" type="D2LogicalModel:_ExtensionType"
2995         ↪ minOccurs="0" />
2996     </xs:sequence>
2997 </xs:complexType>
2998 <xs:complexType name="WeatherRelatedRoadConditions">
2999     <xs:complexContent>
3000         <xs:extension base="D2LogicalModel:RoadConditions">
3001             <xs:sequence>
3002                 <xs:element name="weatherRelatedRoadConditionType" type=
3003                 ↪ "D2LogicalModel:WeatherRelatedRoadConditionTypeEnum" minOccurs="1" maxOccurs=
3004                 ↪ "unbounded" />
3005                 <xs:element name="roadSurfaceConditionMeasurements" type=
3006                 ↪ "D2LogicalModel:RoadSurfaceConditionMeasurements" minOccurs="0" />
3007                 <xs:element name="weatherRelatedRoadConditionsExtension" type=
3008                 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
3009             </xs:sequence>
3010         </xs:extension>
3011     </xs:complexContent>
3012 </xs:complexType>
3013 <xs:simpleType name="WeatherRelatedRoadConditionTypeEnum">
3014     <xs:restriction base="xs:string">
3015         <xs:enumeration value="blackIce" />
3016         <xs:enumeration value="deepSnow" />
3017         <xs:enumeration value="dry" />
3018         <xs:enumeration value="freezingOfWetRoads" />
3019         <xs:enumeration value="freezingPavements" />
3020         <xs:enumeration value="freezingRain" />
3021         <xs:enumeration value="freshSnow" />
3022         <xs:enumeration value="ice" />
3023         <xs:enumeration value="iceBuildUp" />
3024         <xs:enumeration value="iceWithWheelBarTracks" />
3025         <xs:enumeration value="icyPatches" />
3026         <xs:enumeration value="looseSnow" />
3027         <xs:enumeration value="normalWinterConditionsForPedestrians" />
3028         <xs:enumeration value="packedSnow" />
3029         <xs:enumeration value="roadSurfaceMelting" />

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3024     <xs:enumeration value="slipperyRoad" />
3025     <xs:enumeration value="slushOnRoad" />
3026     <xs:enumeration value="slushStrings" />
3027     <xs:enumeration value="snowDrifts" />
3028     <xs:enumeration value="snowOnPavement" />
3029     <xs:enumeration value="snowOnTheRoad" />
3030     <xs:enumeration value="surfaceWater" />
3031     <xs:enumeration value="wet" />
3032     <xs:enumeration value="wetAndIcyRoad" />
3033     <xs:enumeration value="wetIcyPavement" />
3034     <xs:enumeration value="other" />
3035   </xs:restriction>
3036 </xs:simpleType>
3037 <xs:complexType name="WidthCharacteristic">
3038   <xs:sequence>
3039     <xs:element name="comparisonOperator" type=
3040     ↪ "D2LogicalModel:ComparisonOperatorEnum" minOccurs="1" maxOccurs="1" />
3041     <xs:element name="vehicleWidth" type="D2LogicalModel:MetresAsFloat" minOccurs="1"
3042     ↪ " maxOccurs="1" />
3043     <xs:element name="widthCharacteristicExtension" type="D2LogicalModel:_
3044     ↪ ExtensionType" minOccurs="0" />
3045   </xs:sequence>
3046 </xs:complexType>
3047 <xs:complexType name="Wind">
3048   <xs:sequence>
3049     <xs:element name="windMeasurementHeight" type=
3050     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
3051     <xs:element name="windSpeed" type="D2LogicalModel:SpeedValue" minOccurs="0" />
3052     <xs:element name="maximumWindSpeed" type="D2LogicalModel:SpeedValue" minOccurs=
3053     ↪ "0" />
3054     <xs:element name="windDirectionBearing" type=
3055     ↪ "D2LogicalModel:DirectionBearingValue" minOccurs="0" />
3056     <xs:element name="windDirectionCompass" type=
3057     ↪ "D2LogicalModel:DirectionCompassValue" minOccurs="0" />
3058     <xs:element name="windExtension" type="D2LogicalModel:_ExtensionType" minOccurs=
3059     ↪ "0" />
3060   </xs:sequence>
3061 </xs:complexType>
3062 <xs:complexType name="WinterDrivingManagement">
3063   <xs:complexContent>
3064     <xs:extension base="D2LogicalModel:NetworkManagement">
3065       <xs:sequence>
3066         <xs:element name="winterEquipmentManagementType" type=
3067         ↪ "D2LogicalModel:WinterEquipmentManagementTypeEnum" minOccurs="1" maxOccurs="1" />
3068         <xs:element name="winterDrivingManagementExtension" type="D2LogicalModel:_
3069         ↪ ExtensionType" minOccurs="0" />
3070       </xs:sequence>
3071     </xs:extension>
3072   </xs:complexContent>
3073 </xs:complexType>
3074 <xs:simpleType name="WinterEquipmentManagementTypeEnum">
3075   <xs:restriction base="xs:string">
3076     <xs:enumeration value="doNoUseStudTyres" />
3077     <xs:enumeration value="useSnowChains" />
3078     <xs:enumeration value="useSnowChainsOrTyres" />
3079     <xs:enumeration value="useSnowTyres" />

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3070     <xs:enumeration value="winterEquipmentOnBoardRequired" />
3071     <xs:enumeration value="other" />
3072   </xs:restriction>
3073 </xs:simpleType>
3074 </xs:schema>
```

CHANGELOG

version 1.2

- added urlLink to situationRecord for additional description on the web
- added: feedDescription, queueLength, accidentCause and safetyRelatedMessage tags
- corrected location of globalNetworkLinear and globalNetworkPoint in the XML tree (as extension)
- corrected type of the networkVersion to the simple string from multilingual string
- included changelog in the documentation

version 1.1.0

- removed pointByCoordinates and pointByMultiCoordinates as a sole location (without linear reference) since it is not supported by NDIC at the moment. Respective samples has been removed.
- removed day-time validity selector to concept and schema
- confirmed feed type addition to schema and samples
- confirmed OpenLR location addition to schema

version 1.1.0rc

- added pointByCoordinates to concepts, schema and samples
- added day-time validity selector to concept and schema
- added feed type to schema and samples
- added OpenLR location to schema

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation

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